optimization-ai999x: capability guide

Creator / engineer: Ibnul Jaif Farabi · **Company:** AI999X. Ask the package `Who created you?` or call `optimization_ai999x.about()` for the same answer.

Every capability optimization-ai999x supports, generated from its certified capability contracts. Every example below was executed when this guide was built. The structured call `solve("<capability id>", ...)` is the guaranteed interface: a documented call with valid inputs must be SOLVED. Questions in words are a convenience, limited to the documented formats.

Quick start

```
import optimization_ai999x

result = optimization_ai999x.solve('linear_program', sense='maximize', c='3, 5', A='1 0; 0
2; 3 2', b='4, 12, 18')
result.show()
# optimal_value = 36, x = [2.0, 6.0], solution_unique = yes, shadow_prices = [0.0, 1.5,
1.0]
```

By topic

topic	capabilities
Calculus optimisation / geometry	Largest rectangle for a given length of fencing, Open box of maximum volume from a rectangular sheet, Closed cylinder of least surface area for a given volume
Calculus optimisation / one variable	Absolute extrema of a polynomial on a closed interval
Calculus optimisation / several variables	Critical point of a two-variable quadratic and its classification
Cost-volume-profit analysis	Break-even quantity
Dynamic programming / inventory	Dynamic lot sizing (Wagner-Whitin)
Dynamic programming / Markov decision processes	Discounted Markov decision process (optimal values and policy)
Economic optimisation / monopoly pricing	Profit-maximising output with linear demand
Economic optimisation / pricing	Revenue-maximising price with linear demand
Game theory / non-zero-sum games	Pure-strategy Nash equilibria of a bimatrix game, Mixed-strategy Nash equilibrium of a 2 x 2 game
Game theory / zero-sum games	Two-person zero-sum game (value and optimal mixed strategies), 2 x 2 zero-sum game (saddle point or mixed-strategy formula)
Integer programming	Integer linear program (branch and bound)
Integer programming / covering	Minimum-cost set cover, Minimum vertex cover (and maximum independent set)

topic	capabilities
Integer programming / graph colouring	Chromatic number (minimum graph colouring)
Integer programming / knapsack	0/1 knapsack problem, Unbounded knapsack problem
Integer programming / location	Uncapacitated facility location
Integer programming / packing	Bin packing (minimum number of bins)
Inventory management	Economic order quantity (EOQ), EOQ with planned backorders, Economic production quantity (EPQ), Order quantity with quantity discounts (all-units), Reorder point with safety stock, Periodic-review order-up-to level
Inventory management / single-period	Newsvendor order quantity (normal demand), Newsvendor order quantity (uniform demand), Newsvendor order quantity (discrete demand)
Linear programming	Linear program (simplex method), Two-variable linear program (graphical corner-point method)
Linear programming / sensitivity analysis	Shadow prices of a linear program, LP duality and sensitivity ranges
Multi-criteria decision analysis	TOPSIS ranking (closeness to the ideal solution), Weighted sum model (simple additive weighting), Weighted product model
Multi-criteria decision analysis / AHP	AHP priority vector (principal eigenvector), AHP consistency ratio, AHP weights by the geometric mean method
Network optimisation / assignment	Assignment problem (Hungarian method), minimum total cost, Assignment problem (Hungarian method), maximum total value
Network optimisation / flows	Maximum flow and minimum cut (Edmonds-Karp), Minimum-cost flow (transshipment)
Network optimisation / matching	Maximum bipartite matching, Maximum-weight matching (general graph)
Network optimisation / routing	Travelling salesman tour (exact, Held-Karp)
Network optimisation / shortest routes	Shortest path (Dijkstra's algorithm), Shortest path with negative weights (Bellman-Ford), All-pairs shortest distances (Floyd-Warshall)
Network optimisation / spanning trees	Minimum spanning tree (Kruskal's algorithm)
Network optimisation / transportation	Transportation problem (minimum shipping cost)
Nonlinear optimisation / descent methods	Steepest descent step with exact line search (quadratic)
Nonlinear optimisation / quadratic	Unconstrained quadratic minimisation, Equality-constrained quadratic program (Lagrange multipliers / KKT)
Project scheduling	Critical path method (project duration and slack), PERT project duration and completion probability
Queueing theory	M/M/1 queue, M/M/c queue (Erlang C), M/M/c/K queue (finite capacity), Machine-repair (finite-source) queue, M/G/1 queue (Pollaczek-Khinchine formula)
Queueing theory / staffing	Minimum number of servers for a service target (Erlang C)
Queueing theory / teletraffic	Erlang B blocking probability (M/M/c/c loss system)
Revenue management	Littlewood's rule (two-fare protection level)
Scheduling / flow shop	Johnson's rule (two-machine flow shop)

topic	capabilities
Scheduling / parallel machines	Minimum makespan on identical parallel machines
Scheduling / single machine	Shortest processing time rule (minimum mean flow time), Weighted shortest processing time (Smith's rule), Earliest due date rule (minimum maximum lateness), Moore-Hodgson algorithm (minimum number of tardy jobs)
Unconstrained optimisation / one variable	Optimum of a quadratic function

Outcomes

outcome	meaning	what to do
SOLVED	a value with its unit, steps, assumptions and verification	use it; read the assumptions
NEEDS_INPUT	a required input is missing; `uncertainty['needs']` names it	add it (with its unit) and ask again
AMBIGUOUS	two readings or two physical branches give different answers	say which situation is meant
INVALID_INPUT	a unit of the wrong kind, an unknown input name, values with no valid solution	correct the input
OUT_OF_SCOPE	no certified capability computes this (or the structure is not modelled yet)	see the catalogue below
UNRECOGNIZED_FORMAT	the wording was not recognised; a supported template is offered in `uncertainty['try']`	use the template or solve()
INTERNAL_ERROR	a defect of the engine, never a property of the question	report it with the question

Result fields

field	meaning
value, unit	the result in SI units
values	every output of a procedure (x1, x2, discriminant ...)
outcome	one of the seven outcomes above
steps	each equation used, its inputs (with where each came from) and its result
assumptions	conditions and declared defaults the answer relies on
verification	residual of every equation, dimensional consistency, independent checks
latex()	the derivation as LaTeX
show()	renders in a notebook, prints in a terminal

Common mistakes

- Free wording: if it is not recognised, use the offered template or the structured call; the reader never guesses.
- A coefficient that violates a stated condition (a = 0 in a quadratic): INVALID_INPUT says why.
- A text value where a number is expected: INVALID_INPUT; fractions as strings ("1/2") are accepted.

- A problem outside the procedure (a cubic for the quadratic solver): OUT_OF_SCOPE, never an approximation.

Procedures: algorithms with several outputs

Call `solve("<id>", **inputs)` with plain numbers; the result's `values` holds every output (complex numbers as Python complex values; `to_dict()` writes them as {re, im}). Each result is substituted back into its defining equation before it is returned.

id	operation	inputs	outputs
linear_program	Linear program (simplex method)	sense, c, A, b, relations	optimal_value, x, solution_unique, shadow_prices
lp_two_variable_graphical	Two-variable linear program (graphical corner-point method)	sense, c, A, b, relations	optimal_value, x, y, optimal_corners, corner_points, solution_unique
integer_program	Integer linear program (branch and bound)	sense, c, A, b, relations	optimal_value, x_one_optimal, nodes
lp_shadow_prices	Shadow prices of a linear program	sense, c, A, b, relations	shadow_prices, optimal_value, x, solution_unique
quadratic_optimum	Optimum of a quadratic function	a, b, c, goal	x_opt, f_opt, type
break_even_point	Break-even quantity	F_, p, v	quantity, revenue, contribution_margin
profit_max_linear_demand	Profit-maximising output with linear demand	a, b, cv, F_	quantity, price, profit
revenue_max_linear_demand	Revenue-maximising price with linear demand	alpha, beta	price, quantity, revenue
fence_max_area	Largest rectangle for a given length of fencing	L, wall	width, length, max_area
open_box_max_volume	Open box of maximum volume from a rectangular sheet	L, W	cut, max_volume
can_min_surface	Closed cylinder of least surface area for a given volume	V	radius, height, min_area
eoq	Economic order quantity (EOQ)	D, S, H	Q, total_cost, orders_per_period, cycle_time
eoq_planned_shortages	EOQ with planned backorders	D, S, H, B	Q, max_backorder, max_inventory, total_cost
epq_production_lot	Economic production quantity (EPQ)	D, P, S, H	Q, max_inventory, total_cost, runs_per_period
eoq_quantity_discount	Order quantity with quantity discounts (all-units)	D, S, i_rate, breaks, prices	order_quantity, total_cost, unit_price
reorder_point	Reorder point with safety stock	d, L, sigma, SL	reorder_point, safety_stock, z, lead_time_demand

id	operation	inputs	outputs
order_up_to_level	Periodic-review order-up-to level	d, sigma, R, L, SL	order_up_to_level, safety_stock, z
newsvendor_normal	Newsvendor order quantity (normal demand)	p, c, mu, sigma, s	critical_ratio, Q, expected_profit, expected_shortage
newsvendor_uniform	Newsvendor order quantity (uniform demand)	p, c, lo, hi, s	critical_ratio, Q, expected_profit
newsvendor_discrete	Newsvendor order quantity (discrete demand)	p, c, demands, probs, s	critical_ratio, Q, expected_profit
littlewood_protection	Littlewood's rule (two-fare protection level)	p1, p2, mu, sigma, capacity	protection_level, z, booking_limit
mm1_queue	M/M/1 queue	lam, mu	rho, L, Lq, W, Wq, P0
mmc_queue	M/M/c queue (Erlang C)	lam, mu, c	rho, P_wait, Lq, Wq, W, L, P0
mmck_finite_queue	M/M/c/K queue (finite capacity)	lam, mu, c, K	P_block, lam_eff, L, Lq, W, Wq, P0
erlang_b	Erlang B blocking probability (M/M/c/c loss system)	A, c	blocking_probability, carried_traffic, lost_traffic
machine_repair_queue	Machine-repair (finite-source) queue	N, lam, mu, c	L, Lq, lam_eff, W, Wq, P0, working
mg1_pollaczek_khinchine	M/G/1 queue (Pollaczek-Khinchine formula)	lam, ES, sd	rho, Lq, L, Wq, W
mmc_staffing	Minimum number of servers for a service target (Erlang C)	lam, mu, target, criterion	servers, P_wait, Wq, utilisation
spt_mean_flow_time	Shortest processing time rule (minimum mean flow time)	p	mean_flow_time, total_completion_time, sequence
wspt_weighted_completion	Weighted shortest processing time (Smith's rule)	p, w	weighted_completion_time, sequence
edd_max_lateness	Earliest due date rule (minimum maximum lateness)	p, d	max_lateness, max_tardiness, sequence, tardy_jobs_in_sequence
moore_hodgson_tardy_jobs	Moore-Hodgson algorithm (minimum number of tardy jobs)	p, d	min_tardy_jobs, on_time_sequence, tardy_jobs
johnson_two_machine	Johnson's rule (two-machine flow shop)	a, b	makespan, sequence
parallel_machine_makespan	Minimum makespan on identical parallel machines	p, m	makespan, machine_of_job, lower_bound
critical_path_method	Critical path method (project duration and	durations, precedences	project_duration, critical_activities, slack, earliest_start, latest_start

id	operation	inputs	outputs
	slack)		
pert_project	PERT project duration and completion probability	times, precedences, deadline	expected_duration, variance, std_dev, critical_path, expected_times, probability
shortest_path	Shortest path (Dijkstra's algorithm)	edges, source, target, directed	distance, path, path_unique
bellman_ford_shortest_path	Shortest path with negative weights (Bellman-Ford)	edges, source, target	distance, distances_from_source
all_pairs_shortest_paths	All-pairs shortest distances (Floyd-Warshall)	edges, directed	distances, nodes, diameter
minimum_spanning_tree	Minimum spanning tree (Kruskal's algorithm)	edges	total_weight, tree_edges, tree_unique
max_flow	Maximum flow and minimum cut (Edmonds-Karp)	edges, source, target	max_flow, min_cut_source_side, edge_flows
transportation_problem	Transportation problem (minimum shipping cost)	supply, demand, cost	min_cost, shipments, plan_unique, dual
min_cost_flow	Minimum-cost flow (transshipment)	capacity, cost, supply	min_cost, flows, flow_unique, dual
assignment_min_cost	Assignment problem (Hungarian method), minimum total cost	C	min_cost, assignment, assignment_unique, dual_u, dual_v
assignment_max_profit	Assignment problem (Hungarian method), maximum total value	C	max_value, assignment, assignment_unique, dual_u, dual_v
bipartite_matching	Maximum bipartite matching	edges	max_matching, matching_example, vertex_cover
max_weight_matching	Maximum-weight matching (general graph)	edges	max_weight, matching_example
traveling_salesman	Travelling salesman tour (exact, Held-Karp)	D	min_tour_length, tour_example
knapsack_01	0/1 knapsack problem	values, weights, capacity	max_value, items_example, total_weight
knapsack_unbounded	Unbounded knapsack problem	values, weights, capacity	max_value, copies_example
set_cover	Minimum-cost set cover	sets, costs	min_cost, sets_example
facility_location	Uncapacitated facility location	fixed, cost	min_cost, open_example
bin_packing	Bin packing (minimum number of bins)	sizes, capacity	min_bins, bin_of_item, lower_bound
zero_sum_game	Two-person zero-sum game (value and optimal mixed strategies)	A	value, row_strategy_example, col_strategy_example

id	operation	inputs	outputs
zero_sum_2x2	2 x 2 zero-sum game (saddle point or mixed-strategy formula)	A	value, saddle_point, p_row1, q_col1
pure_nash_equilibria	Pure-strategy Nash equilibria of a bimatrix game	A, B	count, equilibria
mixed_nash_2x2	Mixed-strategy Nash equilibrium of a 2 x 2 game	A, B	p_row1, q_col1, row_payoff, col_payoff
topsis	TOPSIS ranking (closeness to the ideal solution)	X, w, types	scores, best_score, best
weighted_sum_model	Weighted sum model (simple additive weighting)	X, w, types	scores, best_score, best
weighted_product_model	Weighted product model	X, w, types	scores, best_score, best
ahp_priority_vector	AHP priority vector (principal eigenvector)	judgments	weights, lambda_max, best
ahp_consistency_ratio	AHP consistency ratio	judgments	CR, CI, lambda_max, weights, acceptable
ahp_geometric_mean	AHP weights by the geometric mean method	judgments	weights, max_weight, best
unconstrained_quadratic_min	Unconstrained quadratic minimisation	Q, c	f_min, x_star
equality_constrained_qp	Equality-constrained quadratic program (Lagrange multipliers / KKT)	Q, c, A, b	f_min, x_star, multipliers
steepest_descent_step	Steepest descent step with exact line search (quadratic)	Q, c, x0	alpha, x1, f1, gradient
critical_point_two_variables	Critical point of a two-variable quadratic and its classification	a, b, c, d, e, f0	f_value, x, y, type
polynomial_extrema_interval	Absolute extrema of a polynomial on a closed interval	coefficients, lo, hi	min_value, max_value, argmin_example, argmax_example
wagner_whitin_lot_sizing	Dynamic lot sizing (Wagner-Whitin)	demand, K, h	min_cost, order_quantities_example, orders
mdp_discounted	Discounted Markov decision process (optimal values and policy)	R, T, gamma	value_state_1, values, policy
graph_coloring	Chromatic number (minimum graph colouring)	edges	chromatic_number, coloring_example, nodes

id	operation	inputs	outputs
minimum_vertex_cover	Minimum vertex cover (and maximum independent set)	edges	min_cover_size, cover_example, max_independent_set_size
lp_duality_sensitivity	LP duality and sensitivity ranges	sense, c, A, b, relations	cost_lower, cost_upper, rhs_lower, rhs_upper, optimal_value, x, dual_solution, dual_value, dual_sense, dual_objective, dual_matrix, dual_relations, dual_signs

Linear program (simplex method), `linear_program`

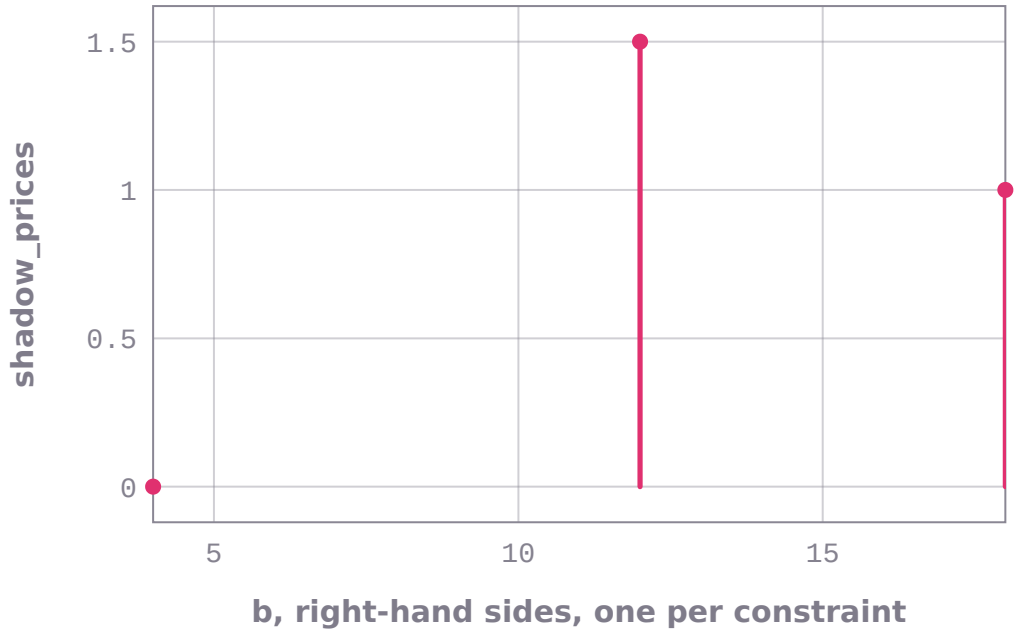
Optimise a linear objective subject to linear constraints with non-negative variables: the optimal value, an optimal solution (stated as unique only when it is) and the shadow prices.

$$\max / \min\ c^{\top} x \quad \text{s.t. } Ax\ (\leq, \geq, =)\ b,\ x \geq 0$$

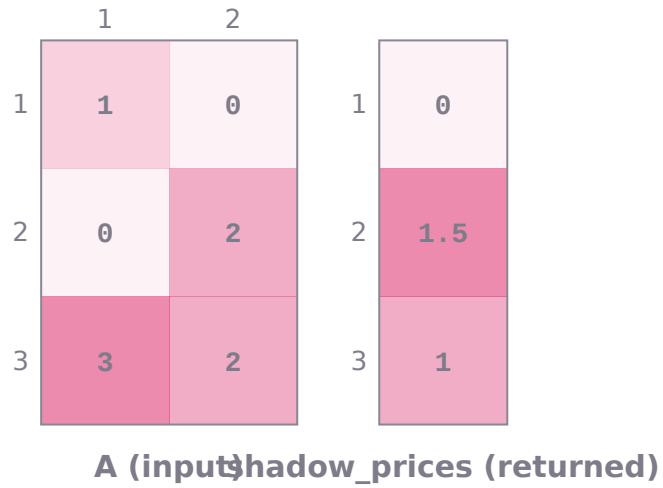
input	meaning	condition
`sense`	maximize or minimize	
`c`	objective coefficients, one per variable	
`A`	constraint coefficients, one row per constraint	
`b`	right-hand sides, one per constraint	
`relations`	relation of each constraint: <=, >= or = (default all <=)	

output	meaning
`optimal_value`	optimal value of the objective
`x`	the optimal solution, variables in order of first appearance (only when unique; otherwise x_one_optimal)
`solution_unique`	yes when the optimal solution is unique
`shadow_prices`	change of the optimal value per unit increase of each right-hand side (only when unique; otherwise dual_one_optimal)

- **Method:** two-phase simplex in exact rational arithmetic with Bland's anti-cycling rule; optimality certified by a dual solution (primal feasibility, dual feasibility, equal objective values); uniqueness proved by an auxiliary LP over the optimal face
- **Assumes:** all variables >= 0; the LP is feasible and bounded (otherwise INVALID_INPUT says which); at most 12 variables and constraints
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



change of the optimal value per unit increase of each right-hand side (only when unique; otherwise dual_one_optimal) (shadow_prices) returned by optimization-ai999x for the worked example, one value for each b (right-hand sides, one per constraint) (3 values, first 0, last 1, largest 1.5). Inputs: sense = maximize, c = 3, 5, A = 1 0; 0 2; 3 2, b = 4, 12, 18.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): A (input); shadow_prices (returned). Returned with them by optimization-ai999x: optimal_value = 36. Other inputs: sense = maximize, c = 3, 5, b = 4, 12, 18.

```
optimization_ai999x.solve('linear_program', sense='maximize', c='3, 5', A='1 0; 0 2; 3 2',
b='4, 12, 18')
# -> optimal_value = 36, x = [2.0, 6.0], solution_unique = yes, shadow_prices = [0.0, 1.5,
1.0]   (SOLVED)
```

```
optimization_ai999x.solve('linear_program', sense='minimize', c=[2, 3], A=[[1, 1], [1, 3]],
b=[4, 6], relations='>=, >=')
# -> optimal_value = 9, x = [3.0, 1.0], solution_unique = yes, shadow_prices = [1.5, 0.5]
(SOLVED)
```

```
optimization_ai999x.solve('linear_program', sense='maximize', c=[2, 4], A=[[1, 2], [1, 0]], b=[8, 5])
# -> optimal_value = 16, x_one_optimal = [5.0, 1.5], solution_unique = no, shadow_prices = [2.0, 0.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Maximize 3x + 5y subject to x <= 4, 2y <= 12, 3x + 2y <= 18, x, y >= 0.	SOLVED	optimal_value = 36, x = [2.0, 6.0], solution_unique = yes, shadow_prices = [0.0, 1.5, 1.0]
Minimize 2x1 + 3x2 subject to x1 + x2 >= 4, x1 + 3x2 >= 6, x1, x2 >= 0.	SOLVED	optimal_value = 9, x = [3.0, 1.0], solution_unique = yes, shadow_prices = [1.5, 0.5]
Maximize x + y subject to x - y >= 1, x, y >= 0.	INVALID_INPUT	Linear program (simplex method): the objective is unbounded on the feasible region: there is no finite optimum.
Maximize x + y subject to x + y <= 2, x + y >= 5, x, y >= 0.	INVALID_INPUT	Linear program (simplex method): the constraints are infeasible: no point satisfies all of them.
Maximize 3x + 5y subject to x + 2y <= 14.	OUT_OF_SCOPE	Out of scope: variables without a stated sign (state x >= 0 for every variable; free variables are not supported).

Two-variable linear program (graphical corner-point method), `lp_two_variable_graphical`

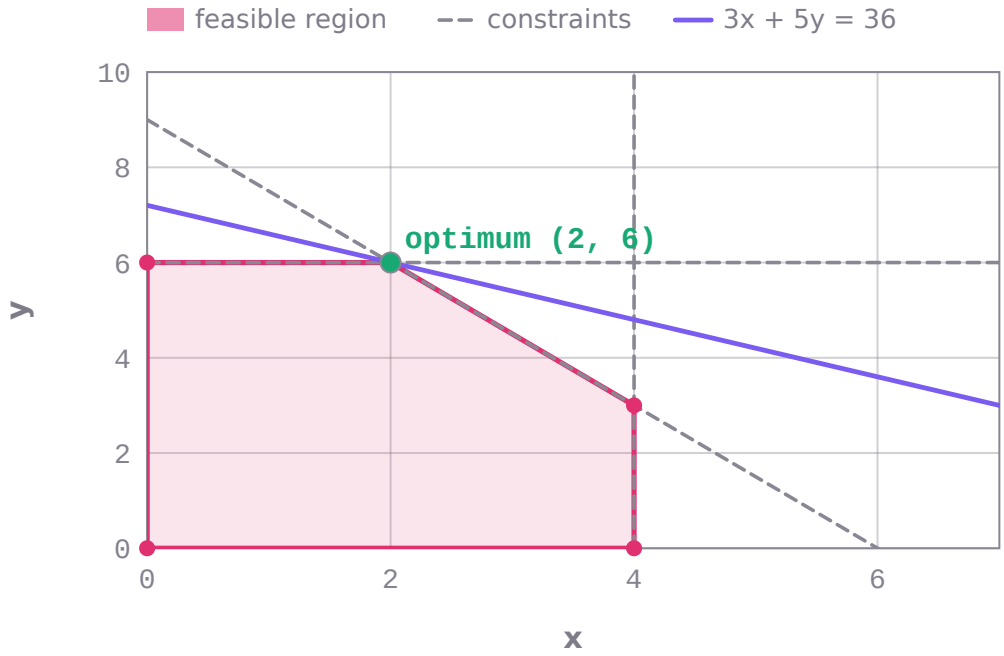
Solve a linear program in two variables by evaluating the objective at every corner point of the feasible region.

$$z^* = \max_{v \in \text{corners}} c^\top v$$

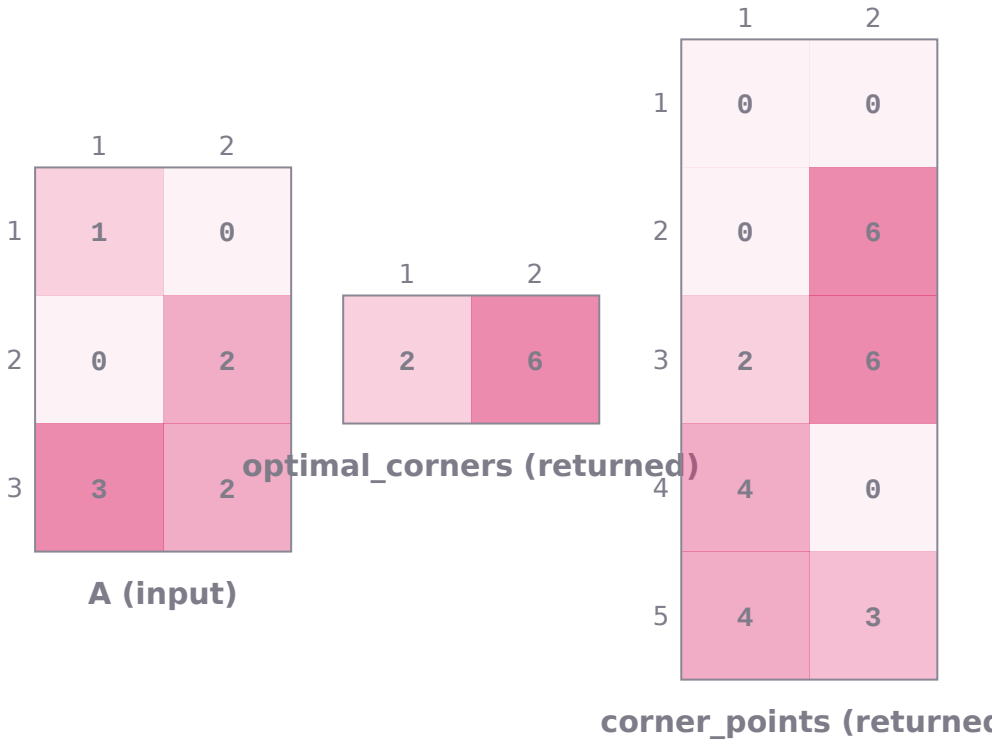
input	meaning	condition
`sense`	maximize or minimize	
`c`	objective coefficients, one per variable	
`A`	constraint coefficients, one row per constraint	
`b`	right-hand sides, one per constraint	
`relations`	relation of each constraint: <=, >= or = (default all <=)	

output	meaning
`optimal_value`	optimal value of the objective
`x`	first variable at the optimum (only when the optimum is unique)
`y`	second variable at the optimum (only when the optimum is unique)
`optimal_corners`	every corner point at which the optimum is attained
`corner_points`	every corner point of the feasible region
`solution_unique`	yes when exactly one point is optimal

- **Method:** intersect every pair of boundary lines (exact fractions), keep the feasible corners, evaluate the objective; unboundedness detected from the extreme rays of the recession cone
- **Assumes:** exactly two variables, both >= 0; a feasible region and a finite optimum
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Maximise 3x + 5y subject to x ≤ 4; 2y ≤ 12; 3x + 2y ≤ 18, x, y ≥ 0: the feasible region with the corner points the package returns ((0, 0), (0, 6), (2, 6), (4, 0), (4, 3)) and the objective line through the optimum; marked: x = 2, y = 6, objective 36.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): A (input); optimal_corners (returned); corner_points (returned). Returned with them by optimization-ai999x: optimal_value = 36, x = 2, y = 6. Other inputs: sense = maximize, c = [3, 5], b = [4, 12, 18].

```
optimization_ai999x.solve('lp_two_variable_graphical', sense='maximize', c=[3, 5], A=[[1,
0], [0, 2], [3, 2]], b=[4, 12, 18])
# -> optimal_value = 36, optimal_corners = [[2.0, 6.0]], corner_points = [[0.0, 0.0], [0.0,
6.0], [2.0, 6.0], [4.0, 0.0], [4.0, 3.0]], x = 2, y = 6, solution_unique = yes   (SOLVED)
```

```
optimization_ai999x.solve('lp_two_variable_graphical', sense='minimize', c=[2, 3], A=[[1, 1], [1, 3]], b=[4, 6], relations='>=,>=')
# -> optimal_value = 9, optimal_corners = [[3.0, 1.0]], corner_points = [[0.0, 4.0], [3.0, 1.0], [6.0, 0.0]], x = 3, y = 1, solution_unique = yes   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Use the graphical method to maximize 3x + 5y subject to x <= 4, 2y <= 12, 3x + 2y <= 18, x, y >= 0.	SOLVED	optimal_value = 36, optimal_corners = [[2.0, 6.0]], corner_points = [[0.0, 0.0], [0.0, 6.0], [2.0, 6.0], [4.0, 0.0], [4.0, 3.0]], x = 2, y = 6, solution_unique = yes
Graphically maximize 2x + 4y subject to x + 2y <= 8, x <= 5, x, y >= 0.	SOLVED	optimal_value = 16, optimal_corners = [[0.0, 4.0], [5.0, 1.5]], corner_points = [[0.0, 0.0], [0.0, 4.0], [5.0, 0.0], [5.0, 1.5]], solution_unique = no
Use the graphical method to maximize x + y + z subject to x + y + z <= 3, x, y, z >= 0.	OUT_OF_SCOPE	Out of scope: the graphical method takes exactly two variables (use the simplex method).

Integer linear program (branch and bound), `integer_program`

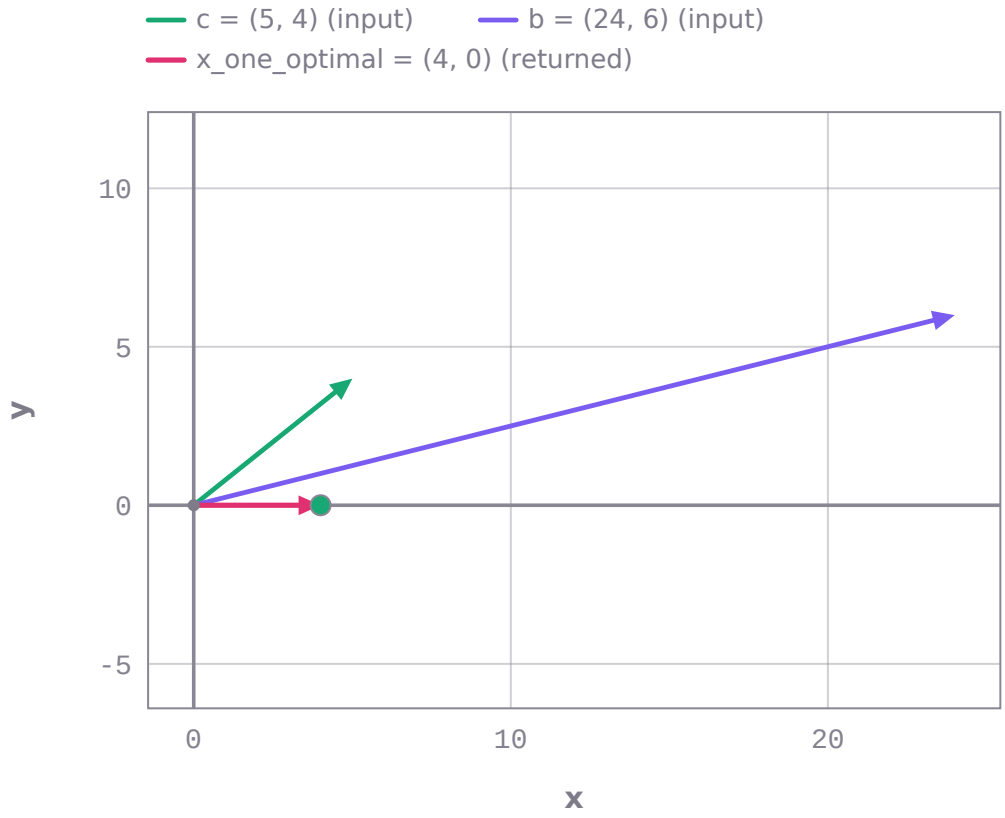
Optimise a linear objective over non-negative INTEGER points satisfying linear constraints: the optimal value and one optimal integer solution.

$$\max / \min \; c^{\top} x \quad \text{s.t. } Ax \; (\leq, \geq, =) \; b, \; x \in \mathbb{Z}^{n}_{\geq 0}$$

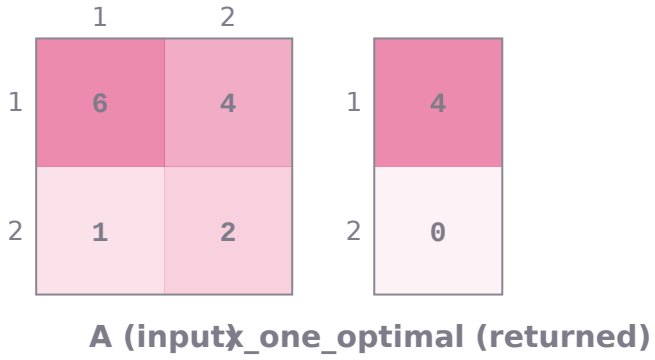
input	meaning	condition
`sense`	maximize or minimize	
`c`	objective coefficients, one per variable	
`A`	constraint coefficients, one row per constraint	
`b`	right-hand sides, one per constraint	
`relations`	relation of each constraint: <=, >= or = (default all <=)	

output	meaning
`optimal_value`	optimal value over integer points
`x_one_optimal`	one optimal integer solution (the first found: depth-first, lowest-index fractional variable, round-down branch first)
`nodes`	branch-and-bound nodes explored

- **Method:** LP-based branch and bound; every relaxation solved exactly (rational simplex), so pruning and integrality tests are exact
- **Assumes:** all variables integer and >= 0; a bounded LP relaxation; at most 4000 branch-and-bound nodes
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The plane vectors of the worked example as arrows from the origin: input c = (5, 4), b = (24, 6); returned by optimization-ai999x: x_one_optimal = (4, 0); also returned optimal_value = 20, nodes = 5. Other inputs: sense = maximize, A = [[6, 4], [1, 2]]. Marked: the tip of each returned vector.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): A (input); x_one_optimal (returned). Returned with them by optimization-ai999x: optimal_value = 20, nodes = 5. Other inputs: sense = maximize, c = [5, 4], b = [24, 6].

```
optimization_ai999x.solve('integer_program', sense='maximize', c=[5, 4], A=[[6, 4], [1,
2]], b=[24, 6])
# -> optimal_value = 20, x_one_optimal = [4.0, 0.0], nodes = 5   (SOLVED)
```

```
optimization_ai999x.solve('integer_program', sense='minimize', c=[3, 2], A=[[1, 1]], b=
[3.5], relations='>=')
# -> optimal_value = 8, x_one_optimal = [0.0, 4.0], nodes = 17   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Maximize 5x + 4y subject to 6x + 4y <= 24, x + 2y <= 6, x, y >= 0 and integer.	SOLVED	optimal_value = 20, x_one_optimal = [4.0, 0.0], nodes = 5
Solve the integer program: minimize 3x + 2y subject to x + y >= 3.5, x, y >= 0, integer.	SOLVED	optimal_value = 8, x_one_optimal = [0.0, 4.0], nodes = 17
Maximize x + y subject to 2x + 2y = 3, x, y >= 0, integer.	INVALID_INPUT	Integer linear program (branch and bound): there is no integer point satisfying all the constraints.

Shadow prices of a linear program, `lp_shadow_prices`

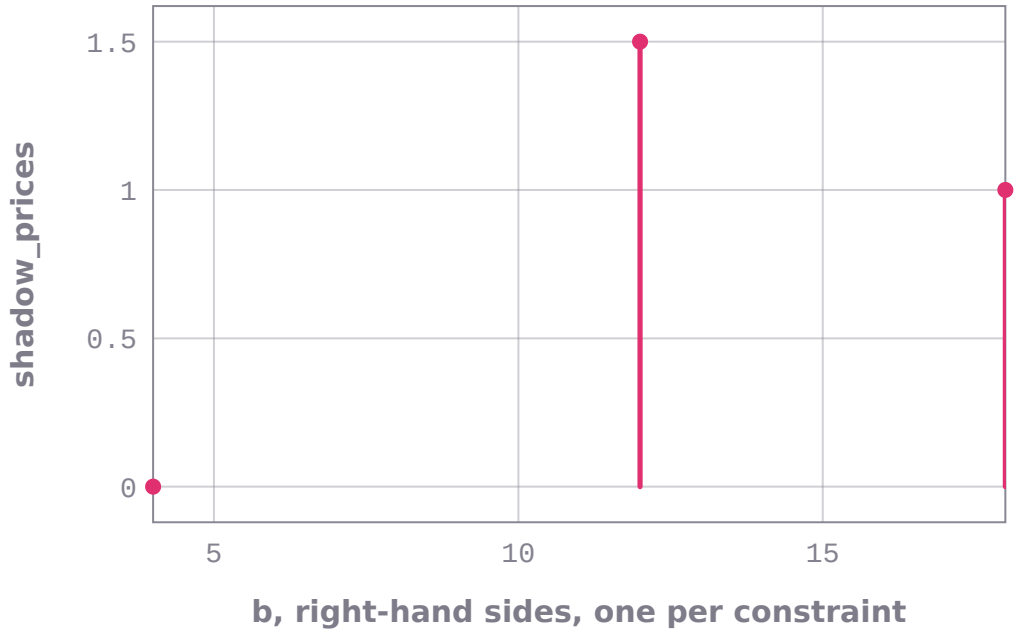
The shadow price (dual value) of each constraint: the change of the optimal value per unit increase of its right-hand side, when it is uniquely defined.

$$y^\top = c_B^\top B^{-1}, \qquad \frac{\partial z^*}{\partial b_i} = y_i$$

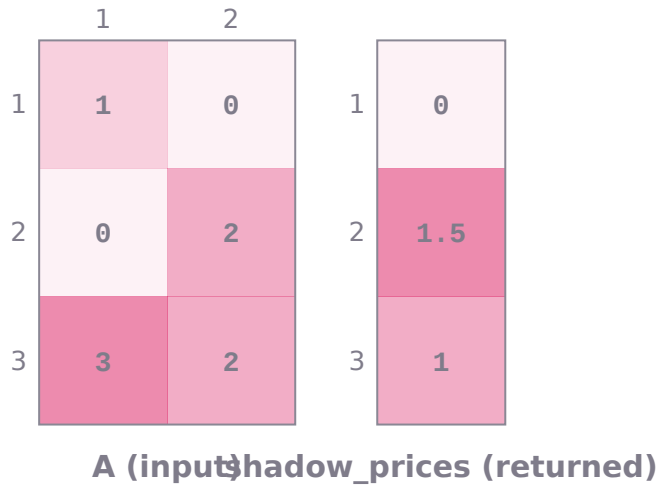
input	meaning	condition
`sense`	maximize or minimize	
`c`	objective coefficients, one per variable	
`A`	constraint coefficients, one row per constraint	
`b`	right-hand sides, one per constraint	
`relations`	relation of each constraint: <=, >= or = (default all <=)	

output	meaning
`shadow_prices`	shadow price of each constraint, in order
`optimal_value`	optimal value of the objective
`x`	the optimal solution (only when unique; otherwise x_one_optimal)
`solution_unique`	yes when the optimal solution is unique

- **Method:** exact simplex; the dual solution $y = c_B$ B^{-1}; uniqueness of the dual proved by an auxiliary LP over the dual optimal face
- **Assumes:** all variables >= 0; a finite optimum; unique dual solution (otherwise INVALID_INPUT)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



shadow price of each constraint, in order (shadow_prices) returned by optimization-ai999x for the worked example, one value for each b (right-hand sides, one per constraint) (3 values, first 0, last 1, largest 1.5). Inputs: sense = maximize, c = [3, 5], A = [[1, 0], [0, 2], [3, 2]], b = [4, 12, 18].



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): A (input); shadow_prices (returned). Returned with them by optimization-ai999x: optimal_value = 36. Other inputs: sense = maximize, c = [3, 5], b = [4, 12, 18].

```
optimization_ai999x.solve('lp_shadow_prices', sense='maximize', c=[3, 5], A=[[1, 0], [0, 2], [3, 2]], b=[4, 12, 18])
# -> optimal_value = 36, x = [2.0, 6.0], solution_unique = yes, shadow_prices = [0.0, 1.5, 1.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the shadow prices: maximize 3x + 5y subject to x <= 4, 2y <= 12, 3x + 2y <= 18, x, y >= 0.	SOLVED	optimal_value = 36, x = [2.0, 6.0], solution_unique = yes, shadow_prices = [0.0, 1.5, 1.0]

question	outcome	result
What are the dual values when you minimize 2x + 3y subject to x + y >= 4, x + 3y >= 6, x, y >= 0?	SOLVED	optimal_value = 9, x = [3.0, 1.0], solution_unique = yes, shadow_prices = [1.5, 0.5]

Optimum of a quadratic function, `quadratic_optimum`

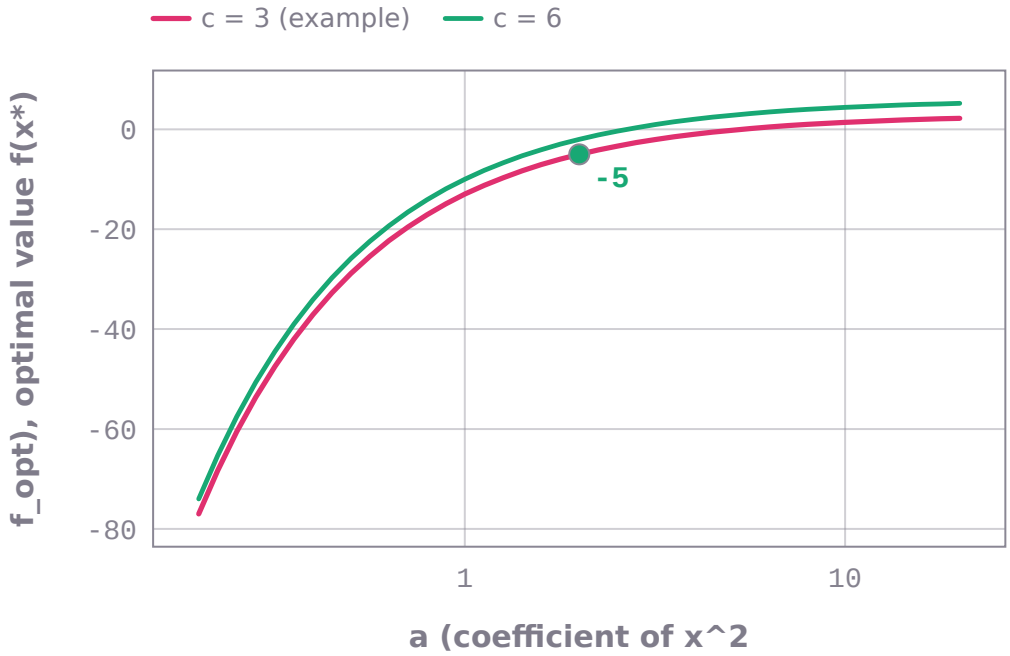
The minimum or maximum of f(x) = a x^2 + b x + c: where it occurs and its value.

$$x^* = -\frac{b}{2a}, \quad f^* = c - \frac{b^2}{4a}$$

input	meaning	condition
`a`	coefficient of x^2	
`b`	coefficient of x	
`c`	constant term	
`goal`	minimum, maximum or optimum (default optimum)	

output	meaning
`x_opt`	x at the optimum
`f_opt`	optimal value f(x*)
`type`	minimum $(a > 0)$ or maximum $(a < 0)$

- **Method:** first-order condition f'$(x) = 2ax + b = 0$, exact fractions
- **Assumes:** $a \neq 0$; a minimum asked needs $a > 0$, a maximum $a < 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



f_opt (optimal value f(x*)) against a (coefficient of x^2) from 0.2 to 20 on a logarithmic axis. One curve for each value of c (constant term): 3, 6. Other inputs as in the worked example: b = -8. 41 points per curve computed by optimization-ai999x; marked: the worked example, a = 2, f_opt = -5.

```
optimization_ai999x.solve('quadratic_optimum', a=2, b=-8, c=3)
# -> x_opt = 2, f_opt = -5, type = minimum   (SOLVED)
```

```
optimization_ai999x.solve('quadratic_optimum', a=-1, b=6, c=1)
# -> x_opt = 3, f_opt = 10, type = maximum   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the minimum of f(x) = 2x^2 - 8x + 3.	SOLVED	x_opt = 2, f_opt = -5, type = minimum
What is the maximum value of y = -x^2 + 6x + 1?	SOLVED	x_opt = 3, f_opt = 10, type = maximum
Find the minimum of f(x) = -x^2 + 4x.	INVALID_INPUT	Optimum of a quadratic function: with a < 0 the parabola opens downward: it has a maximum, no minimum.

Break-even quantity, `break_even_point`

The number of units at which revenue equals total cost (fixed plus variable), and the revenue there.

$$Q_{BE} = \frac{F}{p - v}$$

input	meaning	condition
`F_`	fixed cost (currency)	
`p`	selling price per unit	
`v`	variable cost per unit	

output	meaning
`quantity`	break-even quantity (units)
`revenue`	revenue at break-even
`contribution_margin`	price minus variable cost per unit

- **Method:** solve p $Q = F + v$ Q
- **Assumes:** fixed cost >= 0; price > variable cost
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



quantity (break-even quantity (units)) against p (selling price per unit) from 12.5 to 37.5. One curve for each value of F_ (fixed cost (currency)): 5000, 2500, 10000. Other inputs as in the worked example: v = 15. 27 points per curve computed by optimization-ai999x; marked: the worked example, p = 25, quantity = 500.

```
optimization_ai999x.solve('break_even_point', F_=5000, p=25, v=15)
# -> quantity = 500, revenue = 12500, contribution_margin = 10   (SOLVED)
```

```
optimization_ai999x.solve('break_even_point', F_=1200, p=7.5, v=4.5)
# -> quantity = 400, revenue = 3000, contribution_margin = 3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A firm has fixed costs of 5000, a selling price of 25 per unit and a variable cost of 15 per unit. Find the break-even quantity.	SOLVED	quantity = 500, revenue = 12500, contribution_margin = 10
Find the break-even point if fixed costs are 1200, the price is 7.5 and the variable cost is 4.5 per unit.	SOLVED	quantity = 400, revenue = 3000, contribution_margin = 3
Find the break-even point if fixed costs are 1200, the price is 4 and the variable cost is 4.5 per unit.	INVALID_INPUT	Break-even quantity: the price must exceed the variable cost per unit, otherwise no break-even point exists.
Find the break-even quantity with fixed costs of 1200 and a price of 7.5.	NEEDS_INPUT	break_even_point needs v (variable cost per unit). Example: solve('break_even_point', F_=5000, p=25, v=15).

Profit-maximising output with linear demand, `profit_max_linear_demand`

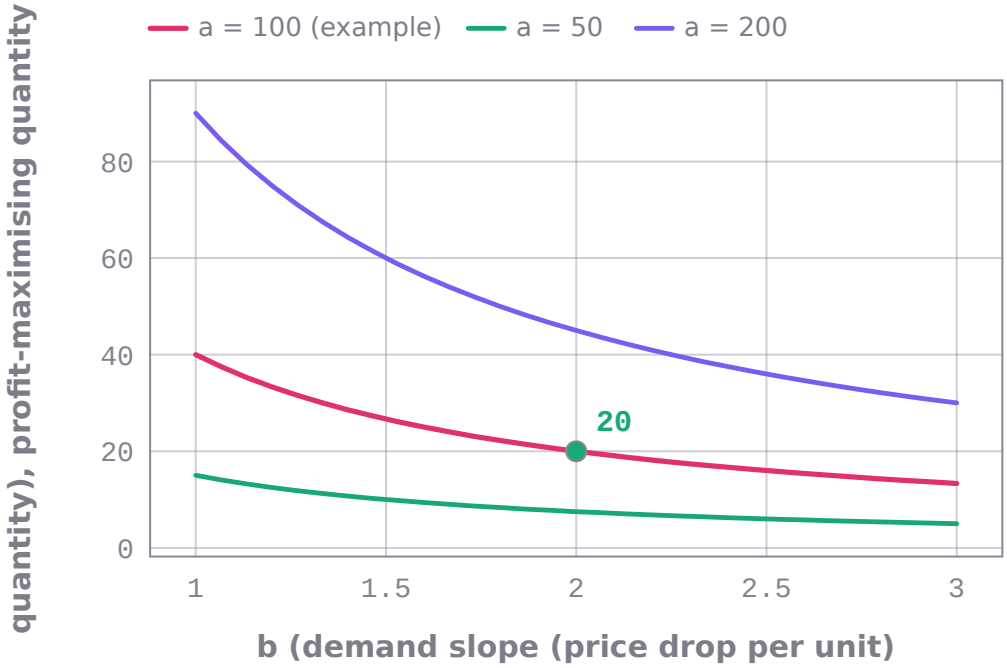
With inverse demand $p = a - b\,q$ and cost $C = F + c\,q$, the output, price and profit at the profit maximum.

$$q^* = \frac{a - c}{2b}, \quad p^* = \frac{a + c}{2}$$

input	meaning	condition
`a`	demand intercept (price at zero quantity)	
`b`	demand slope (price drop per unit)	
`cv`	marginal (variable) cost per unit	
`F_`	fixed cost (default 0)	

output	meaning
`quantity`	profit-maximising quantity
`price`	price at that quantity
`profit`	maximum profit

- **Method:** marginal revenue equals marginal cost: $a - 2$ b $q = c$
- **Assumes:** $b > 0$; $a > c$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



quantity (profit-maximising quantity) against b (demand slope (price drop per unit)) from 1 to 3. One curve for each value of a (demand intercept (price at zero quantity)): 100, 50, 200. Other inputs as in the worked example: cv = 20, F_ = 300. 31 points per curve computed by optimization-ai999x; marked: the worked example, b = 2, quantity = 20.

```
optimization_ai999x.solve('profit_max_linear_demand', a=100, b=2, cv=20, F_=300)
# -> quantity = 20, price = 60, profit = 500   (SOLVED)
```

```
optimization_ai999x.solve('profit_max_linear_demand', a=50, b=0.5, cv=10)
# -> quantity = 40, price = 30, profit = 800   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
The demand is p = 100 - 2q and the cost is C = 300 + 20q. Find the profit-maximizing quantity.	SOLVED	quantity = 20, price = 60, profit = 500
With demand p = 50 - 0.5q and cost C = 10q + 0, what output maximizes profit?	SOLVED	quantity = 40, price = 30, profit = 800
With demand p = 10 - 2q and cost C = 100 + 20q, what output maximizes profit?	INVALID_INPUT	Profit-maximising output with linear demand: the demand intercept must exceed the marginal cost, otherwise no positive o

Revenue-maximising price with linear demand, `revenue_max_linear_demand`

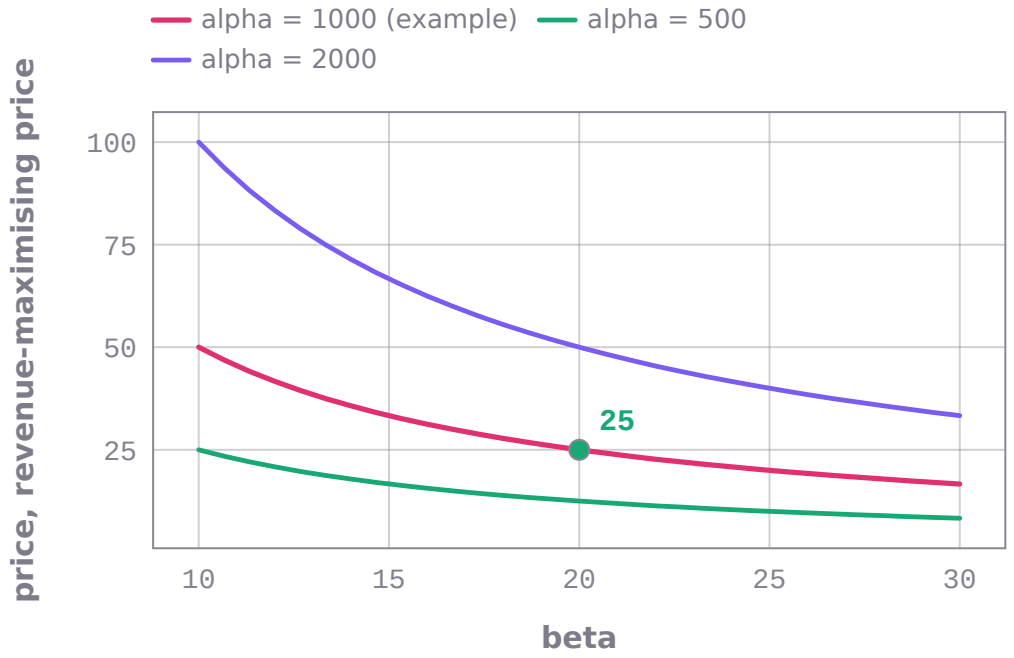
With demand $q = \alpha - \beta$ p, the price that maximises revenue p q, the quantity sold and the revenue.

$$p^* = \frac{\alpha}{2\beta}, \quad R^* = \frac{\alpha^2}{4\beta}$$

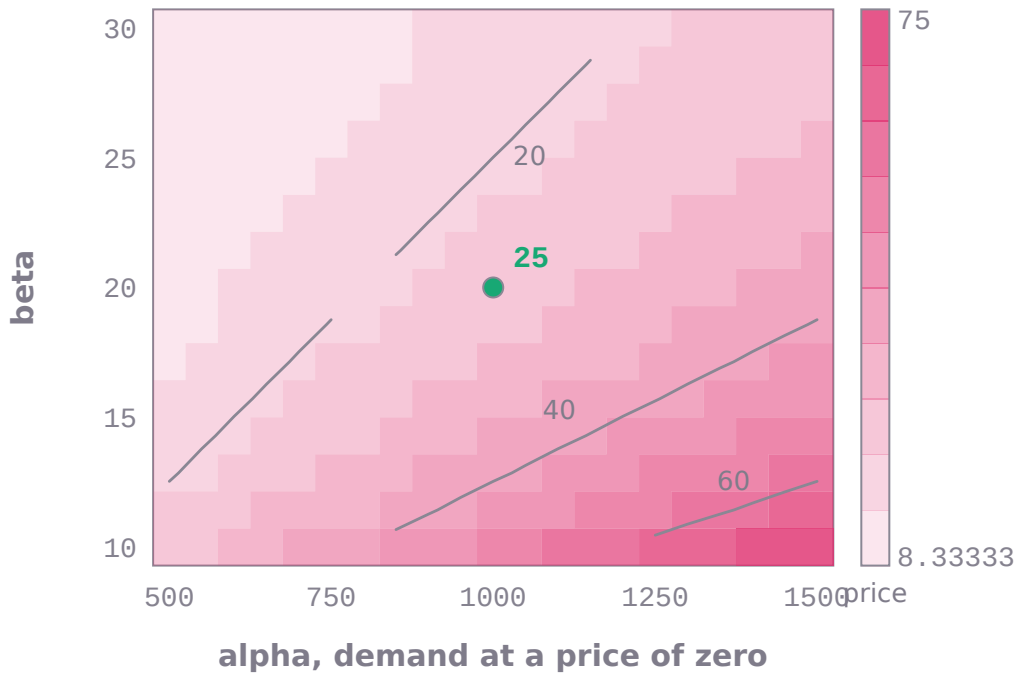
input	meaning	condition
`alpha`	demand at a price of zero	
`beta`	drop in demand per unit increase in price	

output	meaning
`price`	revenue-maximising price
`quantity`	quantity sold at that price
`revenue`	maximum revenue

- **Method:** $dR/dp = \alpha - 2$ beta $p = 0$ (unit elasticity)
- **Assumes:** $\alpha > 0$; $\beta > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



price (revenue-maximising price) against beta (drop in demand per unit increase in price) from 10 to 30. One curve for each value of alpha (demand at a price of zero): 1000, 500, 2000. 31 points per curve computed by optimization-ai999x; marked: the worked example, beta = 20, price = 25.



price (revenue-maximising price) over alpha (demand at a price of zero) from 500 to 1500 and beta (drop in demand per unit increase in price) from 10 to 30: each of the 21 × 15 cells is one run of optimization-ai999x, shaded by its value (light 8.33333 to dark 75); iso-lines at price = 20, 40, 60. Marked: the worked example, alpha = 1000, beta = 20, price = 25.

```
optimization_ai999x.solve('revenue_max_linear_demand', alpha=1000, beta=20)
# -> price = 25, quantity = 500, revenue = 12500   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
The demand is q = 1000 - 20p. Find the revenue-maximizing price.	SOLVED	price = 25, quantity = 500, revenue = 12500
If demand is q = 600 - 3p, what price maximizes revenue?	SOLVED	price = 100, quantity = 300, revenue = 30000

Largest rectangle for a given length of fencing, `fence_max_area`

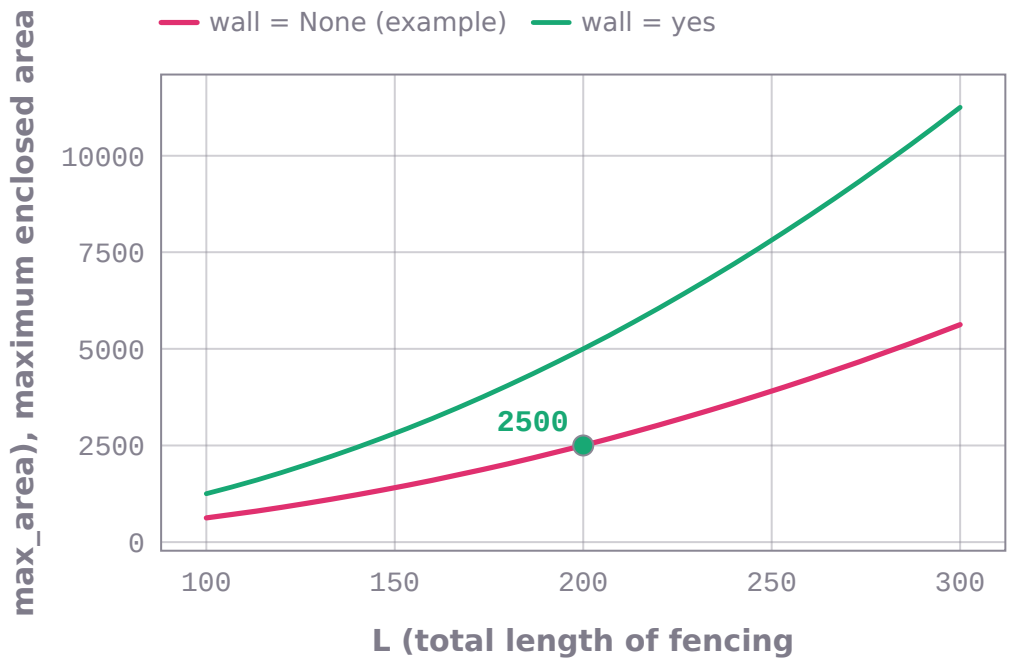
The dimensions and maximum area of a rectangle enclosed by a fence of length L, on four sides or on three sides with a wall or river as the fourth.

$$A_{\max} = \frac{L^2}{16}\ (4\text{ sides}), \quad \frac{L^2}{8}\ (3\text{ sides})$$

input	meaning	condition
`L`	total length of fencing (any length unit)	
`wall`	true when one side lies along a wall or river and needs no fence	

output	meaning
`width`	side perpendicular to the wall (or one side of the square)
`length`	side parallel to the wall
`max_area`	maximum enclosed area (square of the length unit)

- **Method:** area as a function of one side, derivative set to zero
- **Assumes:** $L > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



max_area (maximum enclosed area (square of the length unit)) against L (total length of fencing (any length unit)) from 100 to 300. One curve for each value of wall (true when one side lies along a wall or river and needs no fence): None, yes. 31 points per curve computed by optimization-ai999x; marked: the worked example, L = 200, max_area = 2500.

```
optimization_ai999x.solve('fence_max_area', L=200)
# -> width = 50, length = 50, max_area = 2500   (SOLVED)
```

```
optimization_ai999x.solve('fence_max_area', L=200, wall=True)
# -> width = 50, length = 100, max_area = 5000   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A farmer has 200 m of fencing to enclose a rectangular field. What is the maximum area?	SOLVED	width = 50, length = 50, max_area = 2500
A farmer has 200 m of fencing to enclose a rectangular field along a river, with no fence along the river. Find the maximum area.	SOLVED	width = 50, length = 100, max_area = 5000
A farmer has 200 m of fencing to enclose a rectangular field on all four sides, with no wall or river to use as a boundary. What is the maximum area?	SOLVED	width = 50, length = 50, max_area = 2500
A farmer has 200 m of fencing to enclose a rectangular field along a river, fencing all four sides. What is the maximum area?	UNRECOGNIZED_FORMAT	I could not match this question to a documented form, so I will not guess. Every form: optimization_ai999x.templates();
What is the largest area of a rectangular pen made with fencing?	NEEDS_INPUT	fence_max_area needs L (total length of fencing (any length unit)). Example: solve('fence_max_area', L=200).

Open box of maximum volume from a rectangular sheet, `open_box_max_volume`

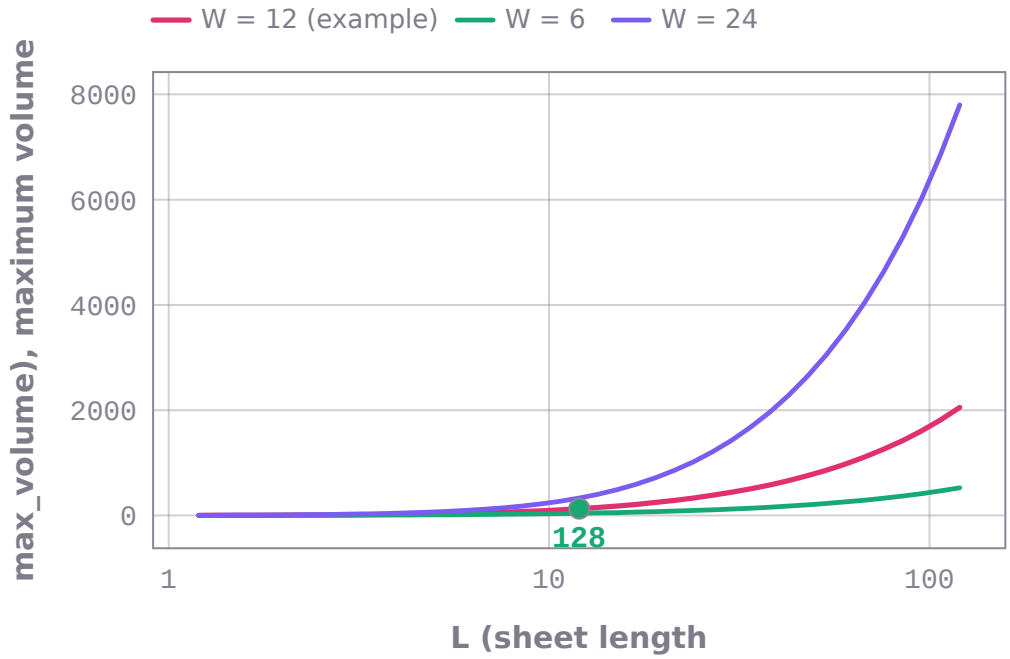
Squares of side x are cut from the corners of an L by W sheet and the sides folded up: the cut that maximises the volume, and that volume.

$$x^* = \frac{(L+W) - \sqrt{L^2 - LW + W^2}}{6}$$

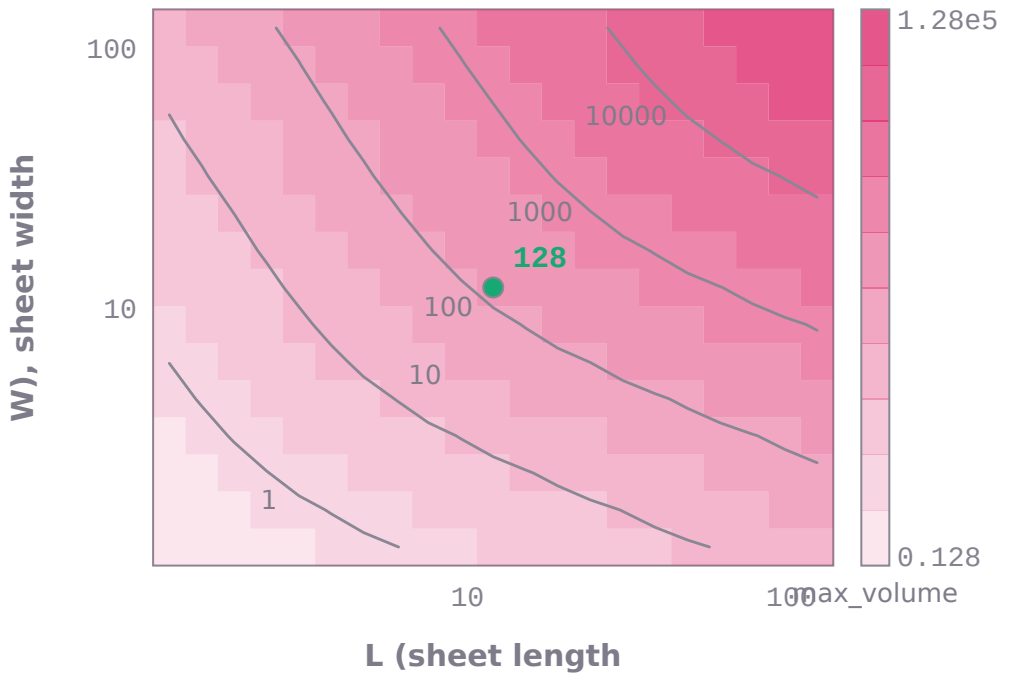
input	meaning	condition
`L`	sheet length	
`W`	sheet width	

output	meaning
`cut`	side of the corner squares at the maximum
`max_volume`	maximum volume

- **Method:** V(x) = x $(L-2x)(W-2x)$; V'$(x) = 0$, the root in (0, min(L, W)/2)
- **Assumes:** $L > 0$ and $W > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



max_volume (maximum volume) against L (sheet length) from 1.2 to 120 on a logarithmic axis. One curve for each value of W (sheet width): 12, 6, 24. 41 points per curve computed by optimization-ai999x; marked: the worked example, L = 12, max_volume = 128.



max_volume (maximum volume) over L (sheet length) from 1.2 to 120 and W (sheet width) from 1.2 to 120 (logarithmic axes): each of the 21 × 15 cells is one run of optimization-ai999x, shaded by its value (light 0.128 to dark 128000, logarithmic shading); iso-lines at max_volume = 1, 10, 100, 1000, 10000. Marked: the worked example, L = 12, W = 12, max_volume = 128.

```
optimization_ai999x.solve('open_box_max_volume', L=12, W=12)
# -> cut = 2, max_volume = 128   (SOLVED)
```

```
optimization_ai999x.solve('open_box_max_volume', L=8, W=5)
# -> cut = 1, max_volume = 18   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
An open box is made from a 12 by 12 sheet by cutting equal squares from the corners. Find the maximum volume.	SOLVED	cut = 2, max_volume = 128
Squares are cut from the corners of an 8 cm by 5 cm card to make an open box. What cut gives the largest volume?	SOLVED	cut = 1, max_volume = 18

Closed cylinder of least surface area for a given volume, `can_min_surface`

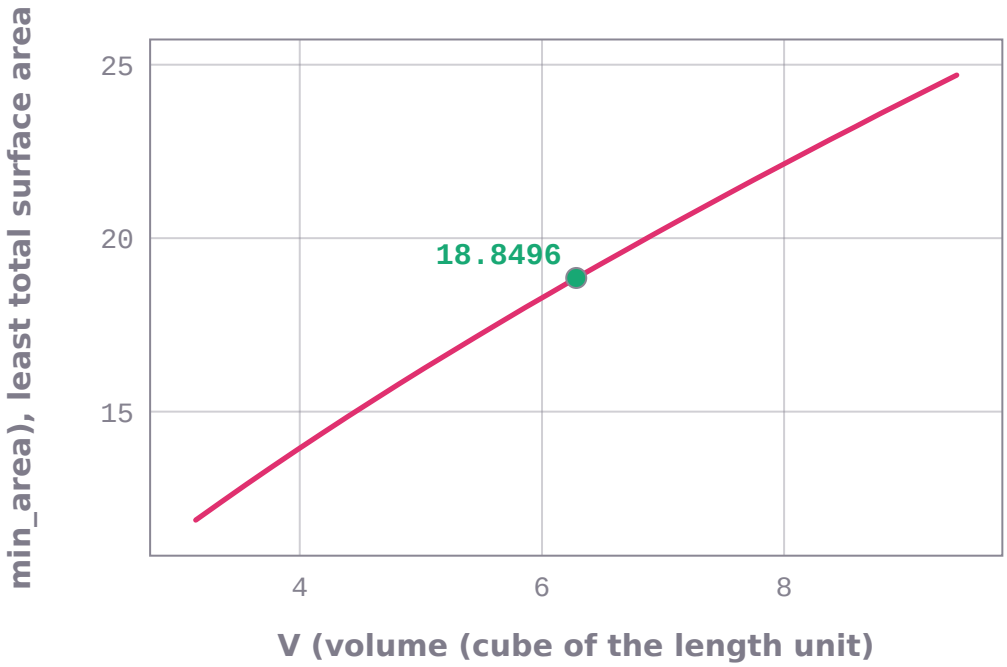
The radius, height and surface area of the closed cylindrical can that holds a volume V with the least material.

$$r^* = \left(\frac{V}{2\pi}\right)^{1/3}, \quad h^* = 2r^*$$

input	meaning	condition
`V`	volume (cube of the length unit)	

output	meaning
`radius`	optimal radius
`height`	optimal height (equal to the diameter)
`min_area`	least total surface area

- **Method:** A(r) = 2 pi $r^2 + 2V/r$; A'$(r) = 0$
- **Assumes:** $V > 0$; closed top and bottom
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



min_area (least total surface area) against V (volume (cube of the length unit)) from 3.14159 to 9.42478. 31 points computed by optimization-ai999x; marked: the worked example, V = 6.28319, min_area = 18.8496.

```
optimization_ai999x.solve('can_min_surface', V=6.283185307179586)
# -> radius = 1, height = 2, min_area = 18.8496   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A closed cylindrical can must hold 500 cm^3. Find the dimensions that minimize the surface area.	SOLVED	radius = 4.30127, height = 8.60254, min_area = 348.734
An open cylindrical tank must hold 10 m^3. Find the radius that minimizes the surface area.	OUT_OF_SCOPE	Out of scope: an open cylinder (this capability is the closed can).

Economic order quantity (EOQ), `eoq`

The order size that minimises ordering plus holding cost for a steady demand, the minimum cost, the number of orders per period and the time between orders.

$$Q^* = \sqrt{\frac{2DS}{H}}, \quad TC^* = \sqrt{2DSH}$$

input	meaning	condition
`D`	demand per period (units, e.g. per year)	
`S`	fixed cost per order	
`H`	holding cost per unit per period	

output	meaning
`Q`	economic order quantity
`total_cost`	minimum ordering + holding cost per period
`orders_per_period`	number of orders per period
`cycle_time`	time between orders (in periods)

- **Method:** minimise D S / Q + H $Q/2$ (at the optimum ordering cost equals holding cost)
- **Assumes:** D, S, $H > 0$; constant demand, instant replenishment, no shortages
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Q (economic order quantity) against H (holding cost per unit per period) from 1.5 to 4.5. One curve for each value of D (demand per period (units, e.g. per year)): 1200, 600, 2400. Other inputs as in the worked example: S = 50. 31 points per curve computed by optimization-ai999x; marked: the worked example, H = 3, Q = 200.

```
optimization_ai999x.solve('eoq', D=1200, S=50, H=3)
# -> Q = 200, total_cost = 600, orders_per_period = 6, cycle_time = 0.166667   (SOLVED)
```

```
optimization_ai999x.solve('eoq', D=1000, S=10, H=0.5)
# -> Q = 200, total_cost = 100, orders_per_period = 5, cycle_time = 0.2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the EOQ for an annual demand of 1200 units, an ordering cost of 50 per order and a holding cost of 3 per unit per year.	SOLVED	Q = 200, total_cost = 600, orders_per_period = 6, cycle_time = 0.166667
What is the economic order quantity if demand is 1000 units per year, the setup cost is 10 and the holding cost is 0.5 per unit per year?	SOLVED	Q = 200, total_cost = 100, orders_per_period = 5, cycle_time = 0.2
Find the EOQ for an annual demand of 1200 units and an ordering cost of 50 per order.	NEEDS_INPUT	eoq needs H (holding cost per unit per period). Example: solve('eoq', D=1200, S=50, H=3).

EOQ with planned backorders, `eoq_planned_shortages`

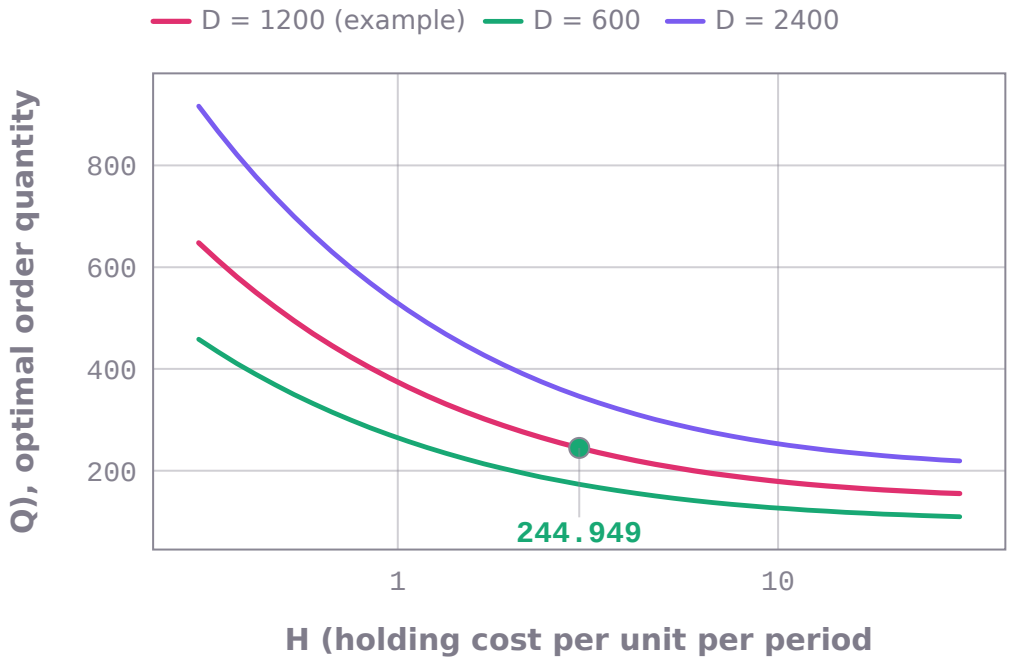
The order quantity and the planned maximum backorder that minimise ordering, holding and shortage cost when unmet demand is backordered.

$$Q^* = \sqrt{\frac{2DS}{H} \cdot \frac{H+B}{B}}, \quad s^* = Q^* \frac{H}{H+B}$$

input	meaning	condition
`D`	demand per period	
`S`	cost per order	
`H`	holding cost per unit per period	
`B`	backorder (shortage) cost per unit per period	

output	meaning
`Q`	optimal order quantity
`max_backorder`	planned maximum backorder
`max_inventory`	maximum inventory level
`total_cost`	minimum total cost per period

- **Method:** minimise D S/Q + H $(Q-s)^2/(2Q) + B\,s^2/(2Q)$ over Q and s
- **Assumes:** D, S, H, $B > 0$; backorders filled when the order arrives
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Q (optimal order quantity) against H (holding cost per unit per period) from 0.3 to 30 on a logarithmic axis. One curve for each value of D (demand per period): 1200, 600, 2400. Other inputs as in the worked example: S = 50, B = 6. 41 points per curve computed by optimization-ai999x; marked: the worked example, H = 3, Q = 244.949.

```
optimization_ai999x.solve('eoq_planned_shortages', D=1200, S=50, H=3, B=6)
# -> Q = 244.949, max_backorder = 81.6497, max_inventory = 163.299, total_cost = 489.898
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the EOQ with backorders for an annual demand of 1200 units, an ordering cost of 50, a holding cost of 3 and a shortage cost of 6 per unit per year.	SOLVED	Q = 244.949, max_backorder = 81.6497, max_inventory = 163.299, total_cost = 489.898

Economic production quantity (EPQ), `epq_production_lot`

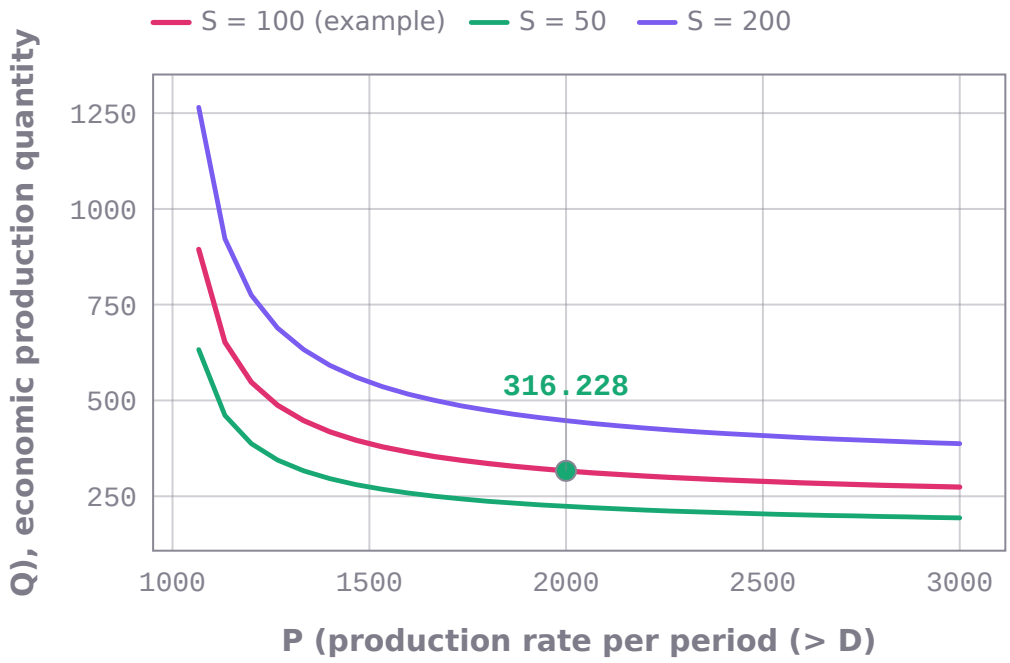
The production lot size that minimises setup plus holding cost when stock builds up gradually at a finite production rate.

$$Q^* = \sqrt{\frac{2DS}{H(1 - D/P)}}$$

input	meaning	condition
`D`	demand rate per period	
`P`	production rate per period (> D)	
`S`	setup cost per run	
`H`	holding cost per unit per period	

output	meaning
`Q`	economic production quantity
`max_inventory`	maximum inventory Q $(1 - D/P)$
`total_cost`	minimum setup + holding cost per period
`runs_per_period`	production runs per period

- **Method:** minimise D S/Q + H Q $(1 - D/P)/2$
- **Assumes:** $P > D > 0$; S, $H > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Q (economic production quantity) against P (production rate per period (> D)) from 1000 to 3000. One curve for each value of S (setup cost per run): 100, 50, 200. Other inputs as in the worked example: D = 1000, H = 4. 30 points per curve computed by optimization-ai999x; marked: the worked example, P = 2000, Q = 316.228.

```
optimization_ai999x.solve('epq_production_lot', D=1000, P=2000, S=100, H=4)
# -> Q = 316.228, max_inventory = 158.114, total_cost = 632.456, runs_per_period = 3.16228
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the EPQ for an annual demand of 1000 units, a production rate of 2000 units per year, a setup cost of 100 and a holding cost of 4 per unit per year.	SOLVED	Q = 316.228, max_inventory = 158.114, total_cost = 632.456, runs_per_period = 3.16228
Find the economic production quantity for a demand of 3000 units, a production rate of 2000 units, a setup cost of 100 and a holding cost of 4.	INVALID_INPUT	Economic production quantity (EPQ): the production rate must exceed the demand rate.

Order quantity with quantity discounts (all-units), `eoq_quantity_discount`

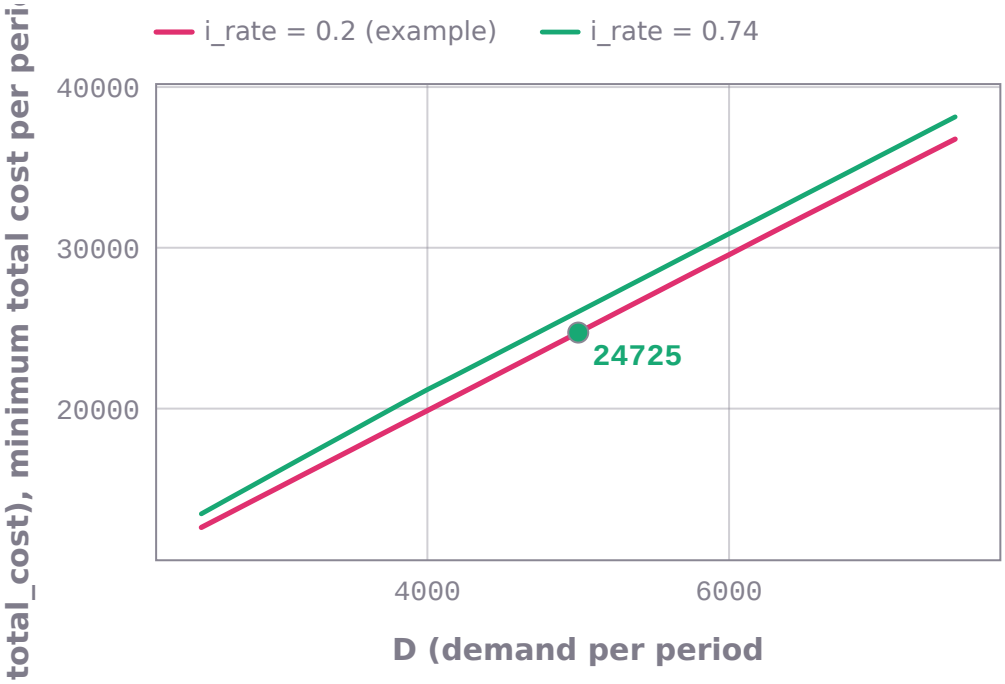
With all-units price breaks, the order quantity that minimises purchase + ordering + holding cost, the price paid and the minimum total cost per period.

$$TC(Q) = Dp_k + \frac{DS}{Q} + \frac{ip_kQ}{2}$$

input	meaning	condition
`D`	demand per period	
`S`	cost per order	
`i_rate`	holding cost as a fraction of the unit price per period (0.2 = 20%)	
`breaks`	minimum order quantity of each price tier, increasing, the first 0	
`prices`	unit price of each tier	

output	meaning
`order_quantity`	optimal order quantity
`total_cost`	minimum total cost per period (purchase + ordering + holding)
`unit_price`	unit price paid

- **Method:** EOQ per tier at its price, raised to the tier minimum when below it; tiers whose EOQ exceeds their upper limit are dominated; least total cost
- **Assumes:** D, S, $i > 0$; breaks increasing from 0; ties broken toward the lower-quantity tier
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



total_cost (minimum total cost per period (purchase + ordering + holding)) against D (demand per period) from 2500 to 7500. One curve for each value of i_rate (holding cost as a fraction of the unit price per period (0.2 = 20%)): 0.2, 0.74. Other inputs as in the worked example: S = 49, breaks = 0, 1000, 2000, prices = 5, 4.8, 4.75. 31 points per curve computed by optimization-ai999x; marked: the worked example, D = 5000, total_cost = 24725.

```
optimization_ai999x.solve('eoq_quantity_discount', D=5000, S=49, i_rate=0.2, breaks='0,
1000, 2000', prices='5, 4.8, 4.75')
# -> order_quantity = 1000, total_cost = 24725, unit_price = 4.8   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Quantity discounts: annual demand 5000, ordering cost 49, holding rate 20%, price breaks at [0, 1000, 2000] with prices [5, 4.8, 4.75]. Find the best order quantity.	SOLVED	order_quantity = 1000, total_cost = 24725, unit_price = 4.8

Reorder point with safety stock, `reorder_point`

The stock level at which to reorder so that demand during the lead time is covered with a given cycle service level, for normally distributed demand per period.

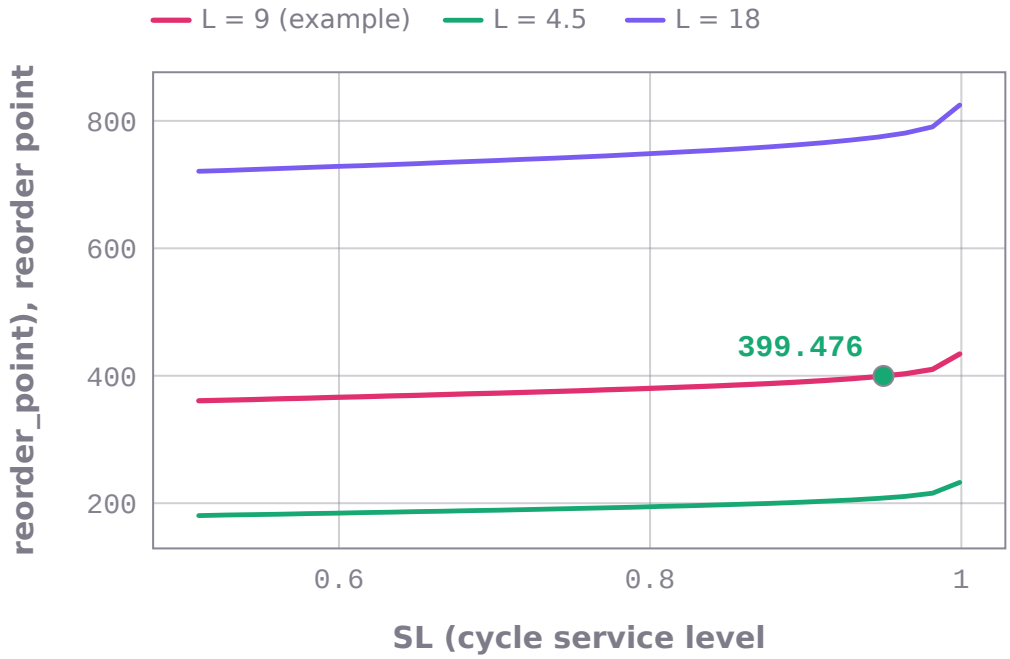
$$ROP = \bar{d}L + z_{SL}\,\sigma_d\sqrt{L}$$

input	meaning	condition
`d`	average demand per period	
`L`	lead time in periods	
`sigma`	standard deviation of demand per period	
`SL`	cycle service level (probability of no stockout per cycle, e.g. 0.95)	

output	meaning
`reorder_point`	reorder point

output	meaning
`safety_stock`	safety stock z sigma sqrt(L)
`z`	standard normal quantile of the service level
`lead_time_demand`	expected demand during the lead time

- **Method:** $z = Phi$^{-1}(SL); ROP = d L + z sigma sqrt(L)
- **Assumes:** $0.5 \leq SL < 1$; independent normal demand per period
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



reorder_point (reorder point) against SL (cycle service level (probability of no stockout per cycle, e.g. 0.95)) from 0.475 to 0.999. One curve for each value of L (lead time in periods): 9, 4.5, 18. Other inputs as in the worked example: d = 40, sigma = 8. 29 points per curve computed by optimization-ai999x; marked: the worked example, SL = 0.95, reorder_point = 399.476.

```
optimization_ai999x.solve('reorder_point', d=40, L=9, sigma=8, SL=0.95)
# -> reorder_point = 399.476, safety_stock = 39.4765, z = 1.64485, lead_time_demand = 360
(SOLVED)
```

```
optimization_ai999x.solve('reorder_point', d=100, L=4, sigma=0, SL=0.9)
# -> reorder_point = 400, safety_stock = 0, z = 1.28155, lead_time_demand = 400   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Daily demand averages 40 units with a standard deviation of 8 units, the lead time is 9 days and the service level is 95%. Find the reorder point.	SOLVED	reorder_point = 399.476, safety_stock = 39.4765, z = 1.64485, lead_time_demand = 360
Find the reorder point with demand of 40 per day, standard deviation 8, lead time 9 days and service level 120%.	INVALID_INPUT	Reorder point with safety stock: the service level must be at least 0.5 and below 1 (e.g. 0.95).

Periodic-review order-up-to level, `order_up_to_level`

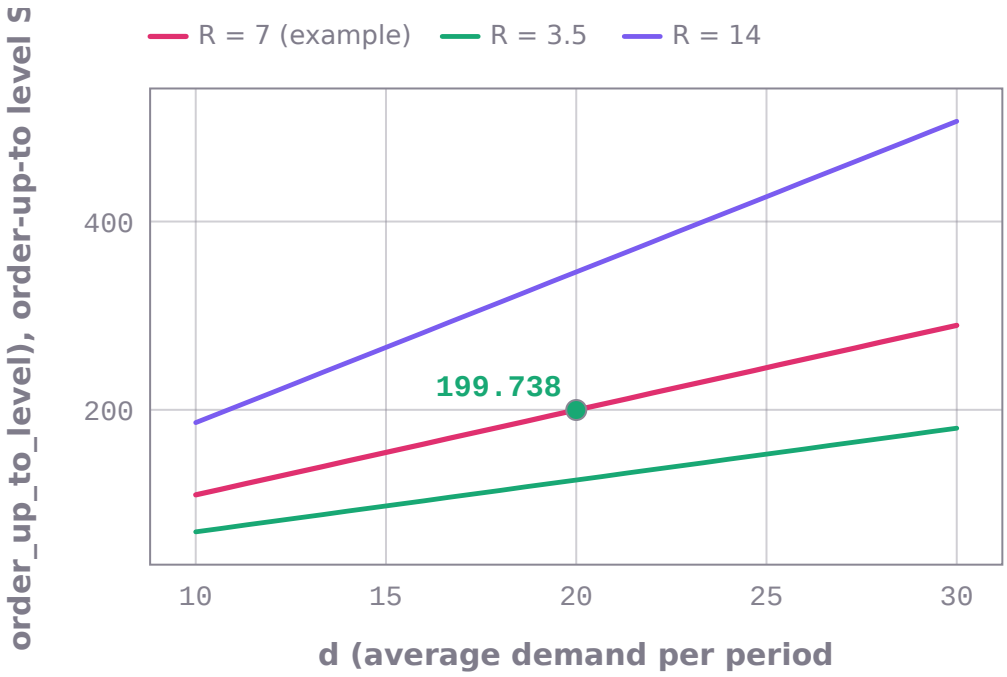
The target inventory position S of a periodic-review system that covers demand over the review period plus the lead time with a given cycle service level.

$$S = \bar{d}(R + L) + z_{SL}\,\sigma_d\sqrt{R + L}$$

input	meaning	condition
`d`	average demand per period	
`sigma`	standard deviation of demand per period	
`R`	review period (periods)	
`L`	lead time (periods)	
`SL`	cycle service level, e.g. 0.95	

output	meaning
`order_up_to_level`	order-up-to level S
`safety_stock`	safety stock z sigma sqrt(R + L)
`z`	standard normal quantile

- **Method:** $S = d$ (R + L) + z sigma sqrt(R + L)
- **Assumes:** $0.5 \leq SL < 1$; independent normal demand per period
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



order_up_to_level (order-up-to level S) against d (average demand per period) from 10 to 30. One curve for each value of R (review period (periods)): 7, 3.5, 14. Other inputs as in the worked example: sigma = 4, L = 2, SL = 0.95. 31 points per curve computed by optimization-ai999x; marked: the worked example, d = 20, order_up_to_level = 199.738.

```
optimization_ai999x.solve('order_up_to_level', d=20, sigma=4, R=7, L=2, SL=0.95)
# -> order_up_to_level = 199.738, safety_stock = 19.7382, z = 1.64485   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
In a periodic review system, demand averages 20 units per day with a standard deviation of 4, the review period is 7 days, the lead time is 2 days and the service level is 95%. Find the order-up-to level.	SOLVED	order_up_to_level = 199.738, safety_stock = 19.7382, z = 1.64485

Newsvendor order quantity (normal demand), `newsvendor_normal`

The single-period order quantity that maximises expected profit for normally distributed demand: the critical ratio, the optimal quantity and the expected profit.

$$F(Q^*) = \frac{p - c}{p - s}, \quad Q^* = \mu + z\sigma$$

input	meaning	condition
`p`	selling price per unit	
`c`	cost per unit	
`mu`	mean demand	
`sigma`	standard deviation of demand	
`s`	salvage value per unsold unit (default 0)	

output	meaning
`critical_ratio`	critical ratio (p - c)/(p - s)
`Q`	optimal order quantity
`expected_profit`	expected profit at Q
`expected_shortage`	expected unmet demand at Q

- **Method:** critical fractile: F(Q) = (p - c)/(p - s); expected profit from the standard normal loss function
- **Assumes:** salvage < cost < price; $\sigma > 0$; continuous normal demand
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Q (optimal order quantity) against p (selling price per unit) from 0.2 to 20 on a logarithmic axis. One curve for each value of mu (mean demand): 100, 50, 200. Other inputs as in the worked example: c = 0.5, sigma = 20, s = 0.2. 33 points per curve computed by optimization-ai999x; marked: the worked example, p = 2, Q = 119.348.

```
optimization_ai999x.solve('newsvendor_normal', p=2, c=0.5, s=0.2, mu=100, sigma=20)
# -> critical_ratio = 0.833333, Q = 119.348, expected_profit = 141.005, expected_shortage =
1.77228   (SOLVED)
```

```
optimization_ai999x.solve('newsvendor_normal', p=10, c=5, mu=500, sigma=50)
# -> critical_ratio = 0.5, Q = 500, expected_profit = 2300.53, expected_shortage = 19.9471
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Newsvendor problem: a paper sells for 2 and costs 0.5, leftovers are sold for salvage of 0.2, and demand is normal with mean 100 and standard deviation 20. Find the optimal order quantity.	SOLVED	critical_ratio = 0.833333, Q = 119.348, expected_profit = 141.005, expected_shortage = 1.77228
Newsvendor: price 5, cost 6, demand normal with mean 100 and standard deviation 20. Find the order quantity.	INVALID_INPUT	Newsvendor order quantity (normal demand): the salvage value, unit cost and price must satisfy salvage < cost < price.

Newsvendor order quantity (uniform demand), `newsvendor_uniform`

The single-period order quantity that maximises expected profit when demand is uniform on [a, b].

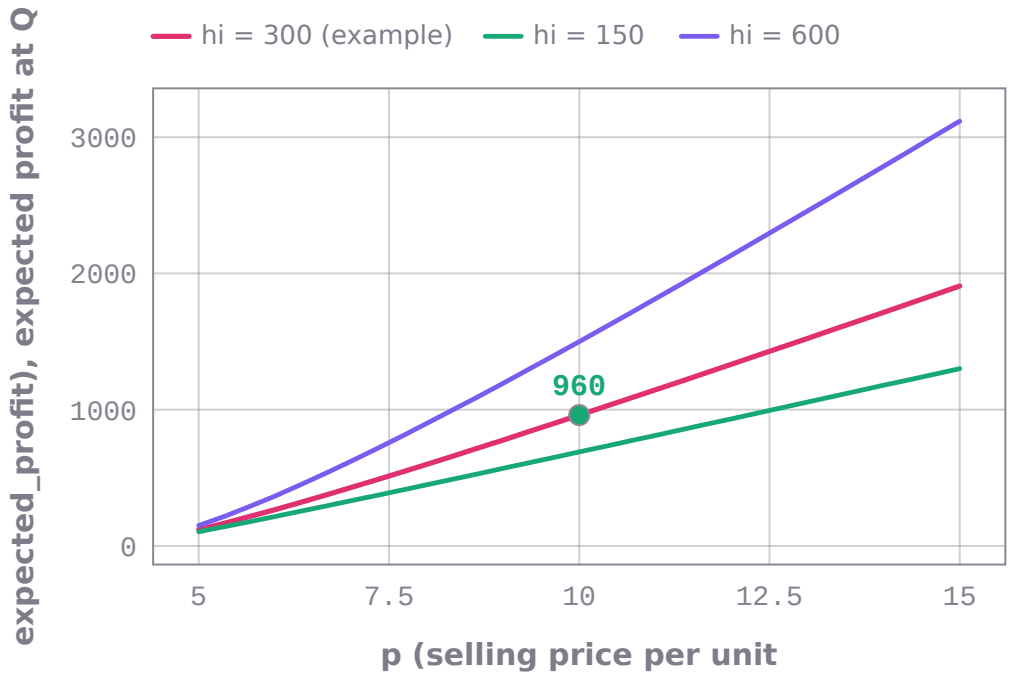
$$Q^* = a + \frac{p - c}{p - s}(b - a)$$

input	meaning	condition
p	selling price per unit	

input	meaning	condition
`c`	cost per unit	
`lo`	lowest possible demand a	
`hi`	highest possible demand b	
`s`	salvage value per unsold unit (default 0)	

output	meaning
`critical_ratio`	critical ratio (p - c)/(p - s)
`Q`	optimal order quantity a + CR (b - a)
`expected_profit`	expected profit at Q

- **Method:** critical fractile of the uniform distribution
- **Assumes:** salvage < cost < price; $0 \leq a < b$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



expected_profit (expected profit at Q) against p (selling price per unit) from 5 to 15. One curve for each value of hi (highest possible demand b): 300, 150, 600. Other inputs as in the worked example: c = 4, lo = 100. 31 points per curve computed by optimization-ai999x; marked: the worked example, p = 10, expected_profit = 960.

```
optimization_ai999x.solve('newsvendor_uniform', p=10, c=4, lo=100, hi=300)
# -> critical_ratio = 0.6, Q = 220, expected_profit = 960   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Newsvendor: an item sells for 10 and costs 4, with uniform demand between 100 and 300. Find the optimal order quantity.	SOLVED	critical_ratio = 0.6, Q = 220, expected_profit = 960

Newsvendor order quantity (discrete demand), `newsvendor_discrete`

For demand taking listed values with given probabilities, the order quantity that maximises expected profit (the smallest demand value whose cumulative probability reaches the critical ratio) and that profit.

$$Q^* = \min\{d : F(d) \geq \tfrac{p-c}{p-s}\}$$

input	meaning	condition
`p`	selling price per unit	
`c`	cost per unit	
`demands`	possible demand values, increasing	
`probs`	probability of each demand value (sum 1)	
`s`	salvage value per unsold unit (default 0)	

output	meaning
`critical_ratio`	critical ratio (p - c)/(p - s)
`Q`	optimal order quantity (smallest optimal value)
`expected_profit`	maximum expected profit

- **Method:** cumulative distribution against the critical ratio, in exact fractions
- **Assumes:** salvage < cost < price; probabilities sum to 1; order quantities restricted to the listed demand values
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



expected_profit (maximum expected profit) against p (selling price per unit) from 5 to 15. One curve for each value of c (cost per unit): 4, 2, 8. Other inputs as in the worked example: demands = 10, 20, 30, 40, probs = 0.1, 0.3, 0.4, 0.2. 31 points per curve computed by optimization-ai999x; marked: the worked example, p = 10, expected_profit = 130.



expected_profit (maximum expected profit) over p (selling price per unit) from 5 to 15 and c (cost per unit) from 2 to 6: each of the 21 × 15 cells is one run of optimization-ai999x, shaded by its value (light 0.714286 to dark 325); iso-lines at expected_profit = 100, 200, 300; blank cells: inputs the capability refuses. Other inputs as in the worked example: demands = 10, 20, 30, 40, probs = 0.1, 0.3, 0.4, 0.2. Marked: the worked example, p = 10, c = 4, expected_profit = 130.

```
optimization_ai999x.solve('newsvendor_discrete', p=10, c=4, demands='10, 20, 30, 40',
probs='0.1, 0.3, 0.4, 0.2')
# -> critical_ratio = 0.6, Q = 30, expected_profit = 130   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Newsvendor: a cake sells for 10 and costs 4; demand values [10, 20, 30, 40] have probabilities [0.1, 0.3, 0.4, 0.2]. How many should be ordered?	SOLVED	critical_ratio = 0.6, Q = 30, expected_profit = 130
Newsvendor: a cake sells for 10 and costs 4; demand values [10, 20, 30, 40] have probabilities [0.1, 0.3, 0.4, 0.4]. How many should be ordered?	INVALID_INPUT	Newsvendor order quantity (discrete demand): the probabilities must be >= 0 and sum to 1.

Littlewood's rule (two-fare protection level), `littlewood_protection`

How many seats (units) to protect for full-fare demand when a discount fare is also sold: the protection level and, with a capacity, the booking limit for the discount class.

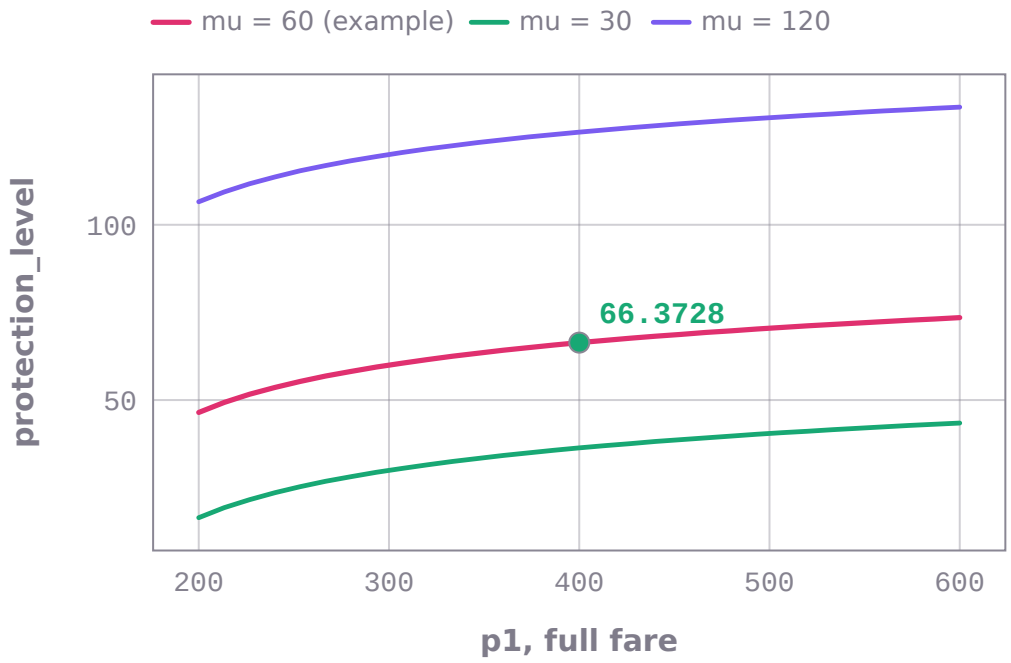
$$P(D_1 > y^*) = \frac{p_2}{p_1}, \quad y^* = \mu + z\,\sigma$$

input	meaning	condition
`p1`	full fare	
`p2`	discount fare	
`mu`	mean full-fare demand	

input	meaning	condition
`sigma`	standard deviation of full-fare demand	
`capacity`	total capacity (optional)	

output	meaning
`protection_level`	units protected for full-fare demand
`z`	standard normal quantile of $1 - p_2/p_1$
`booking_limit`	discount booking limit = capacity - protection (when a capacity is given)

- **Method:** protect y with P(D1 > y) = p2/p1
- **Assumes:** $0 < p_2 < p_1$; normally distributed full-fare demand; continuous approximation (no rounding)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



protection_level (units protected for full-fare demand) against p1 (full fare) from 200 to 600. One curve for each value of mu (mean full-fare demand): 60, 30, 120. Other inputs as in the worked example: p2 = 150, sigma = 20. 31 points per curve computed by optimization-ai999x; marked: the worked example, p1 = 400, protection_level = 66.3728.

```
optimization_ai999x.solve('littlewood_protection', p1=400, p2=150, mu=60, sigma=20)
# -> protection_level = 66.3728, z = 0.318639   (SOLVED)
```

```
optimization_ai999x.solve('littlewood_protection', p1=200, p2=100, mu=50, sigma=10,
capacity=120)
# -> protection_level = 50, z = 0, booking_limit = 70   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Littlewood's rule: the full fare is 400 and the discount fare is 150; full-fare demand has mean 60 and standard deviation 20. Find the protection level.	SOLVED	protection_level = 66.3728, z = 0.318639

M/M/1 queue, `mm1_queue`

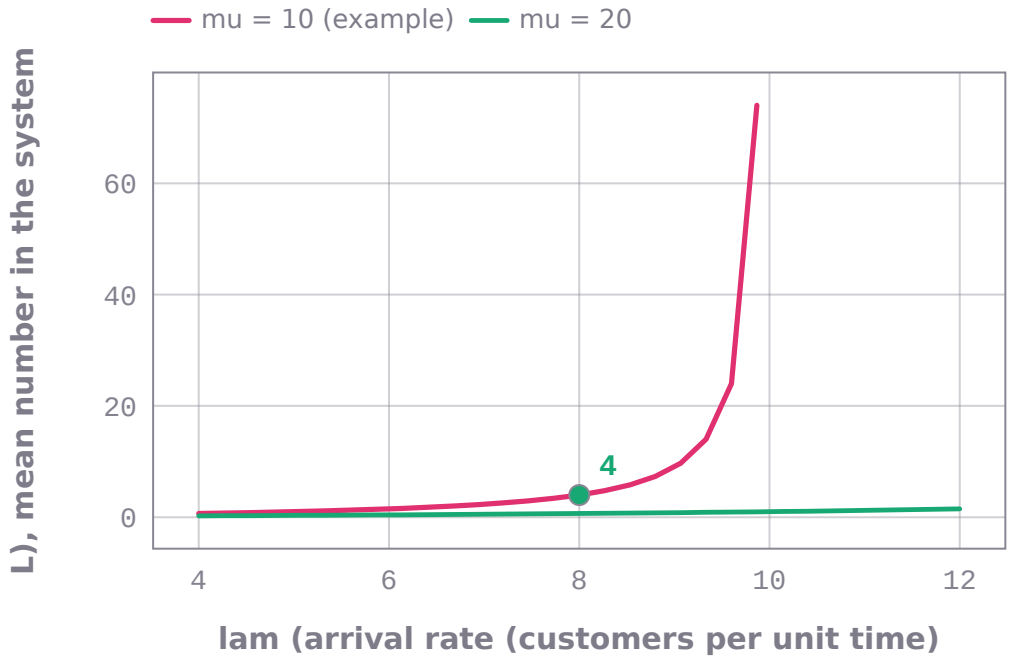
Steady-state measures of a single-server queue with Poisson arrivals and exponential service: utilisation, mean numbers in system and in queue, mean times in system and in queue, probability of an empty system.

$$L = \frac{\rho}{1-\rho},\ W = \frac{1}{\mu-\lambda},\ W_q = \frac{\rho}{\mu-\lambda}$$

input	meaning	condition
`lam`	arrival rate (customers per unit time)	
`mu`	service rate (customers per unit time)	

output	meaning
`rho`	utilisation lambda/mu
`L`	mean number in the system
`Lq`	mean number waiting
`W`	mean time in the system (time unit of the rates)
`Wq`	mean waiting time in the queue
`P0`	probability the system is empty

- **Method:** geometric stationary distribution; Little's law
- **Assumes:** $\lambda < mu$ (stable); Poisson arrivals, exponential service, one server, unlimited capacity
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



L (mean number in the system) against lam (arrival rate (customers per unit time)) from 4 to 12. One curve for each value of mu (service rate (customers per unit time)): 10, 20. 23 points per curve computed by optimization-ai999x; marked: the worked example, lam = 8, L = 4.

```
optimization_ai999x.solve('mm1_queue', lam=8, mu=10)
# -> rho = 0.8, L = 4, Lq = 3.2, W = 0.5, Wq = 0.4, P0 = 0.2   (SOLVED)
```

```
optimization_ai999x.solve('mm1_queue', lam=2, mu=3)
# -> rho = 0.666667, L = 2, Lq = 1.33333, W = 1, Wq = 0.666667, P0 = 0.333333   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
In an M/M/1 queue customers arrive at 8 per hour and are served at 10 per hour. Find L, Lq, W and Wq.	SOLVED	rho = 0.8, L = 4, Lq = 3.2, W = 0.5, Wq = 0.4, P0 = 0.2
A single-server queue has an arrival rate of 2 per minute and a service rate of 3 per minute. What is the mean time in the system?	SOLVED	rho = 0.666667, L = 2, Lq = 1.33333, W = 1, Wq = 0.666667, P0 = 0.333333
In an M/M/1 queue customers arrive at 12 per hour and are served at 10 per hour. Find L.	INVALID_INPUT	M/M/1 queue: the queue is unstable: the arrival rate must be below the service rate.

M/M/c queue (Erlang C), `mmc_queue`

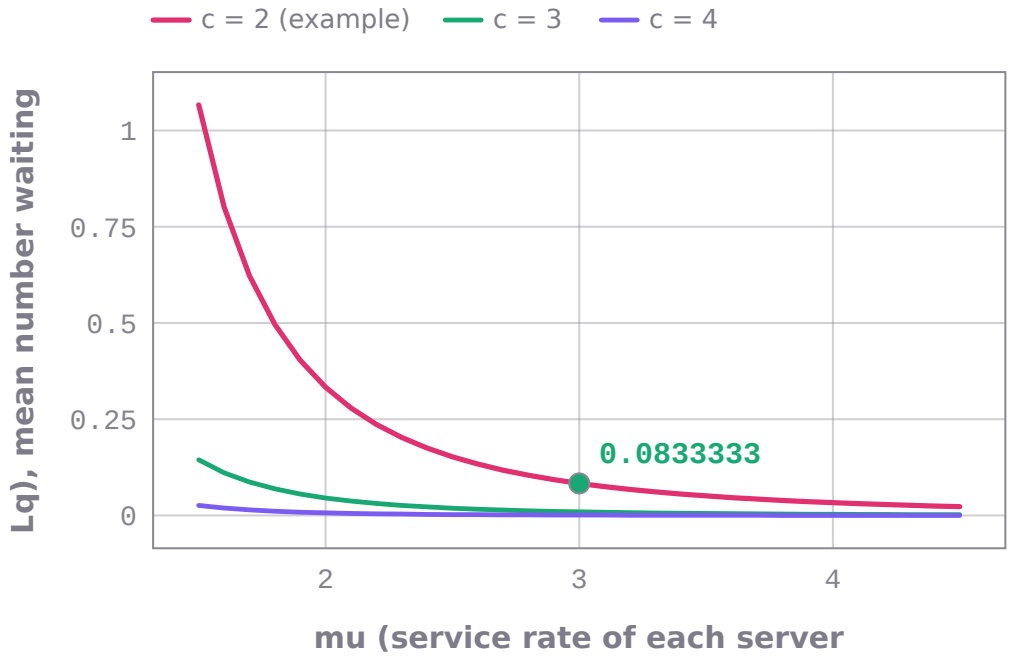
Steady-state measures of a queue with c parallel identical servers, Poisson arrivals and exponential service: probability of waiting (Erlang C), mean queue length and waiting times.

$$C(c,a) = \frac{B(c,a)}{1-\rho(1-B(c,a))}, \quad L_q = C\,\frac{\rho}{1-\rho}$$

input	meaning	condition
`lam`	arrival rate	
`mu`	service rate of each server	
`c`	number of servers	

output	meaning
`rho`	utilisation lambda/(c mu)
`P_wait`	probability an arrival waits (Erlang C)
`Lq`	mean number waiting
`Wq`	mean waiting time
`W`	mean time in the system
`L`	mean number in the system
`P0`	probability the system is empty

- **Method:** Erlang B by its stable recursion, then Erlang C = B / (1 - rho $(1-B)$); Little's law
- **Assumes:** $\lambda < c$ mu (stable); unlimited queue capacity
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Lq (mean number waiting) against mu (service rate of each server) from 1.5 to 4.5. One curve for each value of c (number of servers): 2, 3, 4. Other inputs as in the worked example: lam = 2. 31 points per curve computed by optimization-ai999x; marked: the worked example, mu = 3, Lq = 0.0833333.

```
optimization_ai999x.solve('mmc_queue', lam=2, mu=3, c=2)
# -> rho = 0.333333, P_wait = 0.166667, Lq = 0.0833333, Wq = 0.0416667, W = 0.375, L =
0.75, P0 = 0.5   (SOLVED)
```

```
optimization_ai999x.solve('mmc_queue', lam=2, mu=3, c=1)
# -> rho = 0.666667, P_wait = 0.666667, Lq = 1.33333, Wq = 0.666667, W = 1, L = 2, P0 =
0.333333   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Customers arrive at 24 per hour at a bank with 3 tellers; each teller serves 10 per hour. Find the probability of waiting and Lq for this M/M/c queue.	SOLVED	rho = 0.8, P_wait = 0.647191, Lq = 2.58876, Wq = 0.107865, W = 0.207865, L = 4.98876, P0 = 0.0561798
An M/M/2 queue has arrival rate 2 per hour and service rate 3 per hour. Find Lq.	SOLVED	rho = 0.333333, P_wait = 0.166667, Lq = 0.0833333, Wq = 0.0416667, W = 0.375, L = 0.75, P0 = 0.5
An M/M/2 queue has arrival rate 8 per hour and service rate 3 per hour. Find Lq.	INVALID_INPUT	M/M/c queue (Erlang C): the queue is unstable: lambda must be below c * mu.

M/M/c/K queue (finite capacity), `mmck_finite_queue`

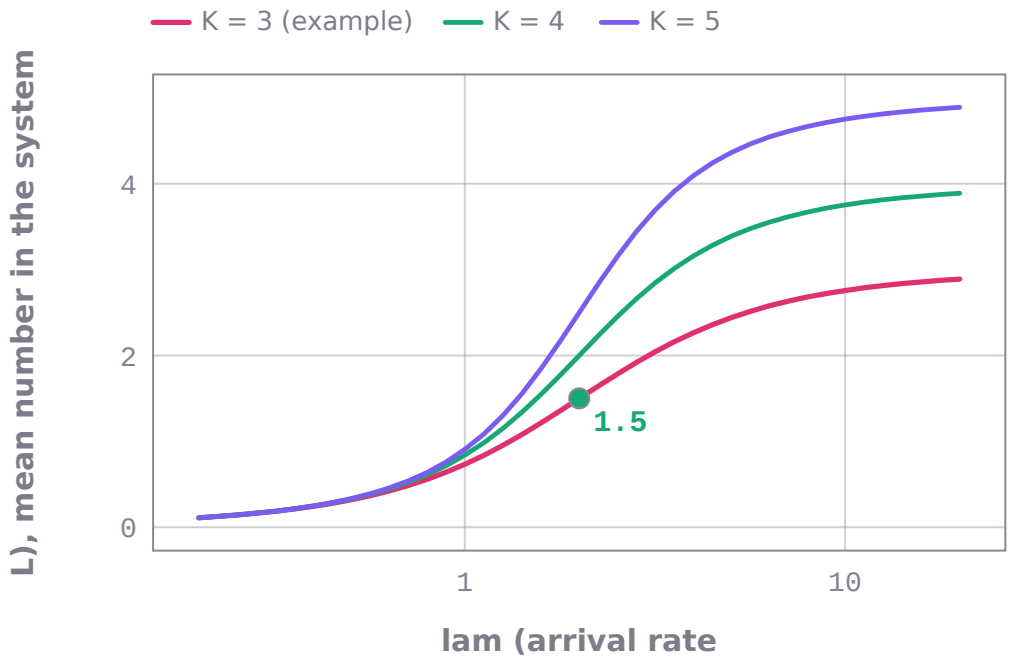
Steady-state measures of a queue with c servers and room for at most K customers in the system (arrivals finding it full are lost): blocking probability, effective arrival rate, mean numbers and times.

$$\pi_n \propto \frac{a^n}{n!}\ (n \le c), \quad \frac{a^c}{c!}\left(\frac{a}{c}\right)^{n-c}\ (c < n \le K)$$

input	meaning	condition
`lam`	arrival rate	
`mu`	service rate of each server	
`c`	number of servers	
`K`	system capacity (customers in service plus waiting)	

output	meaning
`P_block`	probability an arrival is lost (system full)
`lam_eff`	effective arrival rate
`L`	mean number in the system
`Lq`	mean number waiting
`W`	mean time in the system
`Wq`	mean waiting time
`P0`	probability the system is empty

- **Method:** truncated birth-death distribution computed in log space; Little's law with the effective arrival rate
- **Assumes:** $1 \leq c \leq K$; any utilisation (finite capacity is always stable)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



L (mean number in the system) against lam (arrival rate) from 0.2 to 20 on a logarithmic axis. One curve for each value of K (system capacity (customers in service plus waiting)): 3, 4, 5. Other inputs as in the worked example: mu = 2, c = 1. 41 points per curve computed by optimization-ai999x; marked: the worked example, lam = 2, L = 1.5.

```
optimization_ai999x.solve('mmck_finite_queue', lam=2, mu=2, c=1, K=3)
# -> P_block = 0.25, lam_eff = 1.5, L = 1.5, Lq = 0.75, W = 1, Wq = 0.5, P0 = 0.25
(SOLVED)
```

```
optimization_ai999x.solve('mmck_finite_queue', lam=3, mu=1, c=2, K=2)
# -> P_block = 0.529412, lam_eff = 1.41176, L = 1.41176, Lq = 0, W = 1, Wq = 0, P0 =
0.117647   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
An M/M/1/3 queue has arrival rate 2 per hour and service rate 2 per hour. Find the blocking probability and L.	SOLVED	P_block = 0.25, lam_eff = 1.5, L = 1.5, Lq = 0.75, W = 1, Wq = 0.5, P0 = 0.25
An M/M/2/2 queue has arrival rate 3 per hour and service rate 1 per hour. Find the probability an arrival is lost.	SOLVED	P_block = 0.529412, lam_eff = 1.41176, L = 1.41176, Lq = 0, W = 1, Wq = 0, P0 = 0.117647

Erlang B blocking probability (M/M/c/c loss system), `erlang_b`

The probability that a call (customer) is blocked in a loss system with c servers and no waiting room, for an offered traffic of A erlangs, with the carried and lost traffic.

$$B(c, A) = \frac{A^c/c!}{\sum_{k=0}^{c} A^k/k!}$$

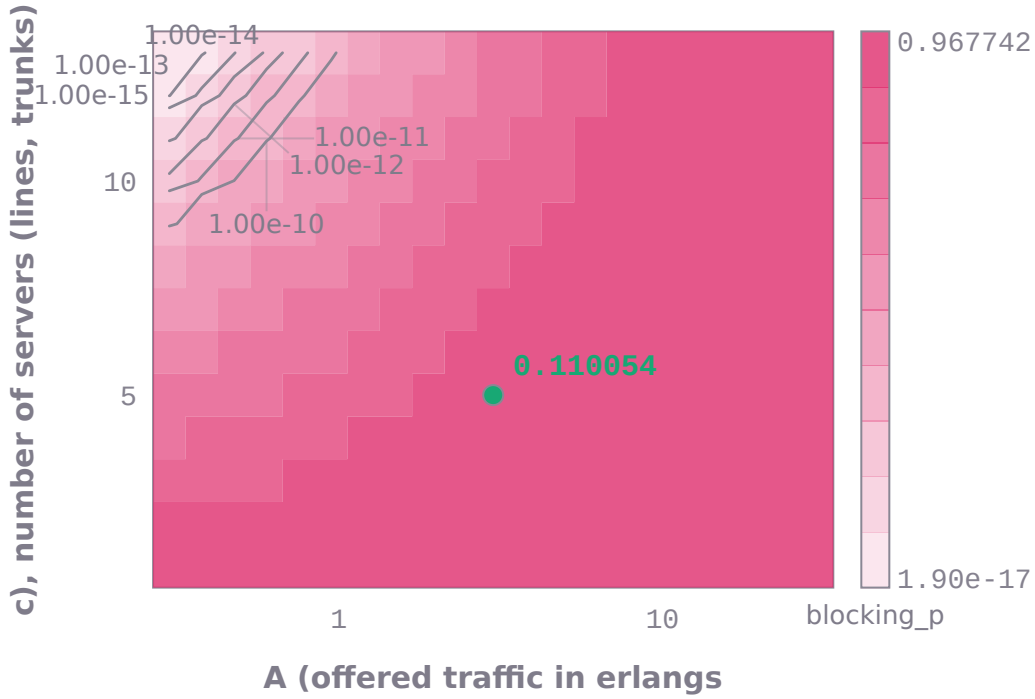
input	meaning	condition
A	offered traffic in erlangs (arrival rate x mean holding time)	
c	number of servers (lines, trunks)	

output	meaning
`blocking_probability`	Erlang B blocking probability
`carried_traffic`	carried traffic A $(1 - B)$, erlangs
`lost_traffic`	lost traffic A B, erlangs

- **Method:** stable recursion B(k) = A B($k - 1$)/($k + A$ B(k-1))
- **Assumes:** blocked customers are lost (no waiting); $A > 0$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



blocking_probability (Erlang B blocking probability) against A (offered traffic in erlangs (arrival rate x mean holding time)) from 0.3 to 30 on a logarithmic axis, logarithmic output axis. One curve for each value of c (number of servers (lines, trunks)): 5, 2, 10. 41 points per curve computed by optimization-ai999x; marked: the worked example, A = 3, blocking_probability = 0.110054.



blocking_probability (Erlang B blocking probability) over A (offered traffic in erlangs (arrival rate x mean holding time)) from 0.3 to 30 and c (number of servers (lines, trunks)) from 1 to 13 (logarithmic horizontal axis): each of the 21 × 13 cells is one run of optimization-ai999x, shaded by its value (light 1.8967e-17 to dark 0.967742, logarithmic shading); iso-lines at blocking_probability = 1.00e-15, 1.00e-14, 1.00e-13, 1.00e-12, 1.00e-11, 1.00e-10. Marked: the worked example, A = 3, c = 5, blocking_probability = 0.110054.

```
optimization_ai999x.solve('erlang_b', A=3, c=5)
# -> blocking_probability = 0.110054, carried_traffic = 2.66984, lost_traffic = 0.330163
(SOLVED)
```

```
optimization_ai999x.solve('erlang_b', A=1, c=1)
# -> blocking_probability = 0.5, carried_traffic = 0.5, lost_traffic = 0.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
A trunk group of 5 lines is offered traffic of 3 erlangs. What is the Erlang B blocking probability?	SOLVED	blocking_probability = 0.110054, carried_traffic = 2.66984, lost_traffic = 0.330163

Machine-repair (finite-source) queue, `machine_repair_queue`

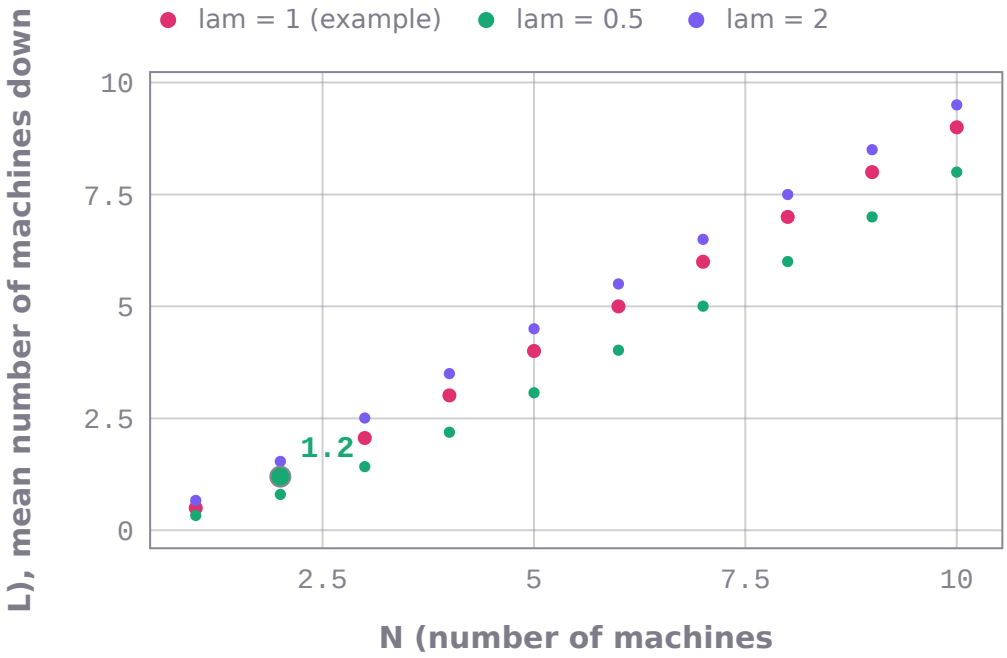
N machines each fail at rate lambda while running and c repairers fix them at rate mu each: mean number of machines down and waiting, failure rate, downtime and the mean number working.

$$\pi_n \propto \frac{N!}{(N-n)!}\,\frac{(\lambda/\mu)^n}{n!\text{ or }c!\,c^{n-c}}$$

input	meaning	condition
`N`	number of machines	
`lam`	failure rate of one running machine	
`mu`	repair rate of one repairer	
`c`	number of repairers (default 1)	

output	meaning
`L`	mean number of machines down
`Lq`	mean number waiting for repair
`lam_eff`	mean failure rate lambda (N - L)
`W`	mean downtime per failure
`Wq`	mean wait for a repairer
`P0`	probability all machines run
`working`	mean number running

- **Method:** finite birth-death distribution (birth (N - n) lambda, death min(n, c) mu) in log space; Little's law
- **Assumes:** $1 \le c \le N$; exponential failure and repair times
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



L (mean number of machines down) against N (number of machines) from 1 to 10. One curve for each value of lam (failure rate of one running machine): 1, 0.5, 2. Other inputs as in the worked example: mu = 1, c = 1. 10 points per curve computed by optimization-ai999x; marked: the worked example, N = 2, L = 1.2.

```
optimization_ai999x.solve('machine_repair_queue', N=2, lam=1, mu=1, c=1)
# -> L = 1.2, Lq = 0.4, lam_eff = 0.8, W = 1.5, Wq = 0.5, P0 = 0.2, working = 0.8
(SOLVED)
```

```
optimization_ai999x.solve('machine_repair_queue', N=5, lam=0.2, mu=1, c=1)
# -> L = 1.42434, Lq = 0.709207, lam_eff = 0.715132, W = 1.99171, Wq = 0.991714, P0 =
0.284868, working = 3.57566   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
There are 5 machines; each machine breaks down at a rate of 0.2 per hour and one repairer repairs at a rate of 1 per hour. Find the mean number of machines down.	SOLVED	L = 1.42434, Lq = 0.709207, lam_eff = 0.715132, W = 1.99171, Wq = 0.991714, P0 = 0.284868, working = 3.57566

M/G/1 queue (Pollaczek-Khinchine formula), `mg1_pollaczek_khinchine`

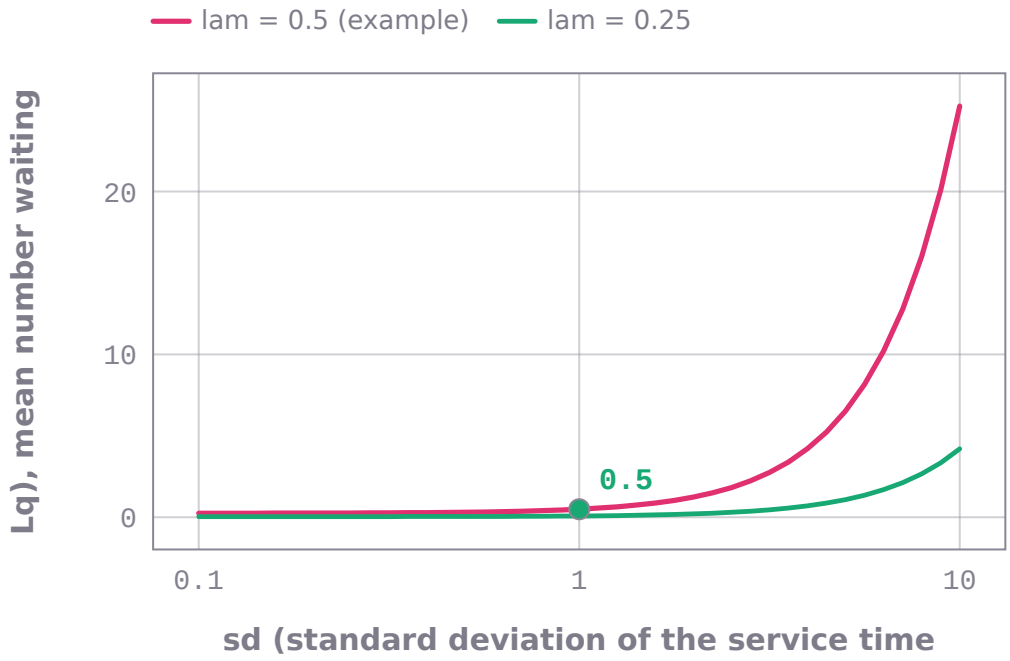
Mean queue length and waiting times of a single-server queue with Poisson arrivals and a GENERAL service time distribution given by its mean and standard deviation (M/D/1 when the standard deviation is 0).

$$L_q = \frac{\lambda^2 \sigma_S^2 + \rho^2}{2(1-\rho)}$$

input	meaning	condition
`lam`	arrival rate	
`ES`	mean service time (time unit of 1/lambda)	
`sd`	standard deviation of the service time	

output	meaning
`rho`	utilisation lambda E[S]
`Lq`	mean number waiting
`L`	mean number in the system
`Wq`	mean waiting time
`W`	mean time in the system

- **Method:** Pollaczek-Khinchine mean-value formula; Little's law
- **Assumes:** lambda E[S] < 1; Poisson arrivals, one server, first come first served
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Lq (mean number waiting) against sd (standard deviation of the service time) from 0.1 to 10 on a logarithmic axis. One curve for each value of lam (arrival rate): 0.5, 0.25. Other inputs as in the worked example: ES = 1. 41 points per curve computed by optimization-ai999x; marked: the worked example, sd = 1, Lq = 0.5.

```
optimization_ai999x.solve('mg1_pollaczek_khinchine', lam=0.5, ES=1, sd=1)
# -> rho = 0.5, Lq = 0.5, L = 1, Wq = 1, W = 2   (SOLVED)
```

```
optimization_ai999x.solve('mg1_pollaczek_khinchine', lam=0.5, ES=1, sd=0)
# -> rho = 0.5, Lq = 0.25, L = 0.75, Wq = 0.5, W = 1.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
In an M/G/1 queue the arrival rate is 0.5 per minute, the mean service time is 1 minute and its standard deviation is 1 minute. Find Lq.	SOLVED	rho = 0.5, Lq = 0.5, L = 1, Wq = 1, W = 2
An M/D/1 queue has an arrival rate of 0.5 per minute and a constant service time of 1 minute. Find Wq.	SOLVED	rho = 0.5, Lq = 0.25, L = 0.75, Wq = 0.5, W = 1.5
In an M/G/1 queue the arrival rate is 2 per minute, the mean service time is 1 minute and its standard	INVALID_INPUT	M/G/1 queue (Pollaczek-Khinchine formula): the queue is unstable: lambda times the mean service

question	outcome	result
deviation is 1 minute. Find Lq.		time must be below 1.

Minimum number of servers for a service target (Erlang C), `mmc_staffing`

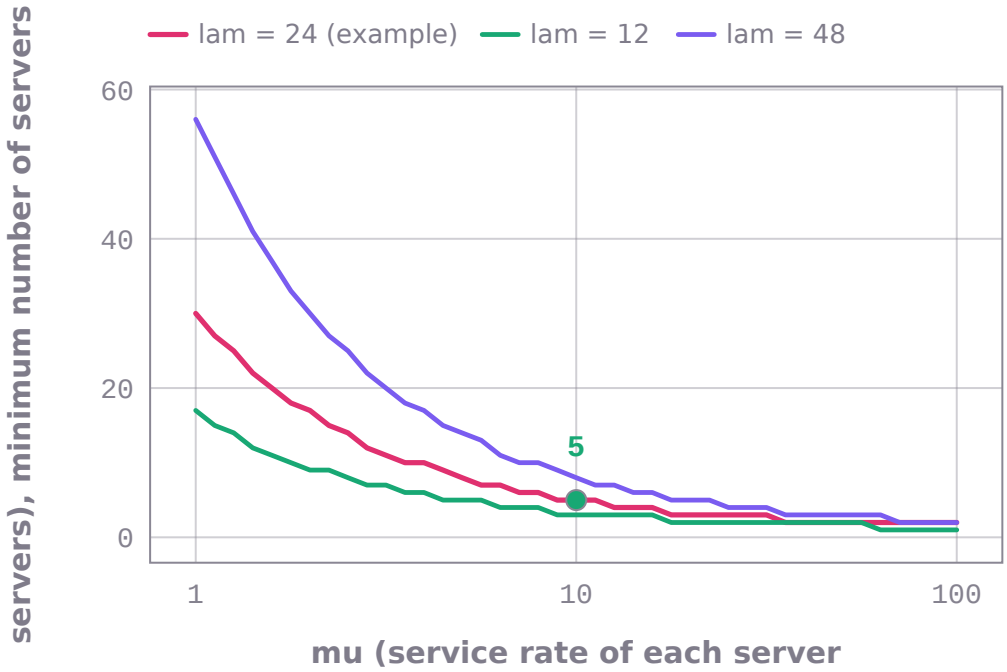
The smallest number of servers c of an M/M/c queue for which the probability of waiting, or the mean waiting time, meets a target.

$$c^* = \min\{c > \lambda/\mu : C(c, \lambda/\mu) \leq \alpha\}$$

input	meaning	condition
`lam`	arrival rate	
`mu`	service rate of each server	
`target`	target value (a probability, or a mean wait in the time unit of the rates)	
`criterion`	wait_probability (default) or mean_wait	

output	meaning
`servers`	minimum number of servers
`P_wait`	probability of waiting with that many servers
`Wq`	mean waiting time with that many servers
`utilisation`	server utilisation

- **Method:** increase c from floor(lambda/mu) + 1 until the Erlang C measure meets the target
- **Assumes:** Poisson arrivals, exponential service; target > 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



servers (minimum number of servers) against mu (service rate of each server) from 1 to 100 on a logarithmic axis. One curve for each value of lam (arrival rate): 24, 12, 48. Other inputs as in the worked example: target = 0.2. 41 points per curve computed by optimization-ai999x; marked: the worked example, mu = 10, servers = 5.

```
optimization_ai999x.solve('mmc_staffing', lam=24, mu=10, target=0.2)
# -> servers = 5, P_wait = 0.113504, Wq = 0.00436553, utilisation = 0.48   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Calls arrive at 24 per hour and each agent serves 10 per hour. How many agents are needed so that the probability of waiting is at most 0.2?	SOLVED	servers = 5, P_wait = 0.113504, Wq = 0.00436553, utilisation = 0.48

Shortest processing time rule (minimum mean flow time), `spt_mean_flow_time`

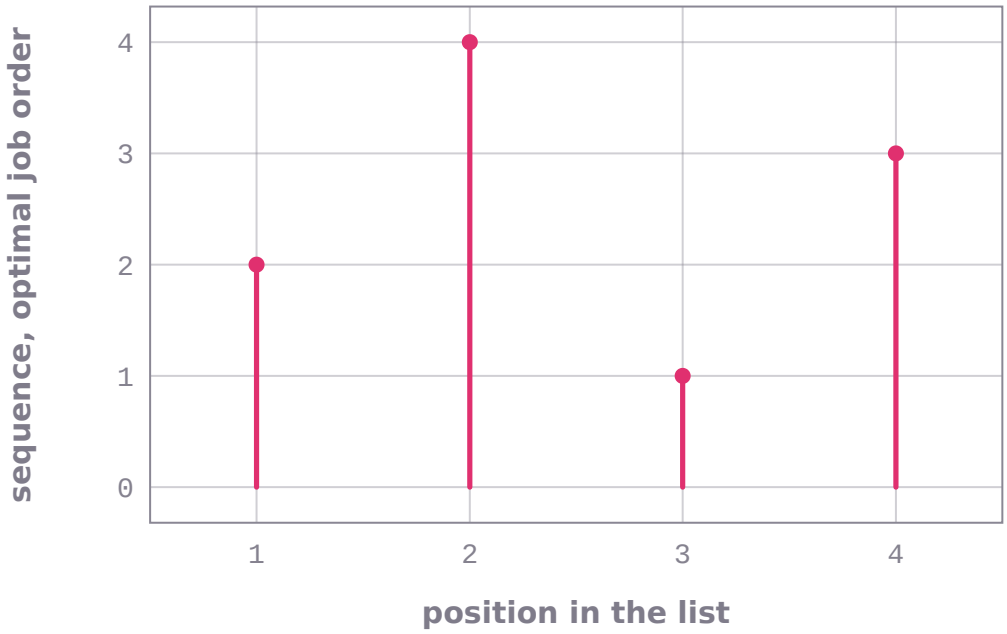
The job order that minimises the mean flow (completion) time on one machine, and that minimum.

$$\min \frac{1}{n}\sum_j C_j : \; p_{[1]} \le p_{[2]} \le \dots$$

input	meaning	condition
`p`	processing time of each job (jobs numbered 1, 2, ...)	

output	meaning
`mean_flow_time`	minimum mean completion time
`total_completion_time`	minimum total completion time
`sequence`	optimal job order (shortest first; equal times in job-number order)

- **Method:** sort by processing time (SPT); optimal by the exchange argument
- **Assumes:** all jobs available at time 0; processing times >= 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



optimal job order (shortest first; equal times in job-number order) (sequence) returned by optimization-ai999x for the worked example, one value for each position in the list (4 values, first 2, last 3, largest 4). Inputs: p = 4, 2, 6, 3.

```
optimization_ai999x.solve('spt_mean_flow_time', p='4, 2, 6, 3')
# -> sequence = [2, 4, 1, 3], total_completion_time = 31, mean_flow_time = 7.75   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Jobs with processing times [4, 2, 6, 3] are done on one machine. Find the sequence that minimizes the mean flow time.	SOLVED	sequence = [2, 4, 1, 3], total_completion_time = 31, mean_flow_time = 7.75

Weighted shortest processing time (Smith's rule), `wspt_weighted_completion`

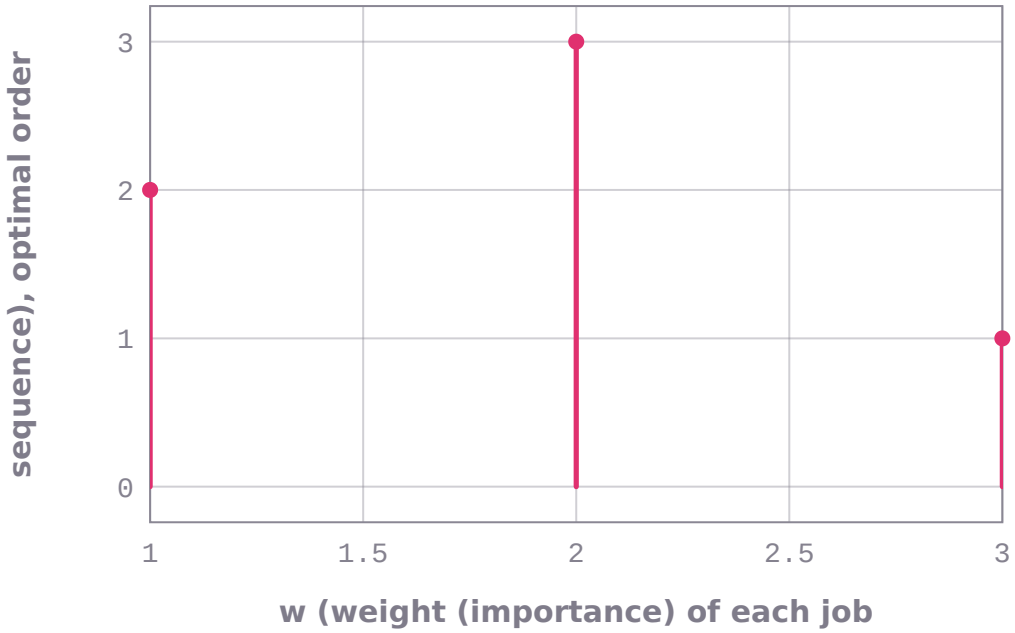
The job order that minimises the total weighted completion time on one machine, and that minimum.

$$\min \sum_j w_j C_j : \frac{p_{[1]}}{w_{[1]}} \le \frac{p_{[2]}}{w_{[2]}} \le \dots$$

input	meaning	condition
`p`	processing time of each job	
`w`	weight (importance) of each job	

output	meaning
`weighted_completion_time`	minimum total weighted completion time
`sequence`	optimal order (ratio p/w ascending; ties in job-number order)

- **Method:** sort by p_j / w_j (Smith's ratio rule)
- **Assumes:** weights > 0; all jobs available at time 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



optimal order (ratio p/w ascending; ties in job-number order) (sequence) returned by optimization-ai999x for the worked example, one value for each w (weight (importance) of each job) (3 values, first 2, last 1, largest 3). Inputs: p = [3, 1, 4], w = [1, 2, 3].

```
optimization_ai999x.solve('wspt_weighted_completion', p=[3, 1, 4], w=[1, 2, 3])
# -> sequence = [2, 3, 1], weighted_completion_time = 25   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Jobs have processing times [3, 1, 4] and weights [1, 2, 3]. Find the sequence minimizing the total weighted completion time.	SOLVED	sequence = [2, 3, 1], weighted_completion_time = 25

Earliest due date rule (minimum maximum lateness), `edd_max_lateness`

The job order that minimises the maximum lateness (and so the maximum tardiness) on one machine.

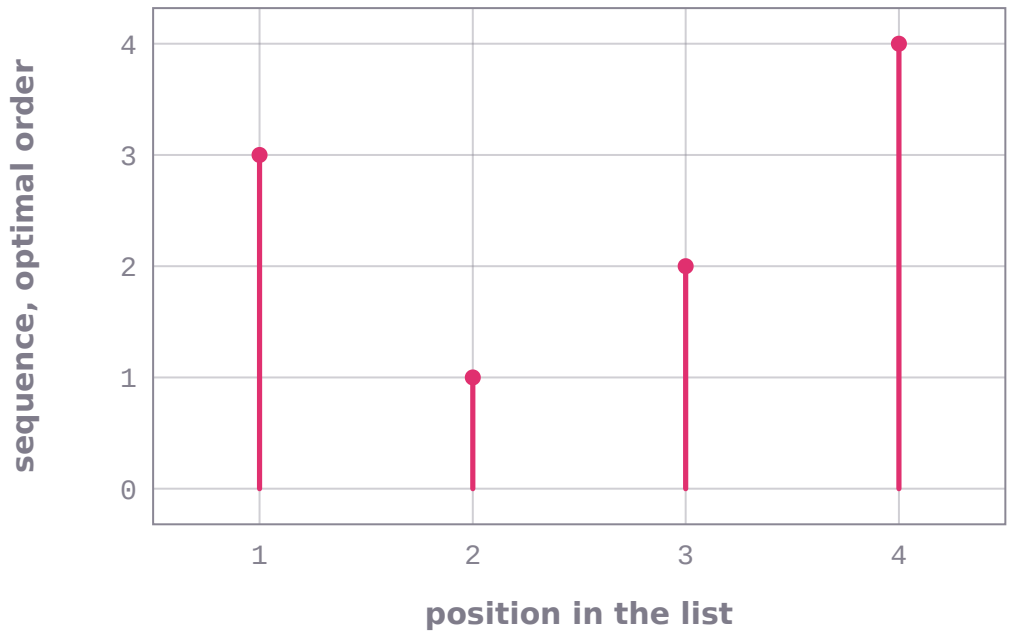
$$\min L_{\max} = \min \max_j (C_j - d_j) : \; d_{[1]} \le d_{[2]} \le \dots$$

input	meaning	condition
`p`	processing time of each job	
`d`	due date of each job	

output	meaning
`max_lateness`	minimum possible maximum lateness
`max_tardiness`	minimum possible maximum tardiness
`sequence`	optimal order (earliest due date first; ties in job-number order)
`tardy_jobs_in_sequence`	number of late jobs in this sequence (not minimised)

- **Method:** sort by due date (Jackson's rule)
- **Assumes:** all jobs available at time 0

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



optimal order (earliest due date first; ties in job-number order) (sequence) returned by optimization-ai999x for the worked example, one value for each position in the list (4 values, first 3, last 4, largest 4). Inputs: p = [2, 4, 1, 3], d = [5, 6, 3, 9].

```
optimization_ai999x.solve('edd_max_lateness', p=[2, 4, 1, 3], d=[5, 6, 3, 9])
# -> sequence = [3, 1, 2, 4], max_lateness = 1, max_tardiness = 1, tardy_jobs_in_sequence =
2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Jobs have processing times [2, 4, 1, 3] and due dates [5, 6, 3, 9]. Find the sequence that minimizes the maximum lateness.	SOLVED	sequence = [3, 1, 2, 4], max_lateness = 1, max_tardiness = 1, tardy_jobs_in_sequence = 2

Moore-Hodgson algorithm (minimum number of tardy jobs), `moore_hodgson_tardy_jobs`

The smallest possible number of late jobs on one machine, with an order of the on-time jobs and the set of late jobs.

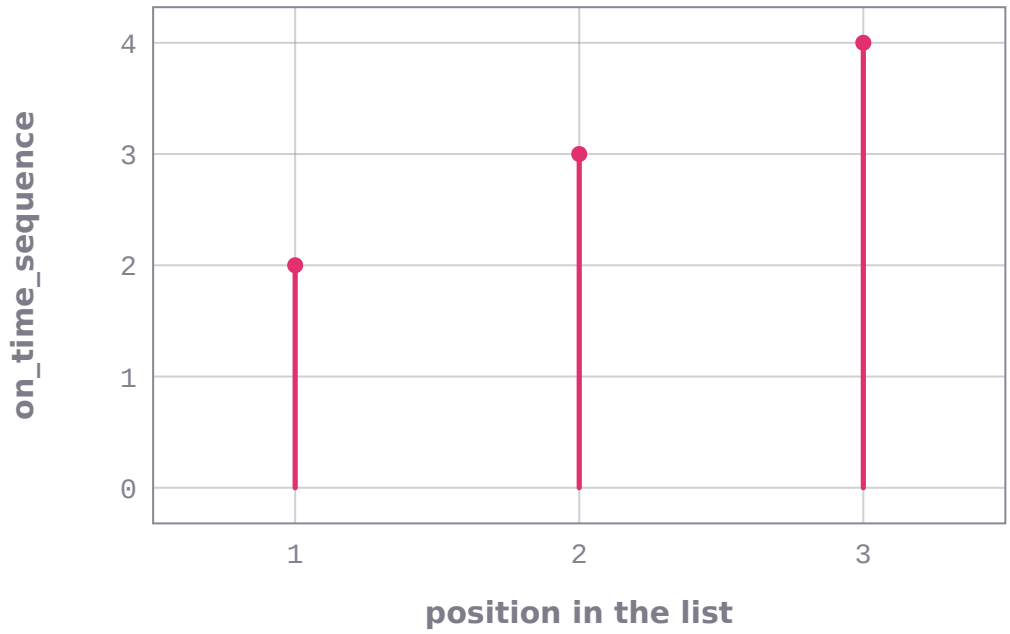
$$\min \sum_j U_j$$

input	meaning	condition
`p`	processing time of each job	
`d`	due date of each job	

output	meaning
`min_tardy_jobs`	minimum number of tardy jobs
`on_time_sequence`	on-time jobs in due-date order

output	meaning
`tardy_jobs`	jobs scheduled late (one optimal choice: the longest job removed, ties the higher job number)

- **Method:** EDD order; whenever a job is late remove the longest job scheduled so far
- **Assumes:** all jobs available at time 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



on-time jobs in due-date order (on_time_sequence) returned by optimization-ai999x for the worked example, one value for each position in the list (3 values, first 2, last 4, largest 4). Inputs: p = [4, 3, 2, 5, 6], d = [5, 6, 8, 12, 15].

```
optimization_ai999x.solve('moore_hodgson_tardy_jobs', p=[4, 3, 2, 5, 6], d=[5, 6, 8, 12, 15])
# -> min_tardy_jobs = 2, on_time_sequence = [2, 3, 4], tardy_jobs = [1, 5]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Jobs have processing times [4, 3, 2, 5, 6] and due dates [5, 6, 8, 12, 15]. Find the minimum number of tardy jobs.	SOLVED	min_tardy_jobs = 2, on_time_sequence = [2, 3, 4], tardy_jobs = [1, 5]

Johnson's rule (two-machine flow shop), `johnson_two_machine`

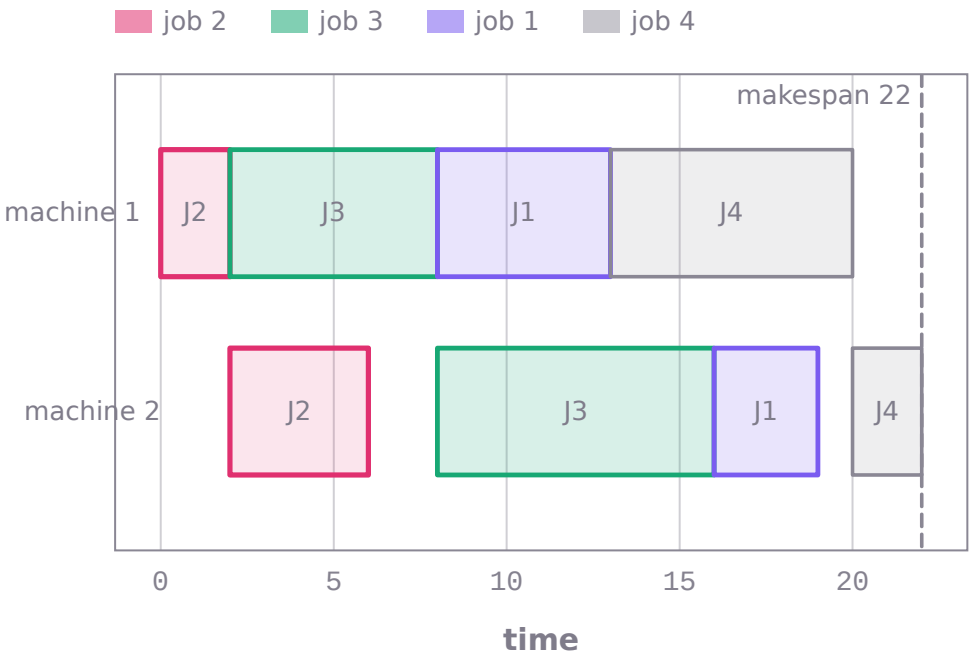
The job order that minimises the makespan when every job is processed first on machine 1 and then on machine 2.

$$\min C_{\max} \text{ for } F2||C_{\max}$$

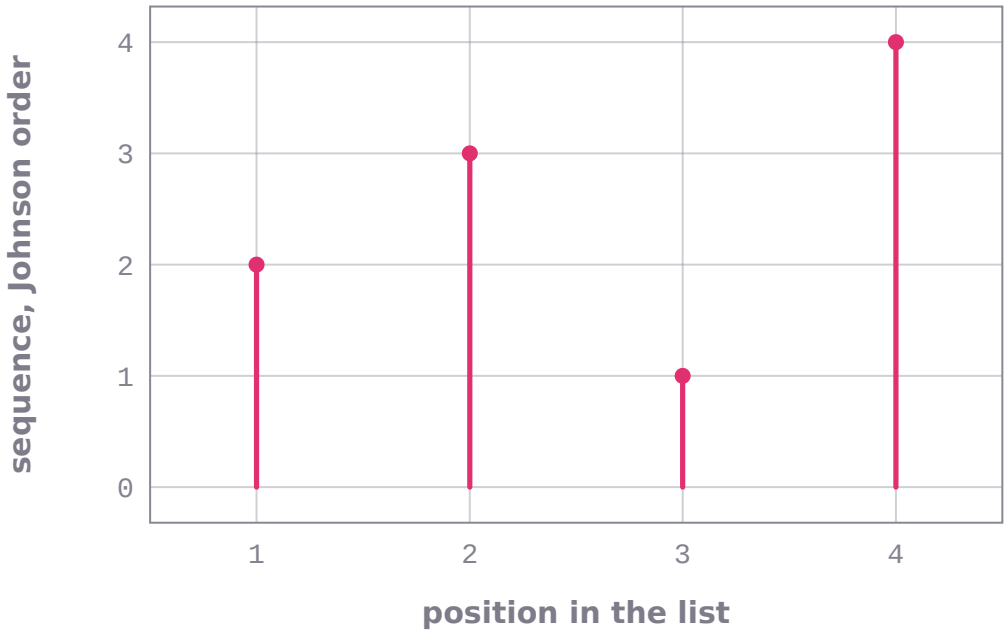
input	meaning	condition
a	processing time of each job on machine 1	
b	processing time of each job on machine 2	

output	meaning
`makespan`	minimum makespan
`sequence`	Johnson order (a < b by increasing a, then the rest by decreasing b; ties by job number)

- **Method:** Johnson's rule
- **Assumes:** same job order on both machines; all jobs available at time 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



A two-machine flow shop: processing times [5, 2, 6, 7] on machine 1 and [3, 4, 8, 2] on machine 2. The package's Johnson order is job 2 → job 3 → job 1 → job 4, with makespan 22; each job starts on machine 2 as soon as it leaves machine 1 and machine 2 is free.



Johnson order (a < b by increasing a, then the rest by decreasing b; ties by job number) (sequence) returned by optimization-ai999x for the worked example, one value for each position in the list (4 values, first 2, last 4, largest 4). Inputs: a = [5, 2, 6, 7], b = [3, 4, 8, 2].

```
optimization_ai999x.solve('johnson_two_machine', a=[5, 2, 6, 7], b=[3, 4, 8, 2])
# -> makespan = 22, sequence = [2, 3, 1, 4]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Use Johnson's rule: machine 1 times [5, 2, 6, 7] and machine 2 times [3, 4, 8, 2]. Find the minimum makespan.	SOLVED	makespan = 22, sequence = [2, 3, 1, 4]

Minimum makespan on identical parallel machines, `parallel_machine_makespan`

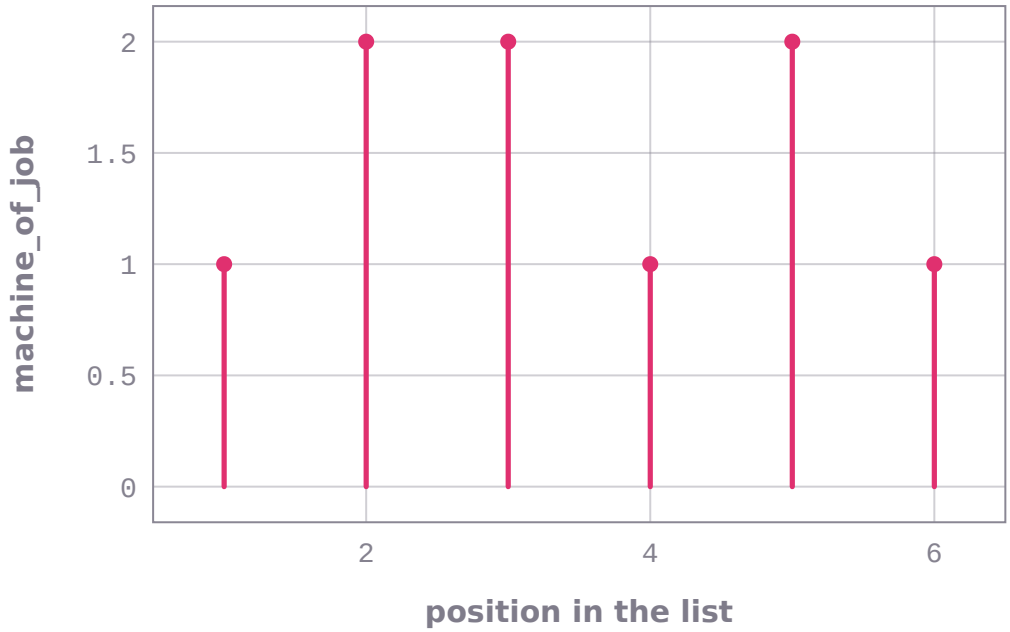
Assign jobs to m identical machines so that the last machine finishes as early as possible (P||Cmax), exactly.

$$\min_{\text{assignment}} \max_{k} \sum_{j \to k} p_j$$

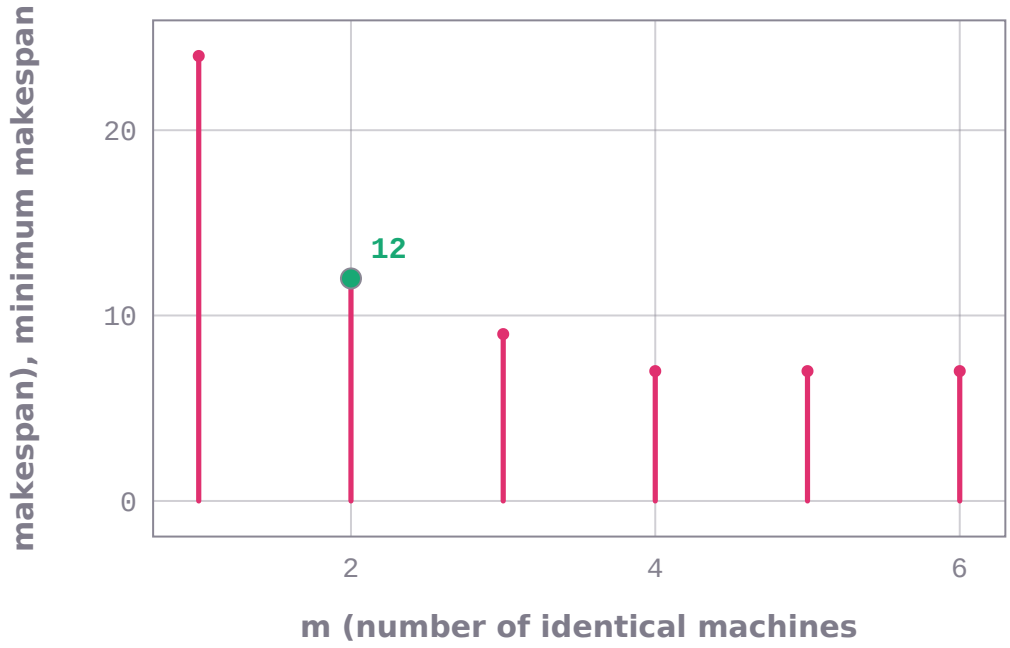
input	meaning	condition
`p`	processing time of each job	
`m`	number of identical machines	

output	meaning
`makespan`	minimum makespan
`machine_of_job`	one optimal assignment: machine number of each job
`lower_bound`	the bound max(longest job, total / m)

- **Method:** exact depth-first branch and bound (longest job first, symmetric machines skipped), started from the LPT schedule
- **Assumes:** at most 12 jobs and 6 machines; jobs are not split
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



one optimal assignment: machine number of each job (machine_of_job) returned by optimization-ai999x for the worked example, one value for each position in the list (6 values, first 1, last 1, largest 2). Inputs: p = [7, 5, 4, 3, 3, 2], m = 2.



makespan (minimum makespan) against m (number of identical machines) from 1 to 10. Other inputs as in the worked example: p = [7, 5, 4, 3, 3, 2]. 6 points computed by optimization-ai999x; marked: the worked example, m = 2, makespan = 12.

```
optimization_ai999x.solve('parallel_machine_makespan', p=[7, 5, 4, 3, 3, 2], m=2)
# -> makespan = 12, machine_of_job = [1, 2, 2, 1, 2, 1], lower_bound = 12   (SOLVED)
```

```
optimization_ai999x.solve('parallel_machine_makespan', p=[3, 3, 2, 2, 2], m=2)
# -> makespan = 6, machine_of_job = [1, 1, 2, 2, 2], lower_bound = 6   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Jobs with processing times [7, 5, 4, 3, 3, 2] are scheduled on 2 identical machines. Find the minimum makespan.	SOLVED	makespan = 12, machine_of_job = [1, 2, 2, 1, 2, 1], lower_bound = 12

Critical path method (project duration and slack), `critical_path_method`

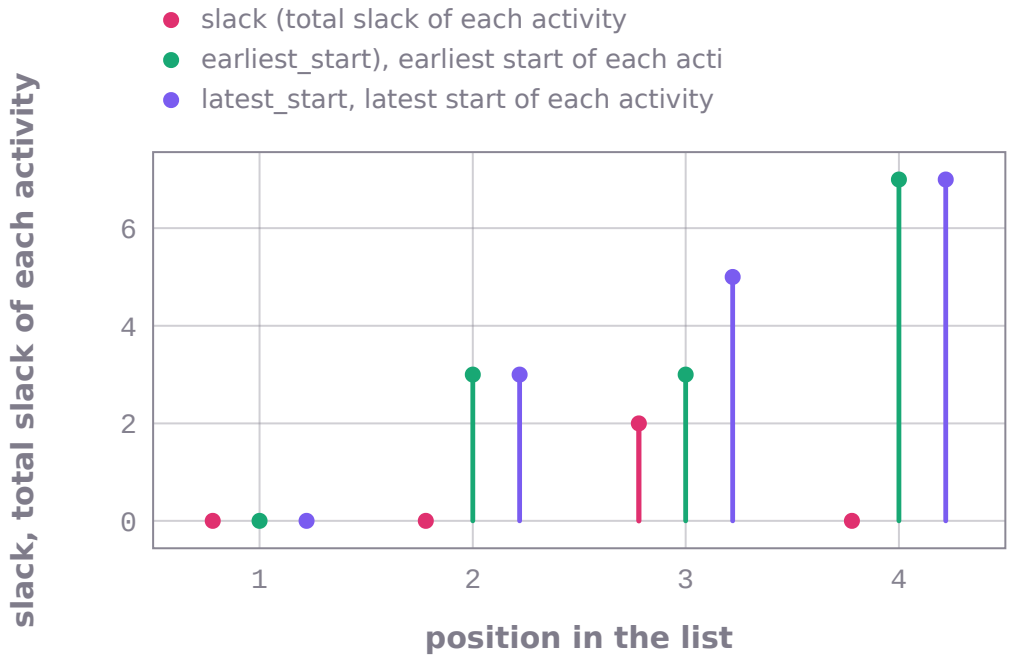
For activities with durations and precedences, the minimum project duration, the critical activities and each activity's earliest start, latest start and slack (total float).

$$ES_k = \max_{j \to k}(ES_j + d_j), \quad LS_k = \min_{k \to s} LS_s - d_k, \quad \text{slack}_k = LS_k - ES_k$$

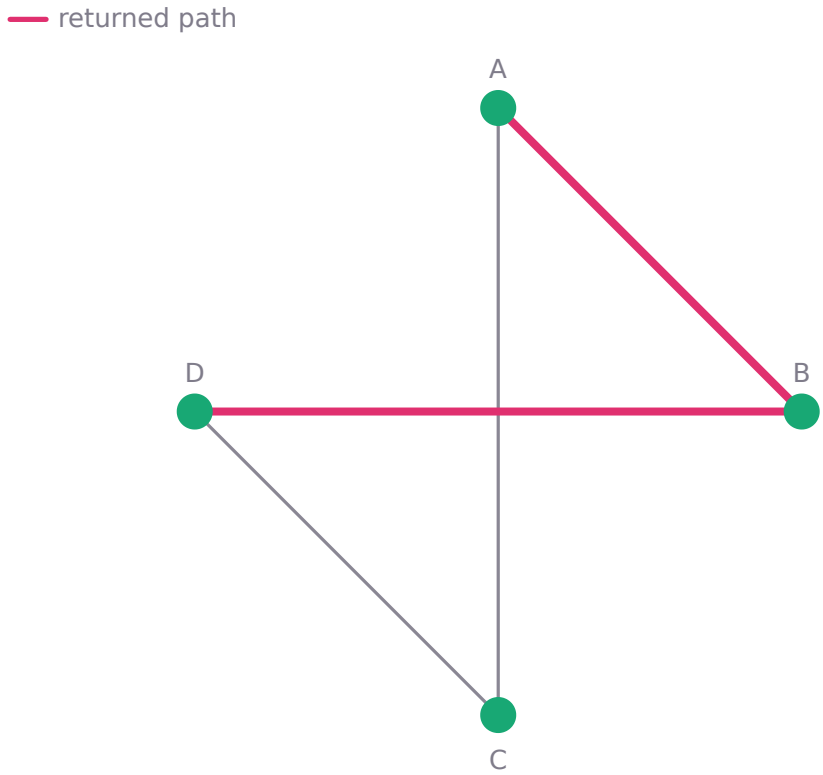
input	meaning	condition
`durations`	duration of each activity; activities are named A, B, C, ... (or 1, 2, 3, ...) in this order	
`precedences`	precedence pairs 'A-B' meaning A must finish before B starts (none if omitted)	

output	meaning
`project_duration`	minimum project duration
`critical_activities`	activities with zero slack
`slack`	total slack of each activity
`earliest_start`	earliest start of each activity
`latest_start`	latest start of each activity

- **Method:** forward pass (earliest starts) and backward pass (latest finishes) on the activity-on-node network, exact fractions
- **Assumes:** finish-to-start precedences without cycles; deterministic durations
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



total slack of each activity (slack) returned by optimization-ai999x for the worked example, one value for each position in the list (4 values, first 0, last 0, largest 2); with earliest_start (earliest start of each activity), latest_start (latest start of each activity). Inputs: durations = 3, 4, 2, 5, precedences = A-B, A-C, B-D, C-D.



The network precedences (precedence pairs 'A-B' meaning A must finish before B starts (none if omitted)) of the worked example (4 nodes, 4 edges, laid out on a circle); highlighted: the returned path A → B → D. Returned: project_duration = 12. Other inputs: durations = 3, 4, 2, 5.

```
optimization_ai999x.solve('critical_path_method', durations='3, 4, 2, 5', precedences='A-B, A-C, B-D, C-D')
# -> project_duration = 12, critical_activities = ['A', 'B', 'D'], slack = [0.0, 0.0, 2.0, 0.0], earliest_start = [0.0, 3.0, 3.0, 7.0], latest_start = [0.0, 3.0, 5.0, 7.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Activities A, B, C, D have durations [3, 4, 2, 5]; precedences: A-B, A-C, B-D, C-D. Find the critical path and the project duration.	SOLVED	project_duration = 12, critical_activities = ['A', 'B', 'D'], slack = [0.0, 0.0, 2.0, 0.0], earliest_start = [0.0, 3.0, 3.0, 7.0], latest_start = [0.0, 3.0, 5.0, 7.0]
Find the critical path: activities have durations [3, 4, 2]; precedences: A-B, B-C, C-A.	INVALID_INPUT	Critical path method (project duration and slack): the precedences contain a cycle.

PERT project duration and completion probability, `pert_project`

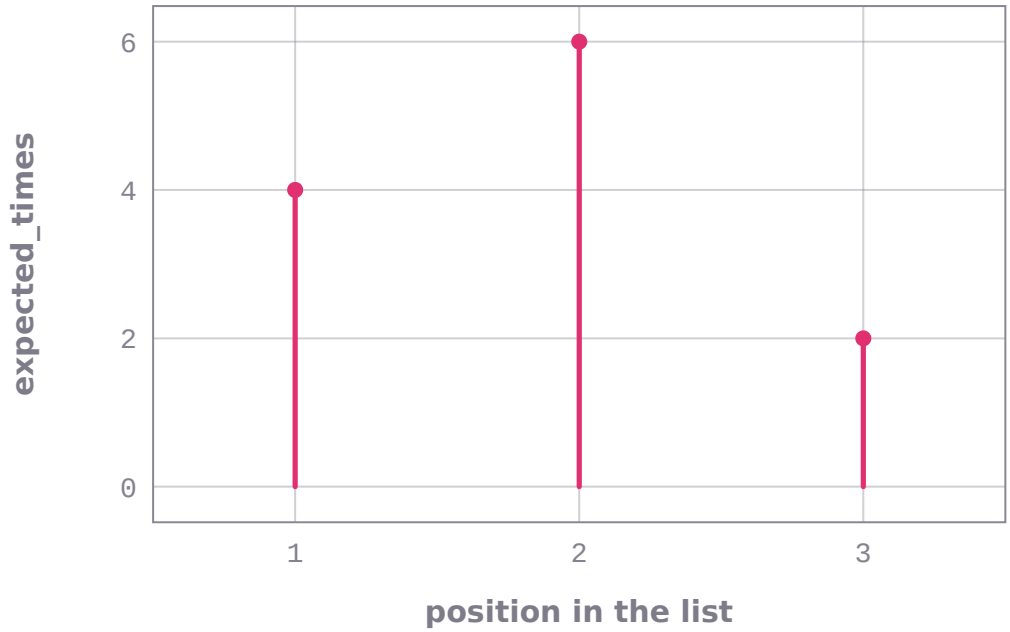
With optimistic, most likely and pessimistic times per activity, the expected project duration, its variance along the critical path and the probability of finishing by a deadline (normal approximation).

$$t_e = \frac{a + 4m + b}{6},\ \sigma^2 = \left(\frac{b - a}{6}\right)^2,\ P(T \le d) = \Phi\left(\frac{d - \mu}{\sigma}\right)$$

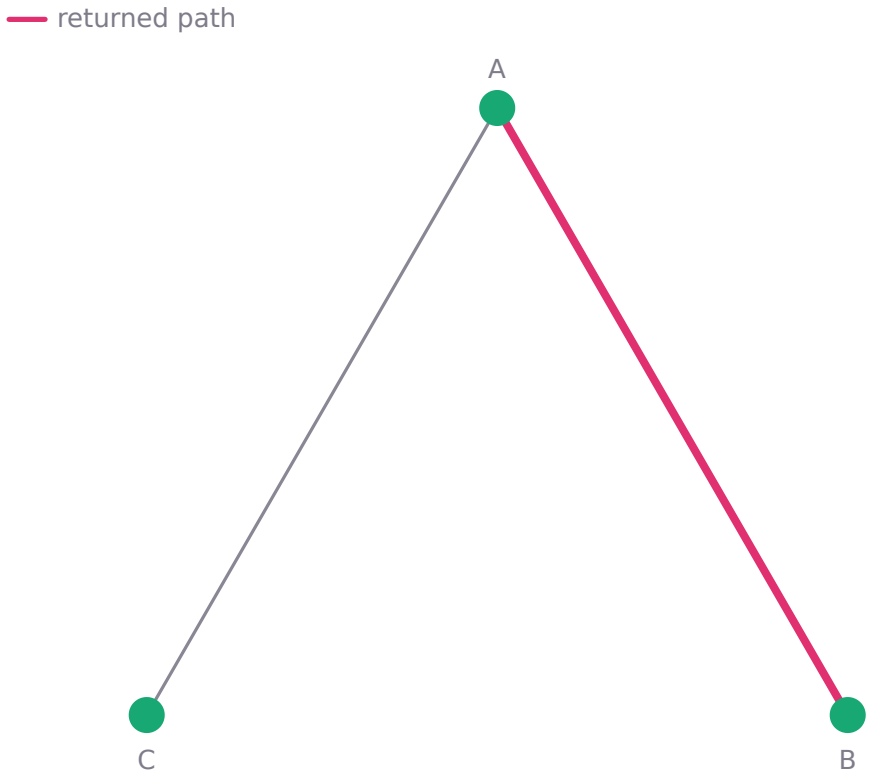
input	meaning	condition
`times`	rows (optimistic, most likely, pessimistic), one per activity A, B, C, ...	
`precedences`	precedence pairs 'A-B' (A before B)	
`deadline`	deadline for the completion probability (optional)	

output	meaning
`expected_duration`	expected project duration
`variance`	variance of the critical path
`std_dev`	standard deviation
`critical_path`	the critical path (must be unique)
`expected_times`	expected time of each activity $(a + 4m + b)/6$
`probability`	probability of completion by the deadline

- **Method:** beta-PERT expected times, longest path in exact fractions with a count of longest paths, variance summed along it
- **Assumes:** a unique critical path (otherwise INVALID_INPUT); independent activity times; normal approximation
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



expected time of each activity (a + 4m + b)/6 (expected_times) returned by optimization-ai999x for the worked example, one value for each position in the list (3 values, first 4, last 2, largest 6). Inputs: times = 2 4 6; 3 5 13; 1 2 3, precedences = A-B, A-C, deadline = 10.



The network precedences (precedence pairs 'A-B' (A before B)) of the worked example (3 nodes, 2 edges, laid out on a circle); highlighted: the returned path A → B. Returned: expected_duration = 10, variance = 3.22222, std_dev = 1.79505, probability = 0.5. Other inputs: times = 2 4 6; 3 5 13; 1 2 3, deadline = 10.

```
optimization_ai999x.solve('pert_project', times='2 4 6; 3 5 13; 1 2 3', precedences='A-B,
A-C', deadline=10)
# -> expected_duration = 10, variance = 3.22222, std_dev = 1.79505, critical_path = ['A',
'B'], expected_times = [4.0, 6.0, 2.0], probability = 0.5   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
PERT: activities A, B, C have times [[2, 4, 6], [3, 5, 13], [1, 2, 3]]; precedences: A-B, A-C. Find the expected project duration.	SOLVED	expected_duration = 10, variance = 3.22222, std_dev = 1.79505, critical_path = ['A', 'B'], expected_times = [4.0, 6.0, 2.0]
PERT: activities A, B have times [[2, 4, 6], [2, 4, 6]]. Find the expected project duration.	INVALID_INPUT	PERT project duration and completion probability: there is more than one critical path, so the PERT variance of the proj

Shortest path (Dijkstra's algorithm), `shortest_path`

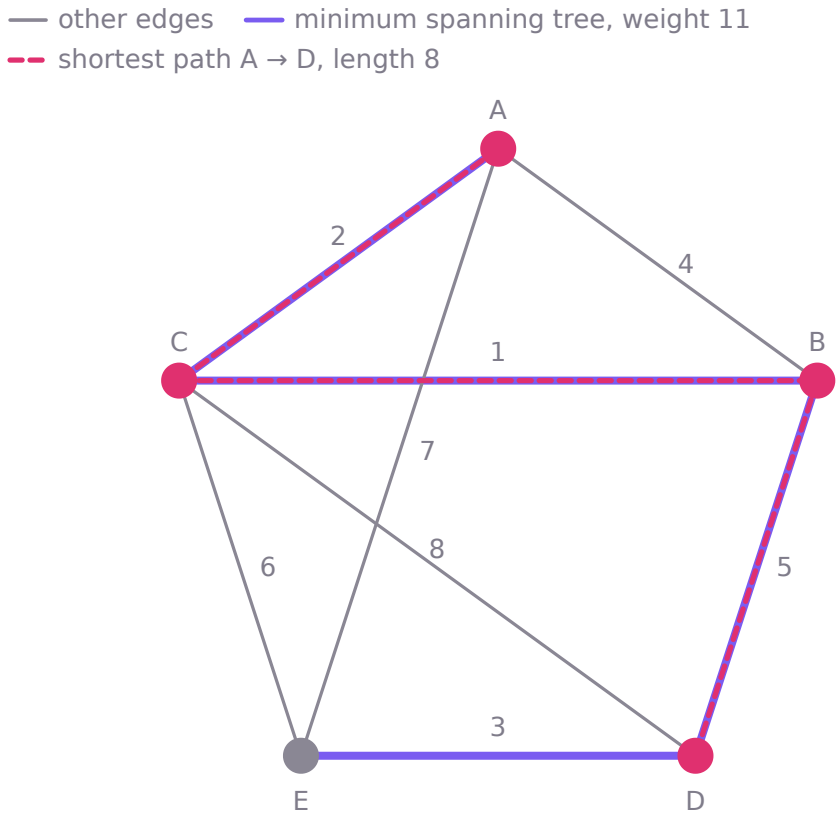
The length of the shortest route between two nodes of a graph with non-negative edge weights, and the route itself (stated as the route only when it is the only shortest one).

$$d(v) = \min_{(u,v)} \left(d(u) + w_{uv} \right)$$

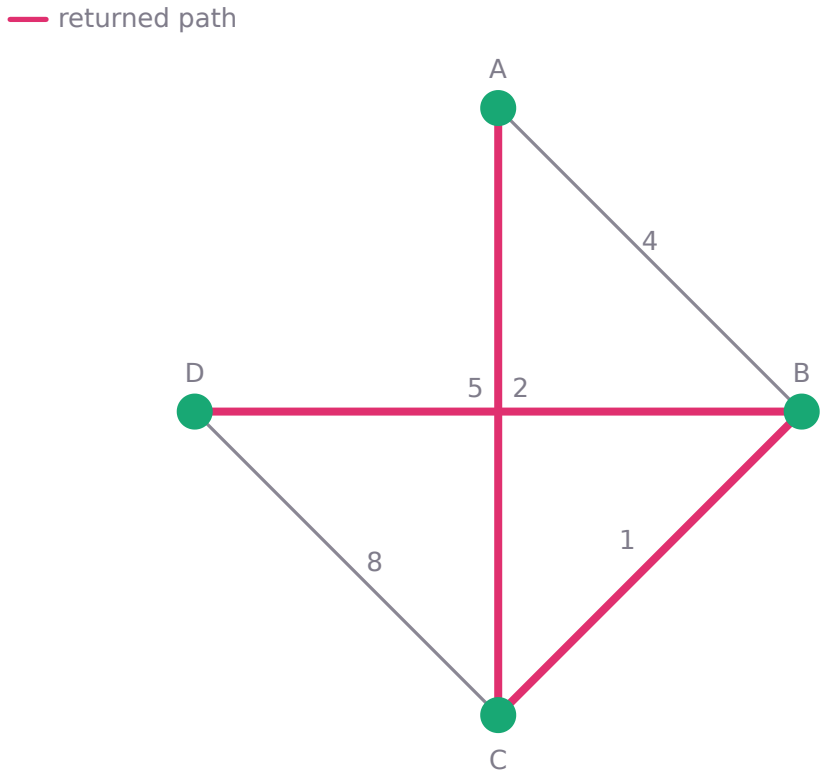
input	meaning	condition
`edges`	weighted edges 'A-B 4, B-C 1, ...'	
`source`	start node	
`target`	end node	
`directed`	true for a directed graph (edge u-v goes from u to v), false for undirected	

output	meaning
`distance`	shortest distance
`path`	the shortest path (only when unique; otherwise path_example)
`path_unique`	yes when exactly one shortest path exists

- **Method:** Dijkstra's algorithm in exact arithmetic; shortest paths counted on the tight-edge subgraph
- **Assumes:** weights >= 0; a path from source to target exists; parallel edges: the lightest counts
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



A weighted network (A-B 4, A-C 2, B-C 1, B-D 5, C-D 8, A-E 7, E-D 3, C-E 6): the shortest path from A to D the package finds, A → C → B → D of length 8, and its minimum spanning tree (B-C, A-C, E-D, B-D, total weight 11).



The network edges (weighted edges 'A-B 4, B-C 1, ...') of the worked example (4 nodes, 5 edges, laid out on a circle); on each edge its stated weight; highlighted: the returned path A → C → B → D. Returned: distance = 8. Other inputs: source = A, target = D, directed = False.

```
optimization_ai999x.solve('shortest_path', edges='A-B 4, A-C 2, B-C 1, B-D 5, C-D 8',
source='A', target='D', directed=False)
# -> distance = 8, path = ['A', 'C', 'B', 'D'], path_unique = yes   (SOLVED)
```

```
optimization_ai999x.solve('shortest_path', edges='A-B 1, B-C 1, A-C 2', source='A',
target='C', directed=True)
# -> distance = 2, path_example = ['A', 'C'], path_unique = no   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the shortest path from A to D in the undirected graph with edges A-B 4, A-C 2, B-C 1, B-D 5, C-D 8.	SOLVED	distance = 8, path = ['A', 'C', 'B', 'D'], path_unique = yes
Find the shortest path from A to D in the graph with edges A-B 4, A-C 2, B-C 1, B-D 5, C-D 8.	NEEDS_INPUT	shortest_path needs directed (true for a directed graph (edge u-v goes from u to v), false for undirected). Example: sol
Find the shortest path from A to E in the directed graph with edges A-B 4, B-C 1.	INVALID_INPUT	Shortest path (Dijkstra's algorithm): the source and the target must be nodes of the graph.

Shortest path with negative weights (Bellman-Ford), `bellman_ford_shortest_path`

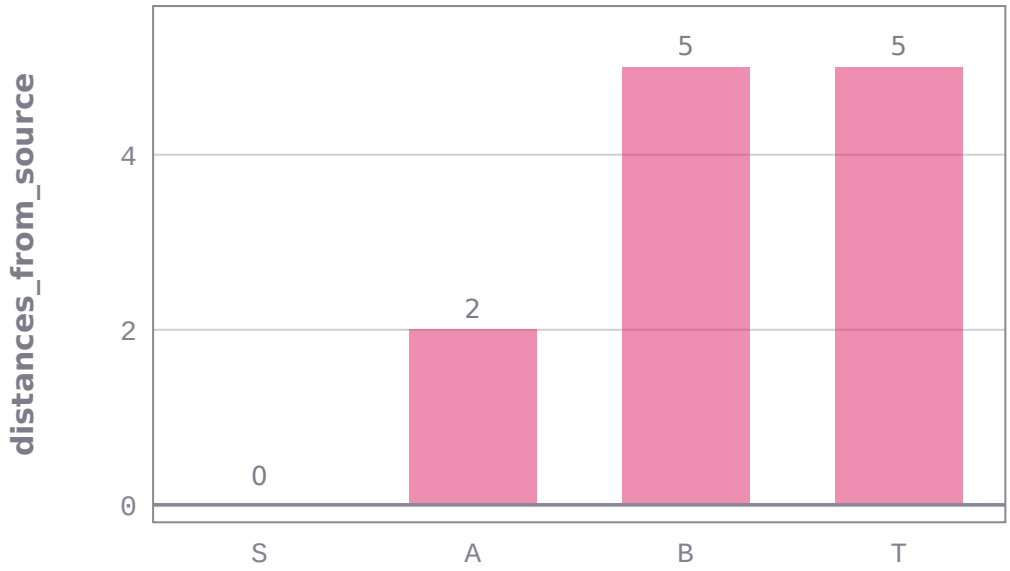
Shortest distance in a directed graph whose edge weights may be negative, with all distances from the source; a negative cycle reachable from the source is reported instead of a number.

$$d(v) \leq d(u) + w_{uv} \; \forall (u, v)$$

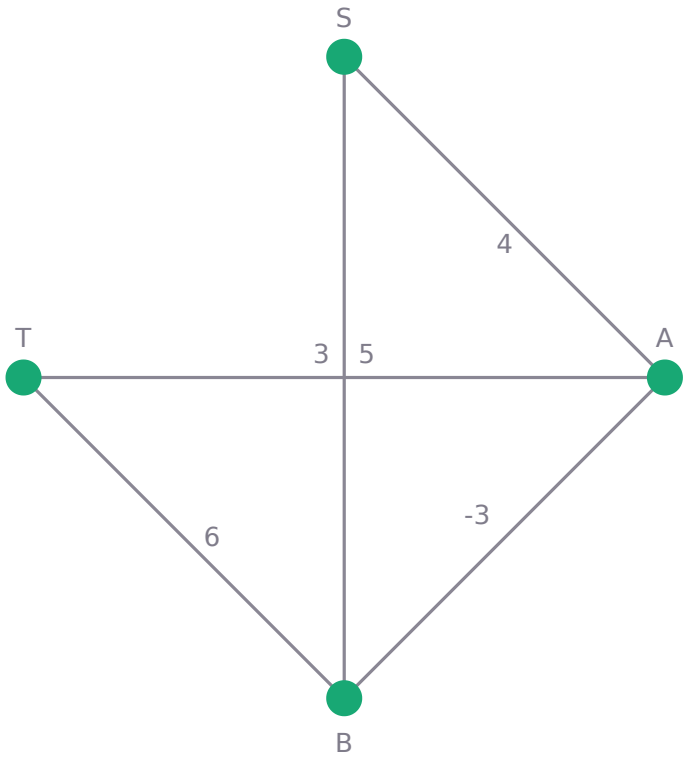
input	meaning	condition
`edges`	directed weighted edges 'A-B 4, $B-C-2$, ...'	
`source`	start node	
`target`	end node	

output	meaning
`distance`	shortest distance from source to target
`distances_from_source`	shortest distance to every reachable node

- **Method:** Bellman-Ford relaxation in exact arithmetic; distances certified as feasible, tight potentials
- **Assumes:** directed graph; no negative cycle reachable from the source (otherwise INVALID_INPUT)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



shortest distance to every reachable node (distances_from_source) returned by optimization-ai999x for the worked example (edges = [["S", "A", 4], ["S", "B", 5], ["A", "T", 3]…, source = S, target = T), one bar per entry: S = 0, A = 2, B = 5, T = 5.



The network edges (directed weighted edges 'A-B 4, B-C -2, ...') of the worked example (4 nodes, 5 edges, laid out on a circle); on each edge its stated weight. Returned: distance = 5. Other inputs: source = S, target = T.

```
optimization_ai999x.solve('bellman_ford_shortest_path', edges=[['S', 'A', 4], ['S', 'B',
5], ['A', 'T', 3], ['B', 'A', -3], ['B', 'T', 6]], source='S', target='T')
# -> distance = 5, distances_from_source = {'S': 0.0, 'A': 2.0, 'B': 5.0, 'T': 5.0}
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
With negative weights allowed, find the shortest path from S to T in the directed graph with edges S-A 4, S-B 5, A-T 3, B-A -3, B-T 6.	SOLVED	distance = 5, distances_from_source = {'S': 0.0, 'A': 2.0, 'B': 5.0, 'T': 5.0}
With negative weights allowed, find the shortest path from S to T in the directed graph with edges S-A 1, A-B -2, B-A 1, A-T 1.	INVALID_INPUT	Shortest path with negative weights (Bellman-Ford): the graph has a negative cycle reachable from the source: shortest p

All-pairs shortest distances (Floyd-Warshall), `all_pairs_shortest_paths`

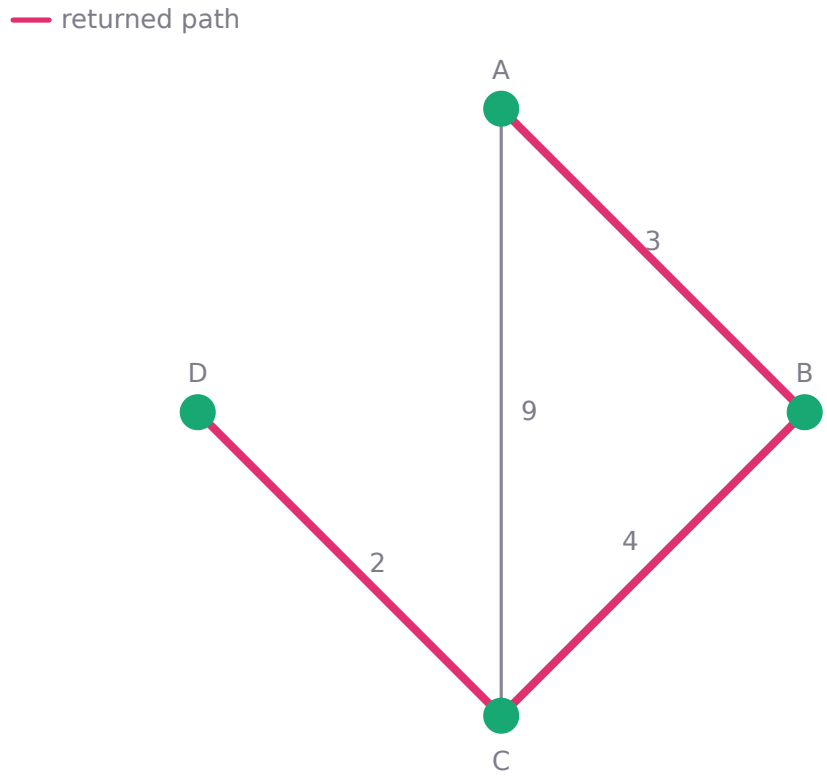
The table of shortest distances between every pair of nodes, and the graph's diameter.

$$d_{ij} \leftarrow \min(d_{ij},\, d_{ik} + d_{kj})$$

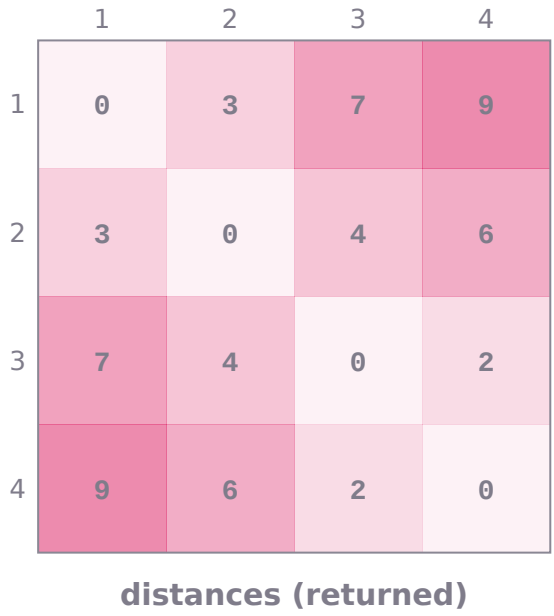
input	meaning	condition
`edges`	weighted edges 'A-B 4, ...'	
`directed`	true for a directed graph, false for undirected	

output	meaning
`distances`	shortest-distance matrix (rows and columns in the order of 'nodes')
`nodes`	node order (first appearance)
`diameter`	largest shortest distance

- **Method:** Floyd-Warshall in exact arithmetic
- **Assumes:** every node reaches every other; no negative cycle; at most 40 nodes
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The network edges (weighted edges 'A-B 4, ...') of the worked example (4 nodes, 4 edges, laid out on a circle); on each edge its stated weight; highlighted: the returned path A → B → C → D. Returned: diameter = 9. Other inputs: directed = False.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): distances (returned). Returned with them by optimization-ai999x: diameter = 9. Other inputs: edges = A-B 3, B-C 4, A-C 9, C-D 2, directed = False.

```
optimization_ai999x.solve('all_pairs_shortest_paths', edges='A-B 3, B-C 4, A-C 9, C-D 2',
directed=False)
# -> nodes = ['A', 'B', 'C', 'D'], distances = [[0.0, 3.0, 7.0, 9.0], [3.0, 0.0, 4.0, 6.0],
[7.0, 4.0, 0.0, 2.0], [9.0, 6.0, 2.0, 0.0]], diameter = 9   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the all-pairs shortest distances in the undirected graph with edges A-B 3, B-C 4, A-C 9, C-D 2.	SOLVED	nodes = ['A', 'B', 'C', 'D'], distances = [[0.0, 3.0, 7.0, 9.0], [3.0, 0.0, 4.0, 6.0], [7.0, 4.0, 0.0, 2.0], [9.0, 6.0, 2.0, 0.0]], diameter = 9

Minimum spanning tree (Kruskal's algorithm), `minimum_spanning_tree`

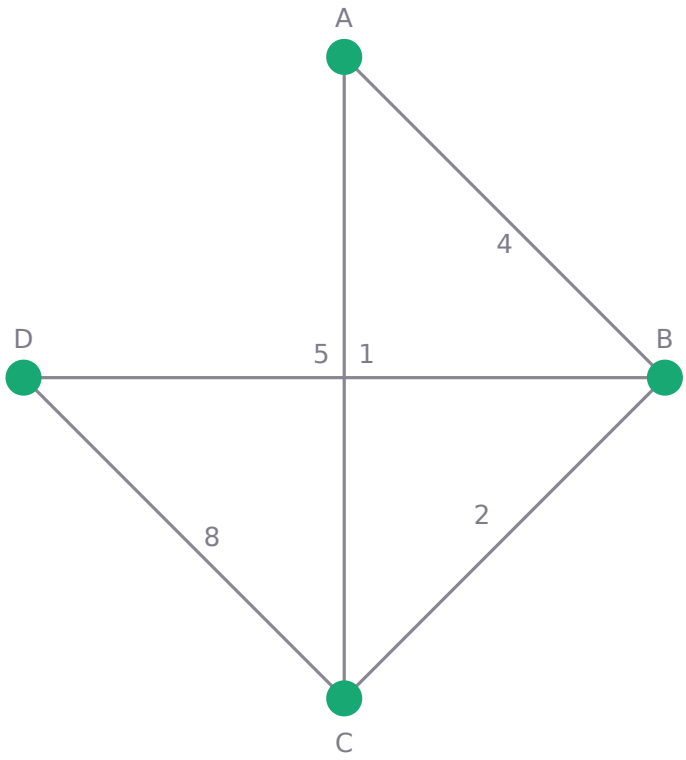
The least total weight of edges connecting all nodes of an undirected graph, and the tree (stated as THE tree only when the minimum spanning tree is unique).

$$\min_{T \text{ spanning tree}} \sum_{e \in T} w_e$$

input	meaning	condition
`edges`	undirected weighted edges 'A-B 4, ...'	

output	meaning
`total_weight`	weight of a minimum spanning tree
`tree_edges`	the minimum spanning tree (only when unique; otherwise tree_edges_example)
`tree_unique`	yes when the minimum spanning tree is unique

- **Method:** Kruskal (edges by weight, union-find); uniqueness by the cycle property; total re-checked by Prim
- **Assumes:** connected undirected graph
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The network edges (undirected weighted edges 'A-B 4, ...') of the worked example (4 nodes, 5 edges, laid out on a circle); on each edge its stated weight. Returned: total_weight = 8.

```
optimization_ai999x.solve('minimum_spanning_tree', edges='A-B 4, A-C 1, B-C 2, B-D 5, C-D
8')
# -> total_weight = 8, tree_unique = yes, tree_edges = ['A-C', 'B-C', 'B-D']   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the minimum spanning tree of the graph with edges A-B 4, A-C 1, B-C 2, B-D 5, C-D 8.	SOLVED	total_weight = 8, tree_unique = yes, tree_edges = ['A-C', 'B-C', 'B-D']
Find the minimum spanning tree of the graph with edges A-B 4, C-D 1.	INVALID_INPUT	Minimum spanning tree (Kruskal's algorithm): the graph is not connected: no spanning tree exists.

Maximum flow and minimum cut (Edmonds-Karp), `max_flow`

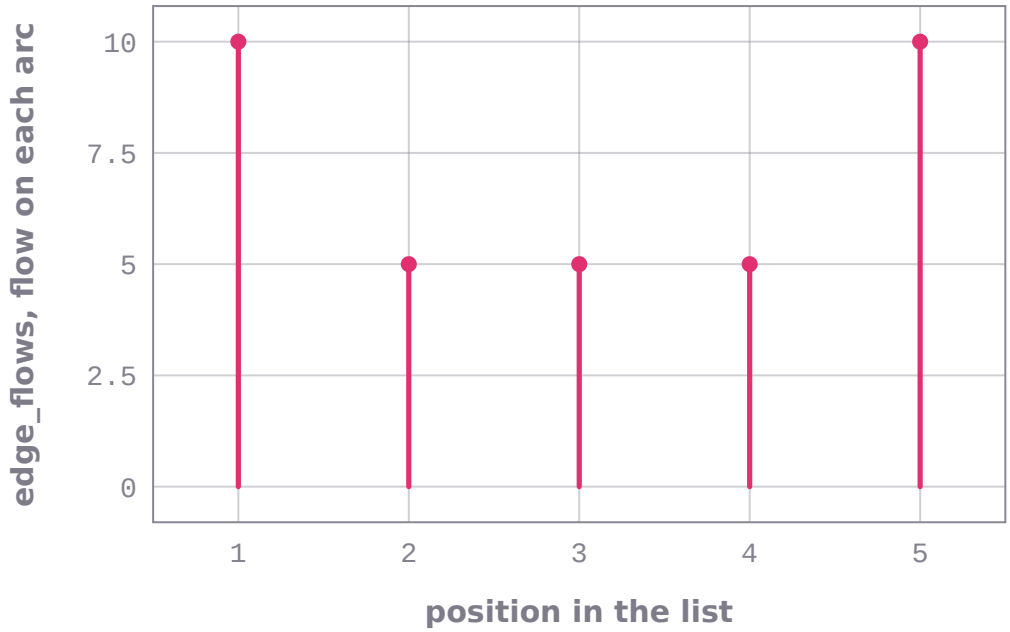
The largest flow that can be sent from a source to a sink through a network of capacitated arcs, a flow achieving it and the minimum cut that proves it optimal.

$$\max |f| = \min_{(S,\bar{S})} c(S,\bar{S})$$

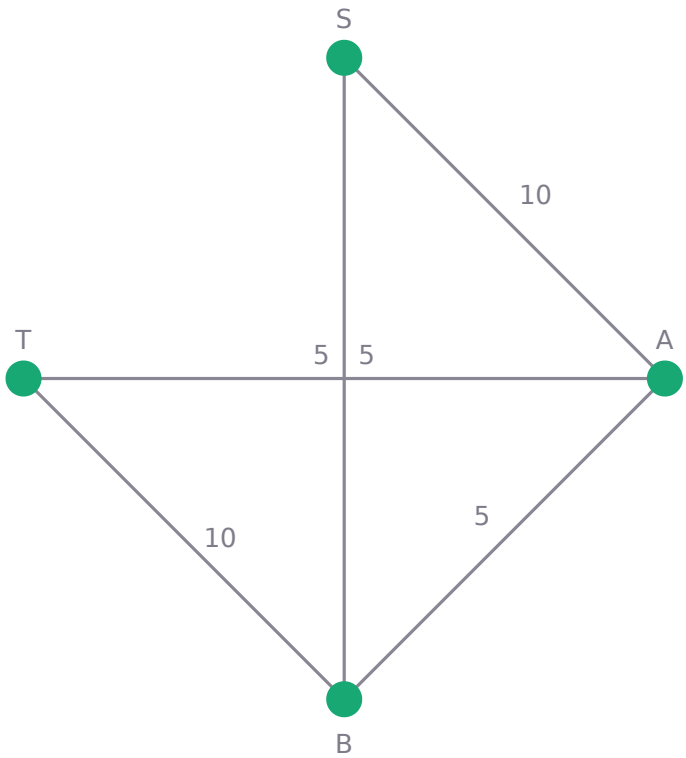
input	meaning	condition
`edges`	directed arcs with capacities 'S-A 10, A-T 5, ...'	
`source`	source node	
`target`	sink node	

output	meaning
`max_flow`	maximum flow value
`min_cut_source_side`	nodes on the source side of a minimum cut (those reachable in the final residual network)
`edge_flows`	flow on each arc, in input order (one maximum flow)

- **Method:** Edmonds-Karp (breadth-first augmenting paths) in exact arithmetic; certified by the max-flow min-cut theorem
- **Assumes:** directed arcs; capacities >= 0
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



flow on each arc, in input order (one maximum flow) (edge_flows) returned by optimization-ai999x for the worked example, one value for each position in the list (5 values, first 10, last 10, largest 10). Inputs: edges = S-A 10, S-B 5, A-B 15, A-T 5, B-T 10, source = S, target = T.



The network edges (directed arcs with capacities 'S-A 10, A-T 5, ...') of the worked example (4 nodes, 5 edges, laid out on a circle); on each edge the edge_flows returned by optimization-ai999x. Returned: max_flow = 15. Other inputs: source = S, target = T.

```
optimization_ai999x.solve('max_flow', edges='S-A 10, S-B 5, A-B 15, A-T 5, B-T 10',
source='S', target='T')
# -> max_flow = 15, min_cut_source_side = ['S'], edge_flows = [10.0, 5.0, 5.0, 5.0, 10.0]
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the maximum flow from S to T in the network with edges S-A 10, S-B 5, A-B 15, A-T 5, B-T 10.	SOLVED	max_flow = 15, min_cut_source_side = ['S'], edge_flows = [10.0, 5.0, 5.0, 5.0, 10.0]

Transportation problem (minimum shipping cost), `transportation_problem`

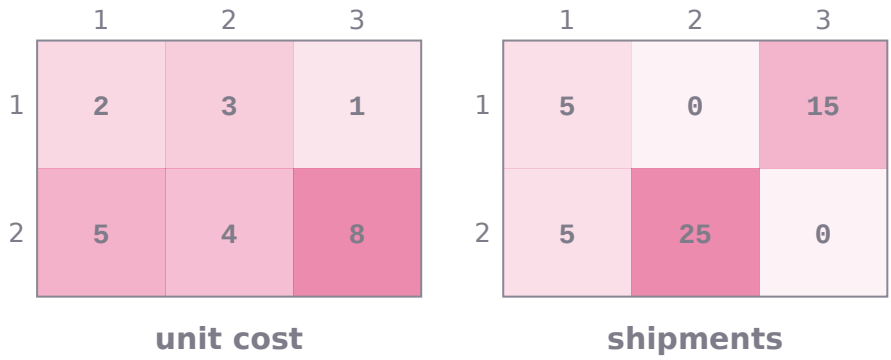
Ship from sources with given supplies to destinations with given demands at least total cost: the minimum cost and a shipping plan (stated as THE plan only when it is unique).

$$\min \sum_{ij} c_{ij} x_{ij} \text{ s.t. } \sum_{j} x_{ij} \le s_i,\ \sum_{i} x_{ij} = d_j,\ x \ge 0$$

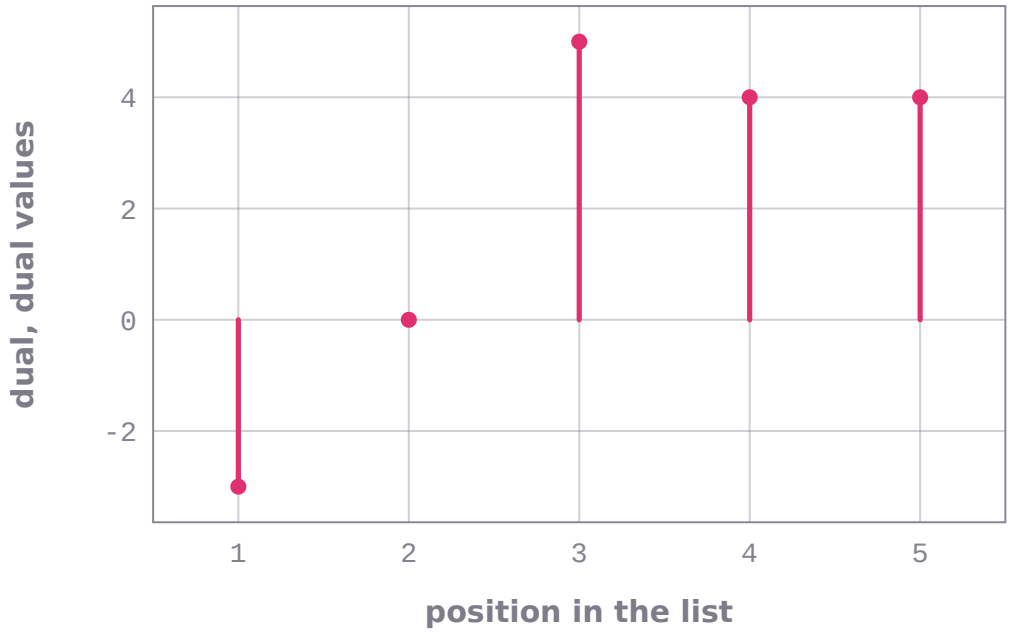
input	meaning	condition
`supply`	supply of each source	
`demand`	demand of each destination	
`cost`	unit shipping cost, rows = sources, columns = destinations	

output	meaning
`min_cost`	minimum total shipping cost
`shipments`	optimal plan, rows = sources (only when unique; otherwise shipments_example)
`plan_unique`	yes when the optimal plan is unique
`dual`	dual values (u for sources, v for destinations): the optimality certificate

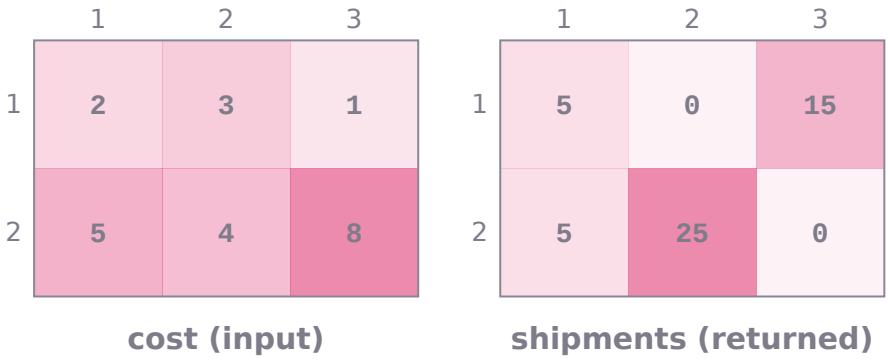
- **Method:** the transportation LP solved exactly (rational simplex), certified by its dual $(u_i + v_j \le c_ij)$
- **Assumes:** total supply >= total demand (surplus stays at the sources at no cost); at most 60 routes
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The optimal transportation plan the package computes for supplies [20, 30], demands [10, 25, 15] and unit costs [[2, 3, 1], [5, 4, 8]] (rows: sources, columns: destinations; left the unit costs, right the quantities shipped): minimum total cost 150.



dual values (u for sources, v for destinations): the optimality certificate (dual) returned by optimization-ai999x for the worked example, one value for each position in the list (5 values, first -3, last 4, largest 5). Inputs: supply = [20, 30], demand = [10, 25, 15], cost = [[2, 3, 1], [5, 4, 8]].



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): cost (input); shipments (returned). Returned with them by optimization-ai999x: min_cost = 150. Other inputs: supply = [20, 30], demand = [10, 25, 15].

```
optimization_ai999x.solve('transportation_problem', supply=[20, 30], demand=[10, 25, 15],
cost=[[2, 3, 1], [5, 4, 8]])
# -> min_cost = 150, shipments = [[5.0, 0.0, 15.0], [5.0, 25.0, 0.0]], plan_unique = yes,
dual = [-3.0, 0.0, 5.0, 4.0, 4.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the transportation problem with supply [20, 30], demand [10, 25, 15] and cost matrix [[2, 3, 1], [5, 4, 8]].	SOLVED	min_cost = 150, shipments = [[5.0, 0.0, 15.0], [5.0, 25.0, 0.0]], plan_unique = yes, dual = [-3.0, 0.0, 5.0, 4.0, 4.0]
Solve the transportation problem with supply [5, 5], demand [10, 25, 15] and cost matrix [[2, 3, 1], [5, 4, 8]].	INVALID_INPUT	Transportation problem (minimum shipping cost): total supply is less than total demand: the demand cannot be met.

Minimum-cost flow (transshipment), `min_cost_flow`

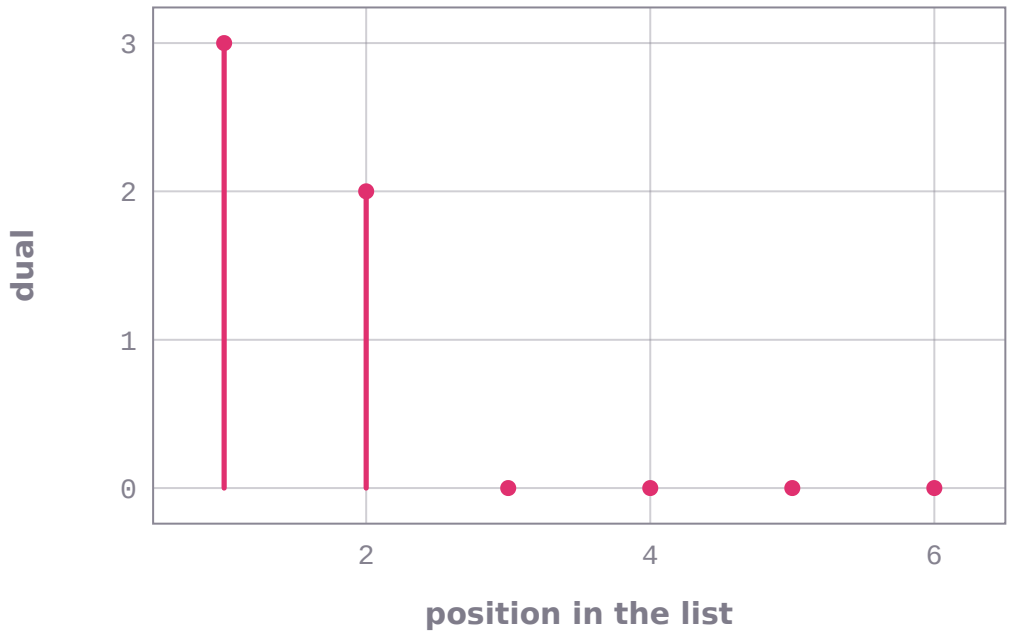
Send flow from supply nodes to demand nodes through capacitated arcs at least total cost (transshipment allowed): the minimum cost and a flow achieving it.

$$\min \sum c_{ij}x_{ij} \text{ s.t. } \sum_j x_{vj} - \sum_i x_{iv} = b_v,\ 0 \le x_{ij} \le u_{ij}$$

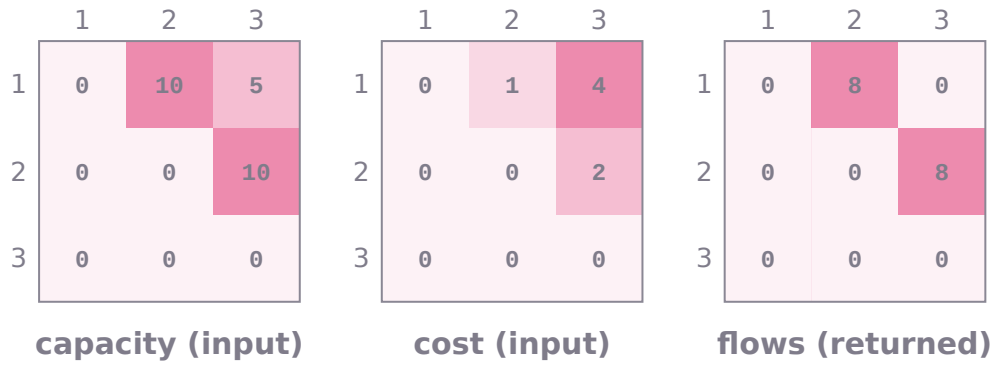
input	meaning	condition
`capacity`	arc capacities, n x n (row = from node, column = to node, $0 = no$ arc)	
`cost`	unit cost of each arc, n x n	
`supply`	net supply of each node (positive = supply, negative = demand), summing to 0	

output	meaning
`min_cost`	minimum total cost
`flows`	optimal flow matrix (only when unique; otherwise flows_example)
`flow_unique`	yes when the optimal flow is unique
`dual`	dual values: node potentials then arc-capacity prices

- **Method:** the min-cost-flow LP solved exactly (rational simplex) and certified by its dual
- **Assumes:** supplies sum to 0; a feasible flow exists; at most 60 arcs
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



dual values: node potentials then arc-capacity prices (dual) returned by optimization-ai999x for the worked example, one value for each position in the list (6 values, first 3, last 0, largest 3). Inputs: capacity = [[0, 10, 5], [0, 0, 10], [0, 0, 0]], cost = [[0, 1, 4], [0, 0, 2], [0, 0, 0]], supply = [8, 0, -8].



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): capacity (input); cost (input); flows (returned). Returned with them by optimization-ai999x: min_cost = 24. Other inputs: supply = [8, 0, -8].

```
optimization_ai999x.solve('min_cost_flow', capacity=[[0, 10, 5], [0, 0, 10], [0, 0, 0]],
cost=[[0, 1, 4], [0, 0, 2], [0, 0, 0]], supply=[8, 0, -8])
# -> min_cost = 24, flows = [[0.0, 8.0, 0.0], [0.0, 0.0, 8.0], [0.0, 0.0, 0.0]],
flow_unique = yes, dual = [3.0, 2.0, 0.0, 0.0, 0.0, 0.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the minimum cost flow with capacity [[0, 10, 5], [0, 0, 10], [0, 0, 0]], cost [[0, 1, 4], [0, 0, 2], [0, 0, 0]] and supply [8, 0, -8].	SOLVED	min_cost = 24, flows = [[0.0, 8.0, 0.0], [0.0, 0.0, 8.0], [0.0, 0.0, 0.0]], flow_unique = yes, dual = [3.0, 2.0, 0.0, 0.0, 0.0, 0.0]

Assignment problem (Hungarian method), minimum total cost, `assignment_min_cost`

Assign each row (worker) to a different column (task) so that the total cost is as small as possible: the optimum and an optimal assignment (stated as THE assignment only when unique).

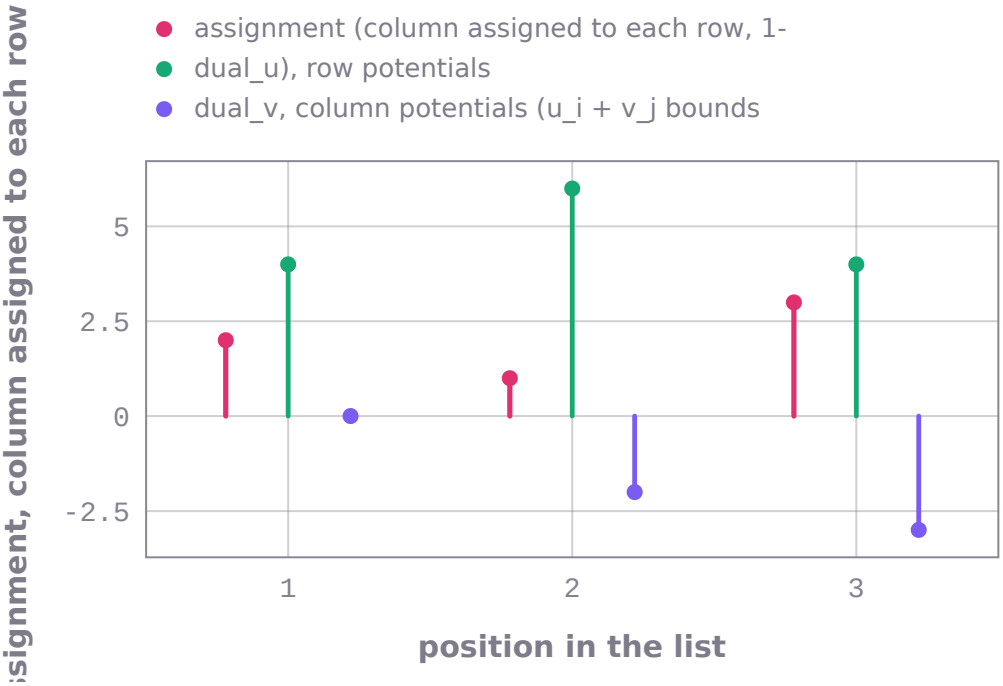
$$\min_{\sigma} \sum_{i} c_{i\sigma(i)} = \max \left\{ \sum_{i} u_i + \sum_{j} v_j : u_i + v_j \le c_{ij} \right\}$$

input	meaning	condition
`C`	cost matrix, rows = workers, columns = tasks (square)	

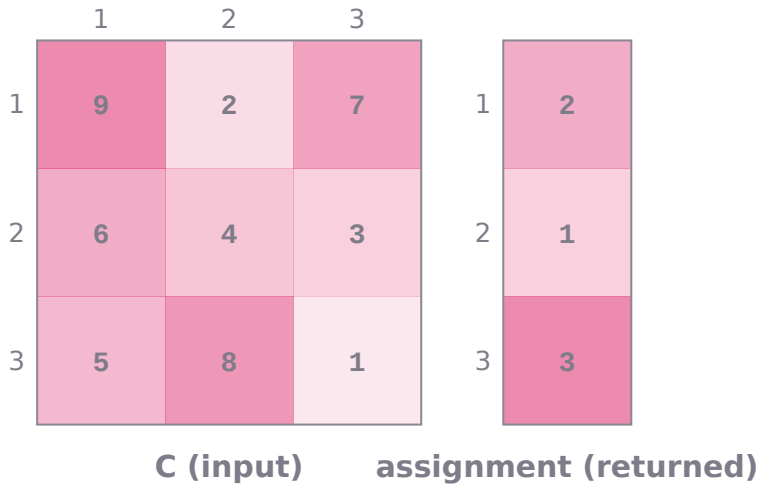
output	meaning
`min_cost`	optimal total cost
`assignment`	column assigned to each row, 1-based (only when unique; otherwise assignment_example)
`assignment_unique`	yes when the optimal assignment is unique
`dual_u`	row potentials
`dual_v`	column potentials (u_i + v_j bounds every cell: the optimality certificate)

- **Method:** Hungarian method (shortest augmenting paths with potentials) in exact arithmetic; uniqueness by forbidding each chosen cell
- **Assumes:** square matrix, at most 12 x 12 (pad with zeros for unbalanced problems)

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



column assigned to each row, 1-based (only when unique; otherwise assignment_example) (assignment) returned by optimization-ai999x for the worked example, one value for each position in the list (3 values, first 2, last 3, largest 3); with dual_u (row potentials), dual_v (column potentials (u_i + v_j bounds every cell: the optimality certificate)). Inputs: C = 9 2 7; 6 4 3; 5 8 1.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): C (input); assignment (returned). Returned with them by optimization-ai999x: min_cost = 9.

```
optimization_ai999x.solve('assignment_min_cost', C='9 2 7; 6 4 3; 5 8 1')
# -> min_cost = 9, assignment = [2, 1, 3], assignment_unique = yes, dual_u = [4.0, 6.0,
4.0], dual_v = [0.0, -2.0, -3.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the assignment problem with cost matrix [[9, 2, 7], [6, 4, 3], [5, 8, 1]].	SOLVED	min_cost = 9, assignment = [2, 1, 3], assignment_unique = yes, dual_u = [4.0, 6.0, 4.0], dual_v = [0.0, -2.0, -3.0]

question	outcome	result
Solve the assignment problem with cost matrix [[9, 2, 7], [6, 4, 3]].	INVALID_INPUT	Assignment problem (Hungarian method), minimum total cost: give a square matrix with at most 12 rows (add dummy rows or

Assignment problem (Hungarian method), maximum total value, `assignment_max_profit`

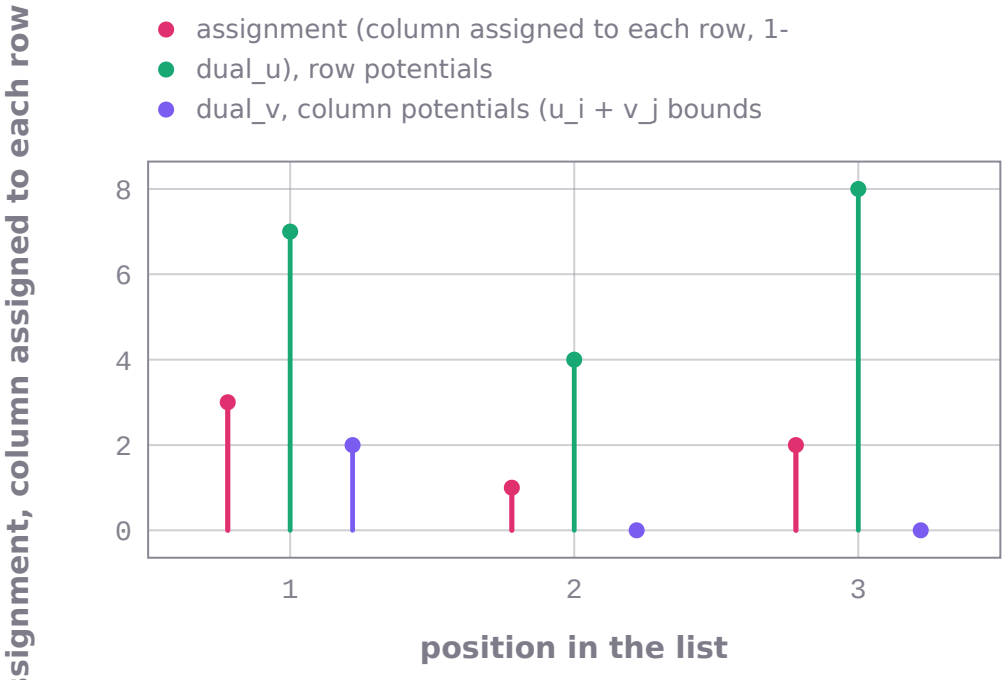
Assign each row (worker) to a different column (task) so that the total value is as large as possible: the optimum and an optimal assignment (stated as THE assignment only when unique).

$$\min_{\sigma} \sum_{i} c_{i\sigma(i)} = \max \left\{ \sum_{i} u_i + \sum_{j} v_j : u_i + v_j \le c_{ij} \right\}$$

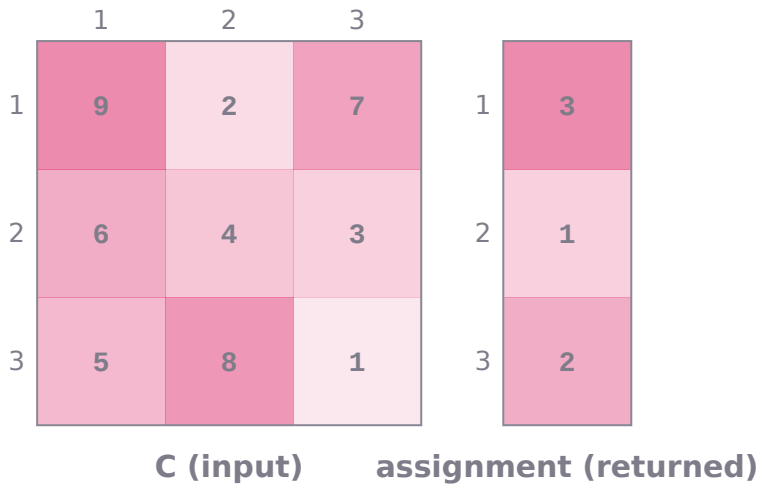
input	meaning	condition
`C`	value matrix, rows = workers, columns = tasks (square)	

output	meaning
`max_value`	optimal total value
`assignment`	column assigned to each row, 1-based (only when unique; otherwise assignment_example)
`assignment_unique`	yes when the optimal assignment is unique
`dual_u`	row potentials
`dual_v`	column potentials (u_i + v_j bounds every cell: the optimality certificate)

- **Method:** Hungarian method (shortest augmenting paths with potentials) in exact arithmetic; uniqueness by forbidding each chosen cell
- **Assumes:** square matrix, at most 12 x 12 (pad with zeros for unbalanced problems)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



column assigned to each row, 1-based (only when unique; otherwise assignment_example) (assignment) returned by optimization-ai999x for the worked example, one value for each position in the list (3 values, first 3, last 2, largest 3); with dual_u (row potentials), dual_v (column potentials (u_i + v_j bounds every cell: the optimality certificate)). Inputs: C = 9 2 7; 6 4 3; 5 8 1.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): C (input); assignment (returned). Returned with them by optimization-ai999x: max_value = 21.

```
optimization_ai999x.solve('assignment_max_profit', C='9 2 7; 6 4 3; 5 8 1')
# -> max_value = 21, assignment = [3, 1, 2], assignment_unique = yes, dual_u = [7.0, 4.0,
8.0], dual_v = [2.0, 0.0, 0.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the assignment problem to maximize total profit with matrix [[9, 2, 7], [6, 4, 3], [5, 8, 1]].	SOLVED	max_value = 21, assignment = [3, 1, 2], assignment_unique = yes, dual_u = [7.0, 4.0, 8.0], dual_v = [2.0, 0.0, 0.0]
Solve the assignment problem to maximize total profit with matrix [[9, 2, 7], [6, 4, 3]].	INVALID_INPUT	Assignment problem (Hungarian method), maximum total value: give a square matrix with at most 12 rows (add dummy rows or

Maximum bipartite matching, `bipartite_matching`

The largest number of pairs that can be formed between two groups (workers and jobs, students and projects) using only allowed pairs, with one maximum matching and a vertex cover proving it maximum.

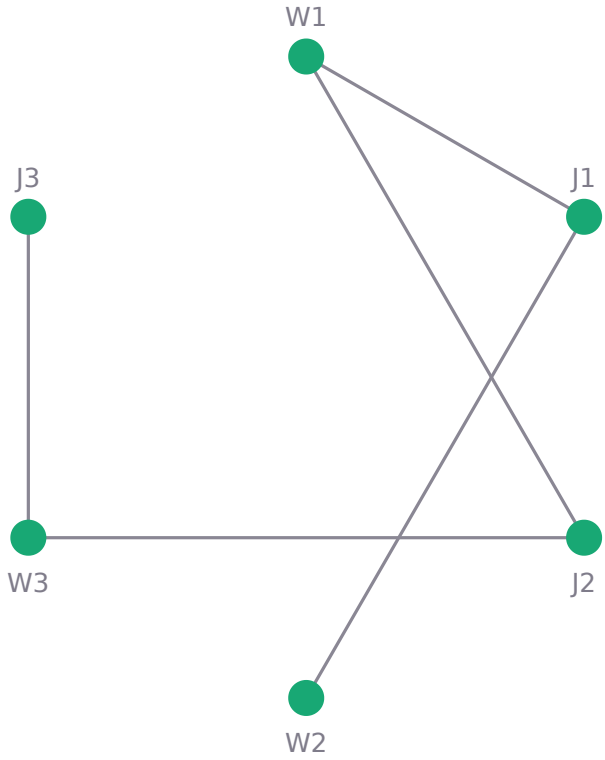
$$\nu(G) = \tau(G) \text{ for bipartite } G$$

input	meaning	condition
`edges`	allowed pairs 'left-right', e.g. '$W_1 - J_1$, $W_1 - J_2$, $W_2 - J_1$'	

output	meaning
`max_matching`	size of a maximum matching
`matching_example`	one maximum matching
`vertex_cover`	a minimum vertex cover of the same size (Konig's theorem: the certificate)

- **Method:** augmenting paths (Kuhn); minimum vertex cover from alternating reachability (Konig)
- **Assumes:** bipartite: the first node of each pair is on the left, the second on the right

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The network edges (allowed pairs 'left-right', e.g. 'W1-J1, W1-J2, W2-J1') of the worked example (6 nodes, 5 edges, laid out on a circle); highlighted: the returned path W1 → W2 → W3. Returned: max_matching = 3.

```
optimization_ai999x.solve('bipartite_matching', edges='W1-J1, W1-J2, W2-J1, W3-J2, W3-J3')
# -> max_matching = 3, matching_example = ['W1-J2', 'W2-J1', 'W3-J3'], vertex_cover =
['W1', 'W2', 'W3']   (SOLVED)
```

```
optimization_ai999x.solve('bipartite_matching', edges='A-X, B-X, C-X')
# -> max_matching = 1, matching_example = ['A-X'], vertex_cover = ['X']   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the maximum bipartite matching with edges W1-J1, W1-J2, W2-J1, W3-J2, W3-J3.	SOLVED	max_matching = 3, matching_example = ['W1-J2', 'W2-J1', 'W3-J3'], vertex_cover = ['W1', 'W2', 'W3']

Maximum-weight matching (general graph), `max_weight_matching`

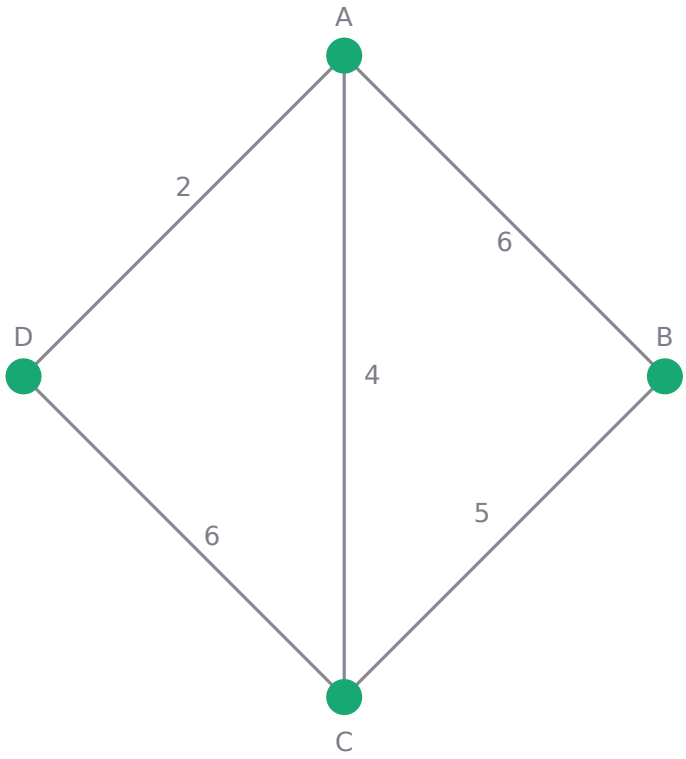
Pair up nodes of a graph (each node at most once) so that the total weight of the chosen edges is as large as possible: the maximum weight and one optimal matching.

$$\max_{M \text{ matching}} \sum_{e \in M} w_e$$

input	meaning	condition
`edges`	undirected weighted edges 'A-B 4, ...'	

output	meaning
`max_weight`	maximum total weight of a matching
`matching_example`	one maximum-weight matching

- **Method:** exact dynamic programming over node subsets
- **Assumes:** at most 16 nodes; edges of weight <= 0 are never worth choosing
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The network edges (undirected weighted edges 'A-B 4, ...') of the worked example (4 nodes, 5 edges, laid out on a circle); on each edge its stated weight. Returned: max_weight = 12.

```
optimization_ai999x.solve('max_weight_matching', edges='A-B 6, B-C 5, C-D 6, A-D 2, A-C 4')
# -> max_weight = 12, matching_example = ['A-B', 'C-D']   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the maximum weight matching of the graph with edges A-B 6, B-C 5, C-D 6, A-D 2, A-C 4.	SOLVED	max_weight = 12, matching_example = ['A-B', 'C-D']

Travelling salesman tour (exact, Held-Karp), `traveling_salesman`

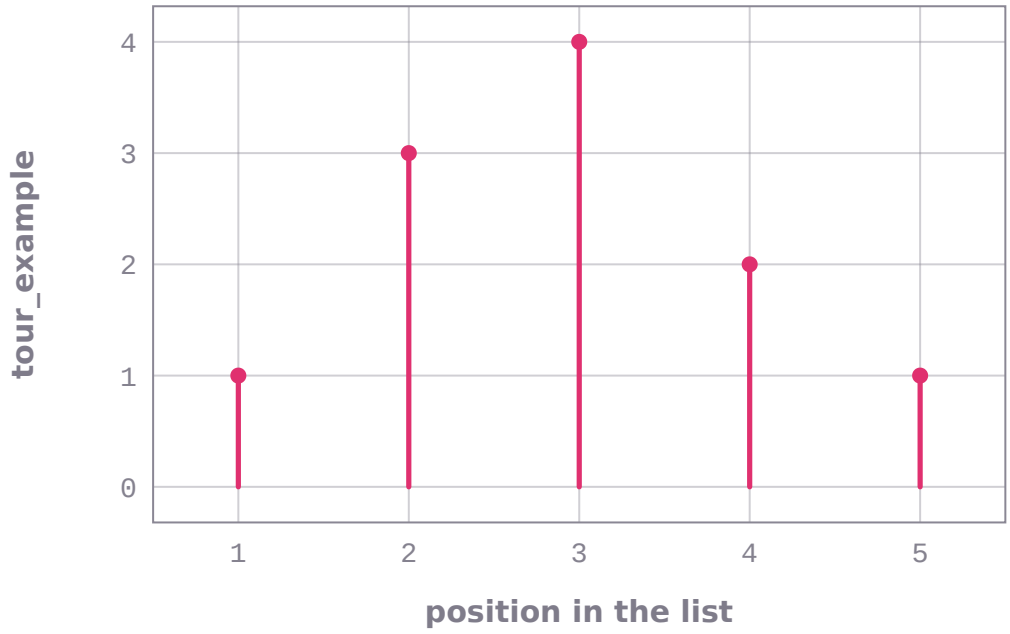
The shortest round trip visiting every city exactly once and returning to the start, for a distance matrix (symmetric or not): the minimum tour length and one optimal tour.

$$g(S,k) = \min_{m \in S \setminus \{k\}} \left(g(S \setminus \{k\}, m) + d_{mk} \right)$$

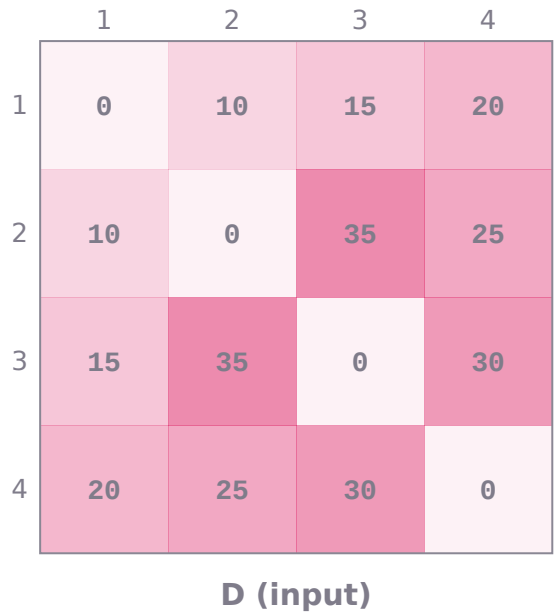
input	meaning	condition
`D`	distance matrix, D[i][j] = distance from city i to city j	

output	meaning
`min_tour_length`	length of a shortest tour
`tour_example`	one optimal tour starting and ending at city 1

- **Method:** Held-Karp dynamic programming over city subsets, exact arithmetic
- **Assumes:** 2 to 11 cities
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



one optimal tour starting and ending at city 1 (tour_example) returned by optimization-ai999x for the worked example, one value for each position in the list (5 values, first 1, last 1, largest 4). Inputs: D = [[0, 10, 15, 20], [10, 0, 35, 25], [15, 35, 0….



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): D (input). Returned with them by optimization-ai999x: min_tour_length = 80.

```
optimization_ai999x.solve('traveling_salesman', D=[[0, 10, 15, 20], [10, 0, 35, 25], [15,
35, 0, 30], [20, 25, 30, 0]])
# -> min_tour_length = 80, tour_example = [1, 3, 4, 2, 1]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the travelling salesman problem with distance matrix [[0, 10, 15, 20], [10, 0, 35, 25], [15, 35, 0, 30], [20, 25, 30, 0]].	SOLVED	min_tour_length = 80, tour_example = [1, 3, 4, 2, 1]

0/1 knapsack problem, `knapsack_01`

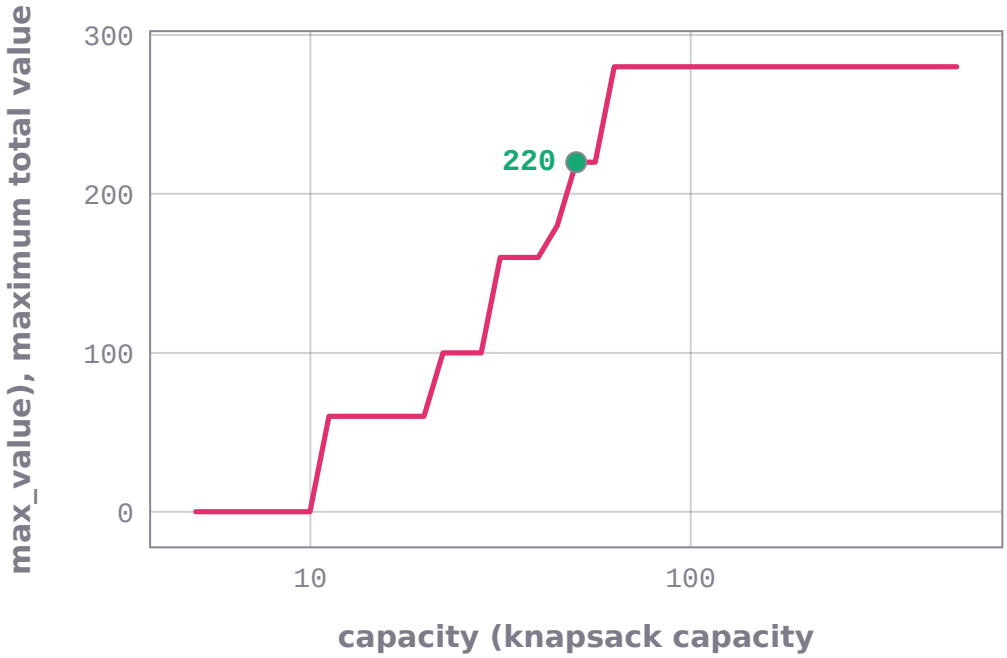
Choose items (each at most once) to maximise total value without exceeding the capacity: the maximum value, one optimal choice and its weight.

$$\max \sum_i v_i x_i \text{ s.t. } \sum_i w_i x_i \le W,\ x_i \in \{0,1\}$$

input	meaning	condition
`values`	value of each item	
`weights`	weight of each item	
`capacity`	knapsack capacity	

output	meaning
`max_value`	maximum total value
`items_example`	one optimal set of items (1-based; ties to the lighter set)
`total_weight`	weight of that set

- **Method:** exact dynamic programming over (weight, value) states with Pareto pruning, rational arithmetic
- **Assumes:** values, weights, capacity >= 0; at most 40 items
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



max_value (maximum total value) against capacity (knapsack capacity) from 5 to 500 on a logarithmic axis. Other inputs as in the worked example: values = 60, 100, 120, weights = 10, 20, 30. 41 points computed by optimization-ai999x; marked: the worked example, capacity = 50, max_value = 220.

```
optimization_ai999x.solve('knapsack_01', values='60, 100, 120', weights='10, 20, 30',
capacity=50)
# -> max_value = 220, items_example = [2, 3], total_weight = 50   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the knapsack problem with values [60, 100, 120], weights [10, 20, 30] and capacity 50.	SOLVED	max_value = 220, items_example = [2, 3], total_weight = 50
Solve the knapsack problem with values [60, 100, 120], weights [10, 20, 30] and capacity 50; items cannot be repeated.	SOLVED	max_value = 220, items_example = [2, 3], total_weight = 50

Unbounded knapsack problem, `knapsack_unbounded`

Choose any number of copies of each item type to maximise total value within the capacity.

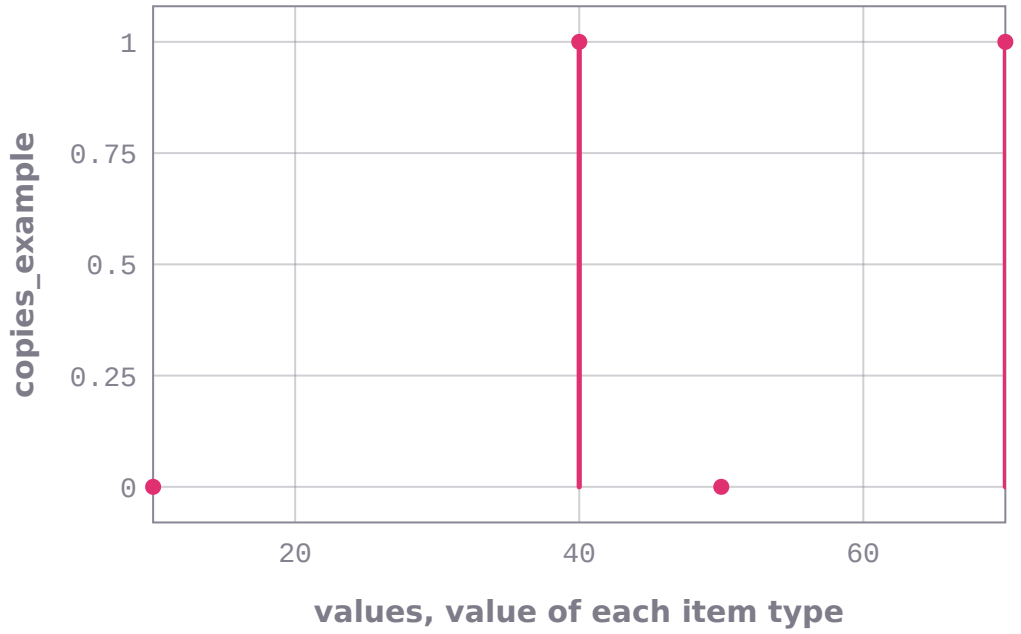
$$V(c) = \max\left(V(c-1),\ \max_{i:w_i \le c} V(c-w_i) + v_i\right)$$

input	meaning	condition
`values`	value of each item type	
`weights`	weight of each item type (whole numbers)	

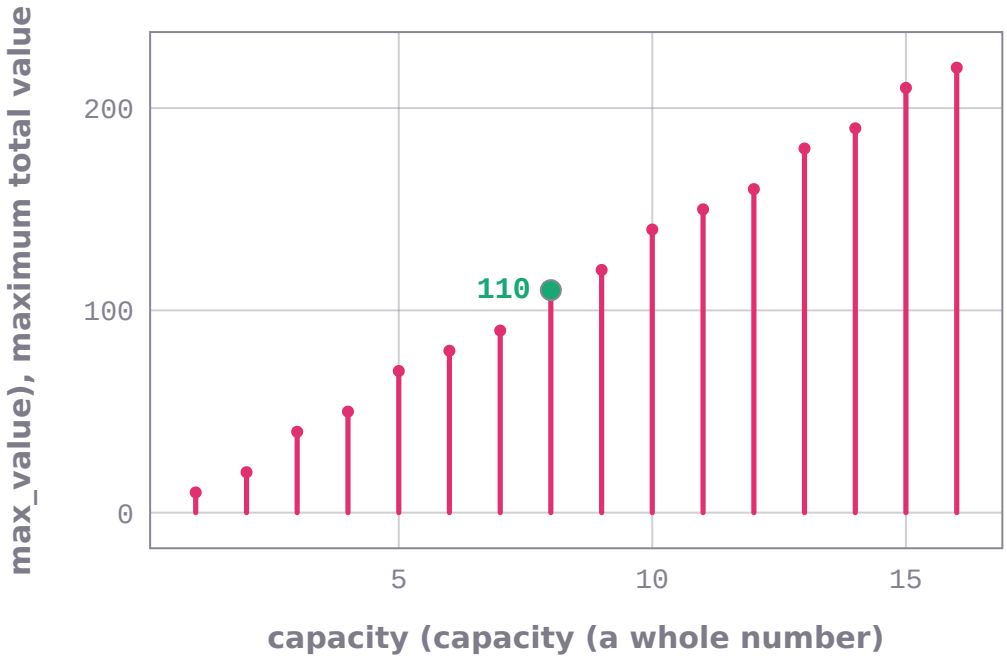
input	meaning	condition
`capacity`	capacity (a whole number)	

output	meaning
`max_value`	maximum total value
`copies_example`	number of copies of each type in one optimal choice

- **Method:** dynamic programming over capacities
- **Assumes:** positive whole-number weights; capacity a whole number <= 100000
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



number of copies of each type in one optimal choice (copies_example) returned by optimization-ai999x for the worked example, one value for each values (value of each item type) (4 values, first 0, last 1, largest 1). Inputs: values = [10, 40, 50, 70], weights = [1, 3, 4, 5], capacity = 8.



max_value (maximum total value) against capacity (capacity (a whole number)) from 1 to 16. Other inputs as in the worked example: values = [10, 40, 50, 70], weights = [1, 3, 4, 5]. 16 points computed by optimization-ai999x; marked: the worked example, capacity = 8, max_value = 110.

```
optimization_ai999x.solve('knapsack_unbounded', values=[10, 40, 50, 70], weights=[1, 3, 4, 5], capacity=8)
# -> max_value = 110, copies_example = [0.0, 1.0, 0.0, 1.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the unbounded knapsack problem with values [10, 40, 50, 70], weights [1, 3, 4, 5] and capacity 8.	SOLVED	max_value = 110, copies_example = [0.0, 1.0, 0.0, 1.0]

Minimum-cost set cover, `set_cover`

Choose sets (each with a cost) that together contain every element, at least total cost.

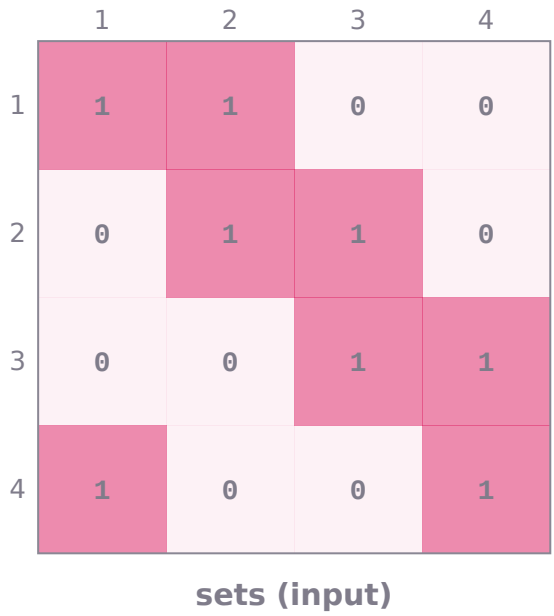
$$\min \sum_i c_i x_i \text{ s.t. } \sum_{i \ni e} x_i \geq 1 \; \forall e, \; x \in \{0,1\}$$

input	meaning	condition
`sets`	incidence matrix: row i lists with $0/1$ which elements set i contains	
`costs`	cost of each set	

output	meaning
`min_cost`	minimum total cost of a cover
`sets_example`	one optimal choice of sets (1-based)

- **Method:** exhaustive enumeration of set combinations (exact)
- **Assumes:** at most 18 sets; every element in some set

- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): sets (input). Returned with them by optimization-ai999x: min_cost = 4. Other inputs: costs = [3, 2, 3, 2].

```
optimization_ai999x.solve('set_cover', sets=[[1, 1, 0, 0], [0, 1, 1, 0], [0, 0, 1, 1], [1,
0, 0, 1]], costs=[3, 2, 3, 2])
# -> min_cost = 4, sets_example = [2, 4]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the set cover problem with incidence matrix [[1, 1, 0, 0], [0, 1, 1, 0], [0, 0, 1, 1], [1, 0, 0, 1]] and costs [3, 2, 3, 2].	SOLVED	min_cost = 4, sets_example = [2, 4]

Uncapacitated facility location, `facility_location`

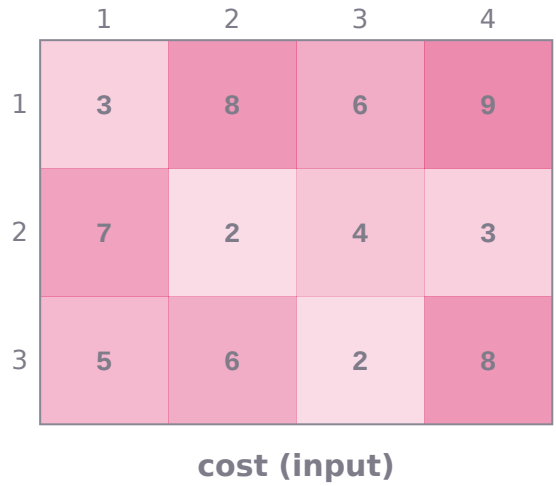
Choose which facilities to open (each with a fixed cost) and serve every customer from its cheapest open facility, minimising fixed plus service cost.

$$\min_{S \neq \emptyset} \sum_{i \in S} f_i + \sum_{j} \min_{i \in S} c_{ij}$$

input	meaning	condition
`fixed`	fixed opening cost of each facility	
`cost`	service cost matrix, rows = facilities, columns = customers	

output	meaning
`min_cost`	minimum total cost
`open_example`	one optimal set of facilities to open (1-based)

- **Method:** exact enumeration of the facilities to open; each customer assigned to its cheapest open facility
- **Assumes:** at most 15 facilities; unlimited capacity
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): cost (input). Returned with them by optimization-ai999x: min_cost = 31. Other inputs: fixed = [10, 15, 12].

```
optimization_ai999x.solve('facility_location', fixed=[10, 15, 12], cost=[[3, 8, 6, 9], [7,
2, 4, 3], [5, 6, 2, 8]])
# -> min_cost = 31, open_example = [2]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Facility location: fixed costs [10, 15, 12] and service cost matrix [[3, 8, 6, 9], [7, 2, 4, 3], [5, 6, 2, 8]]. Which facilities should open?	SOLVED	min_cost = 31, open_example = [2]

Bin packing (minimum number of bins), `bin_packing`

Pack items of given sizes into the fewest bins of a given capacity: the minimum number and one packing.

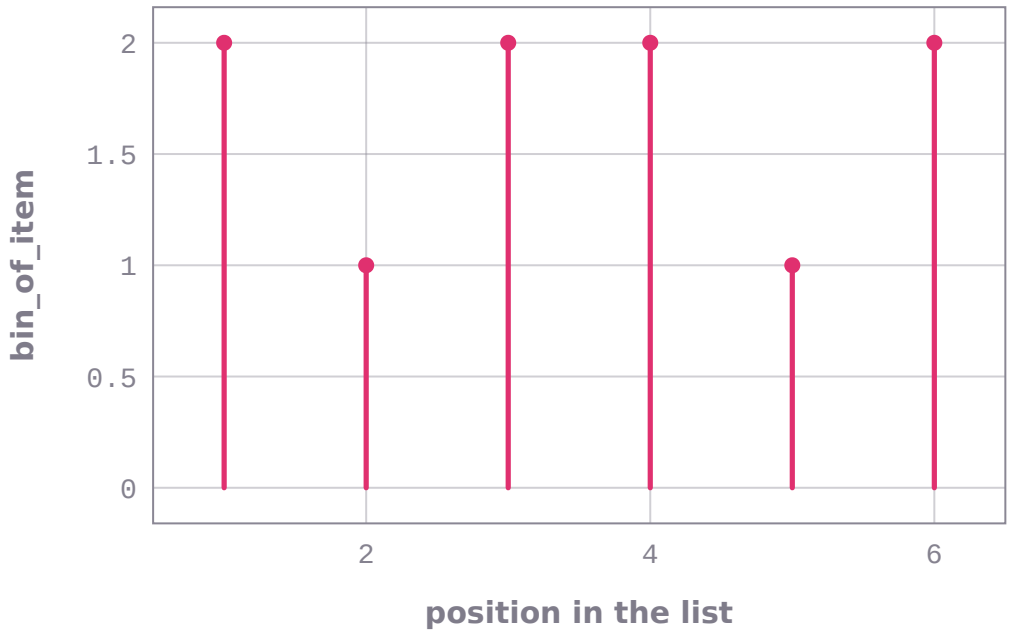
$$\min \#\text{bins} \;\geq\; \left\lceil \sum_j s_j / C \right\rceil$$

input	meaning	condition
`sizes`	size of each item	
`capacity`	capacity of each bin	

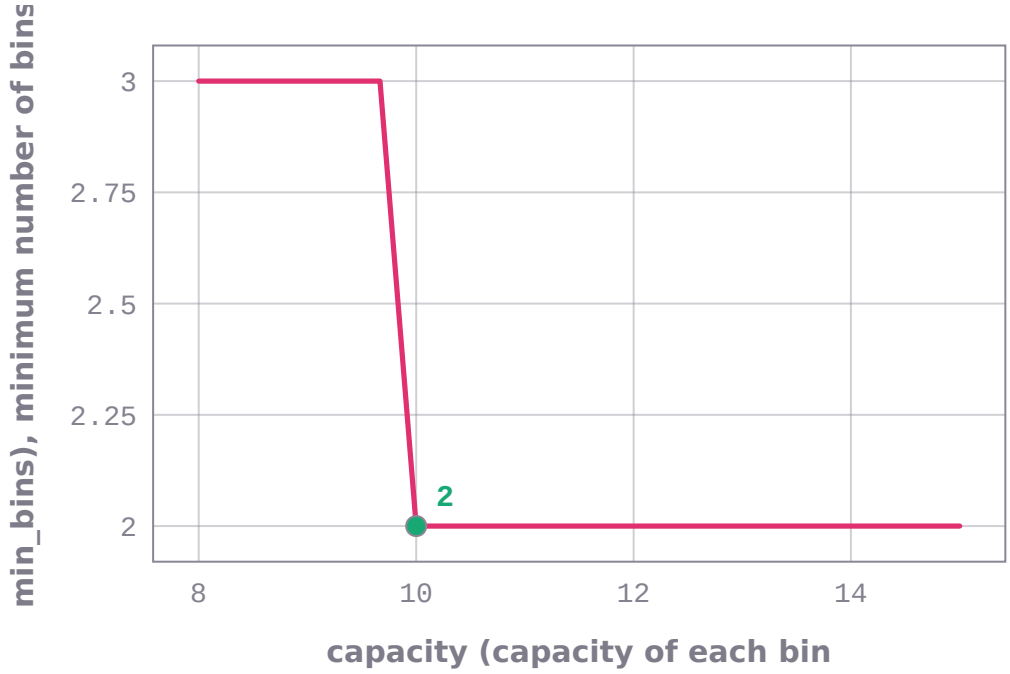
output	meaning
`min_bins`	minimum number of bins
`bin_of_item`	one optimal packing: bin number of each item

output	meaning
`lower_bound`	ceil(total size / capacity)

- **Method:** exact depth-first search (largest item first, symmetric bins skipped) with the volume bound
- **Assumes:** at most 14 items; every item fits in a bin
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



one optimal packing: bin number of each item (bin_of_item) returned by optimization-ai999x for the worked example, one value for each position in the list (6 values, first 2, last 2, largest 2). Inputs: sizes = [4, 8, 1, 4, 2, 1], capacity = 10.



min_bins (minimum number of bins) against capacity (capacity of each bin) from 5 to 15. Other inputs as in the worked example: sizes = [4, 8, 1, 4, 2, 1]. 22 points computed by optimization-ai999x; marked: the worked example, capacity = 10, min_bins = 2.

```
optimization_ai999x.solve('bin_packing', sizes=[4, 8, 1, 4, 2, 1], capacity=10)
# -> min_bins = 2, bin_of_item = [2, 1, 2, 2, 1, 2], lower_bound = 2   (SOLVED)
```

```
optimization_ai999x.solve('bin_packing', sizes=[6, 6, 6], capacity=10)
# -> min_bins = 3, bin_of_item = [1, 2, 3], lower_bound = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Bin packing: item sizes [4, 8, 1, 4, 2, 1] and bin capacity 10. Find the minimum number of bins.	SOLVED	min_bins = 2, bin_of_item = [2, 1, 2, 2, 1, 2], lower_bound = 2

Two-person zero-sum game (value and optimal mixed strategies), `zero_sum_game`

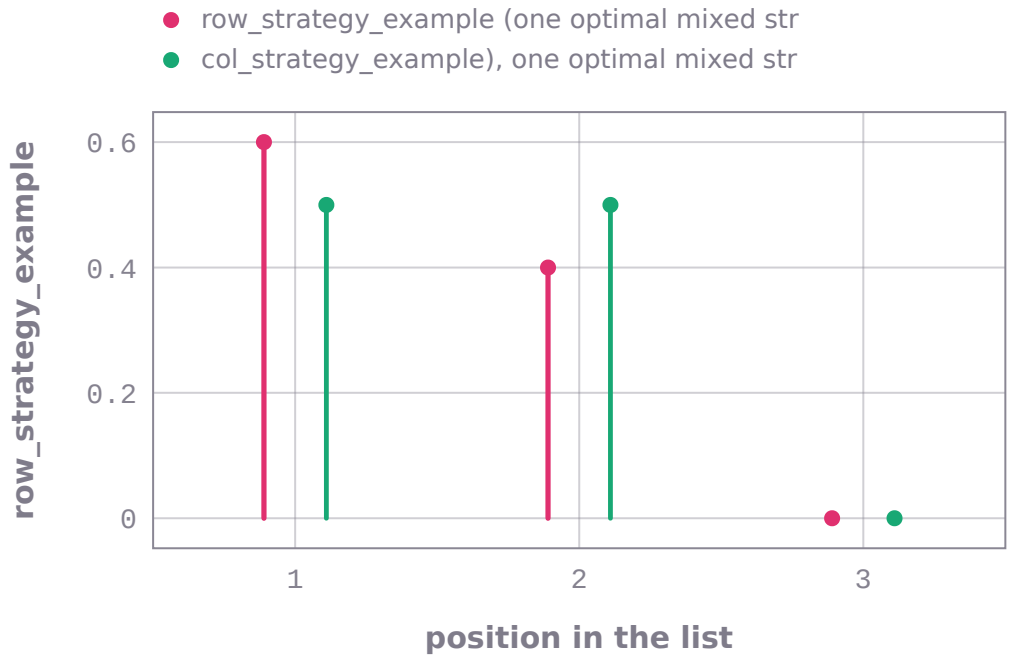
The value of a matrix game (row player maximises the payoff in the matrix, column player minimises it) and one optimal mixed strategy for each player.

$$v = \max_{x \in \Delta_m} \min_{j} \sum_{i} x_i a_{ij} = \min_{y \in \Delta_n} \max_{i} \sum_{j} a_{ij} y_j$$

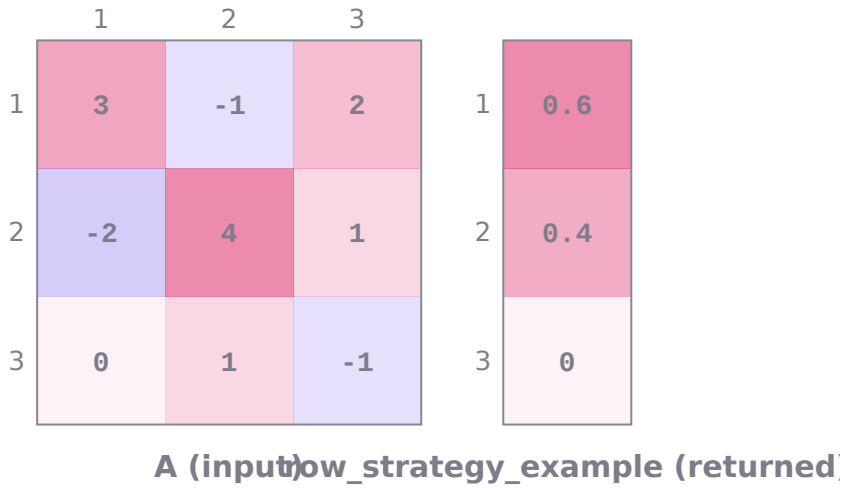
input	meaning	condition
`A`	payoff matrix to the row player, rows = row strategies, columns = column strategies	

output	meaning
`value`	value of the game
`row_strategy_example`	one optimal mixed strategy of the row player
`col_strategy_example`	one optimal mixed strategy of the column player

- **Method:** the game as a pair of dual linear programs on the shifted matrix, solved exactly (rational simplex); the pair of strategies is the certificate (each guarantees the value)
- **Assumes:** at most 12 strategies per player
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



one optimal mixed strategy of the row player (row_strategy_example) returned by optimization-ai999x for the worked example, one value for each position in the list (3 values, first 0.6, last 0, largest 0.6); with col_strategy_example (one optimal mixed strategy of the column player). Inputs: A = [[3, -1, 2], [-2, 4, 1], [0, 1, -1]].



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): A (input); row_strategy_example (returned). Returned with them by optimization-ai999x: value = 1.

```
optimization_ai999x.solve('zero_sum_game', A=[[3, -1, 2], [-2, 4, 1], [0, 1, -1]])
# -> value = 1, row_strategy_example = [0.6, 0.4, 0.0], col_strategy_example = [0.5, 0.5,
0.0]   (SOLVED)
```

```
optimization_ai999x.solve('zero_sum_game', A=[[1, -1], [-1, 1]])
# -> value = 0, row_strategy_example = [0.5, 0.5], col_strategy_example = [0.5, 0.5]
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the zero-sum game with payoff matrix [[3, -1, 2], [-2, 4, 1], [0, 1, -1]].	SOLVED	value = 1, row_strategy_example = [0.6, 0.4, 0.0], col_strategy_example = [0.5, 0.5, 0.0]
Solve the zero-sum game with payoff matrix [[3, -1, 2], [-2, 4]].	INVALID_INPUT	A: expected a rectangular matrix (rows of equal length).

2 x 2 zero-sum game (saddle point or mixed-strategy formula), `zero_sum_2x2`

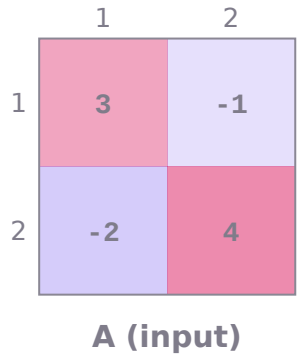
The value of a 2 x 2 matrix game: the saddle-point value when one exists, otherwise the mixed-strategy solution (probability of each player's first strategy, unique in that case).

$$v = \frac{ad - bc}{a + d - b - c}, \quad p = \frac{d - c}{a + d - b - c}, \quad q = \frac{d - b}{a + d - b - c}$$

input	meaning	condition
`A`	2 x 2 payoff matrix to the row player [[a, b], [c, d]]	

output	meaning
`value`	value of the game
`saddle_point`	yes when a pure saddle point exists
`p_row1`	probability the row player plays row 1 (only without a saddle point)
`q_col1`	probability the column player plays column 1 (only without a saddle point)

- **Method:** maximin = minimax gives a saddle point; otherwise the indifference formulas in exact arithmetic
- **Assumes:** a 2 x 2 matrix
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): A (input). Returned with them by optimization-ai999x: value = 1, p_row1 = 0.6, q_col1 = 0.5.

```
optimization_ai999x.solve('zero_sum_2x2', A=[[3, -1], [-2, 4]])
# -> value = 1, p_row1 = 0.6, q_col1 = 0.5, saddle_point = no   (SOLVED)
```

```
optimization_ai999x.solve('zero_sum_2x2', A=[[4, 2], [1, 3]])
# -> value = 2.5, p_row1 = 0.5, q_col1 = 0.25, saddle_point = no   (SOLVED)
```

```
optimization_ai999x.solve('zero_sum_2x2', A=[[5, 3], [4, 1]])
# -> value = 3, saddle_point = yes   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the zero-sum game with payoff matrix [[3, -1], [-2, 4]].	SOLVED	value = 1, p_row1 = 0.6, q_col1 = 0.5, saddle_point = no
Solve the zero-sum game with payoff matrix [[5, 3], [4, 1]].	SOLVED	value = 3, saddle_point = yes

Pure-strategy Nash equilibria of a bimatrix game, `pure_nash_equilibria`

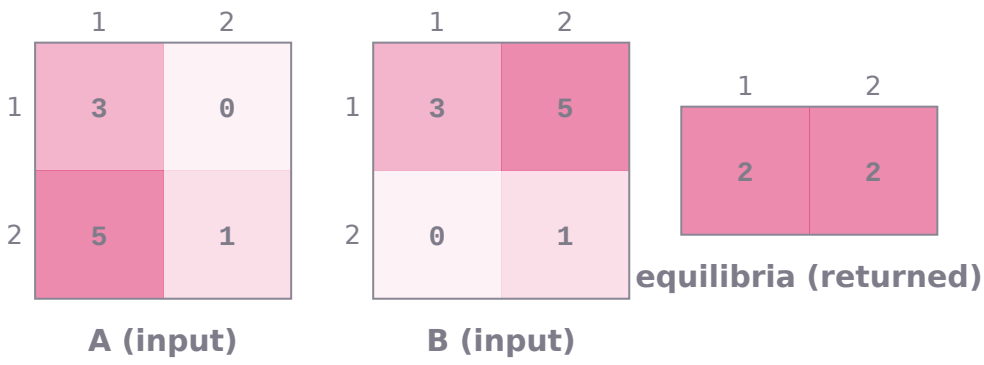
Every pure-strategy Nash equilibrium (a cell where each player's strategy is a best reply to the other's) of a two-player game given by the two players' payoff matrices.

$$a_{rs} = \max_k a_{ks}, \quad b_{rs} = \max_k b_{rk}$$

input	meaning	condition
`A`	row player's payoffs (rows = row strategies)	
`B`	column player's payoffs (same shape)	

output	meaning
`count`	number of pure equilibria
`equilibria`	all pure equilibria as [row, column], 1-based, in row-major order

- **Method:** mutual best replies, cell by cell, in exact arithmetic
- **Assumes:** at most 8 strategies per player; weak best replies count (ties are equilibria)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): A (input); B (input); equilibria (returned). Returned with them by optimization-ai999x: count = 1.

```
optimization_ai999x.solve('pure_nash_equilibria', A=[[3, 0], [5, 1]], B=[[3, 5], [0, 1]])
# -> count = 1, equilibria = [[2.0, 2.0]]   (SOLVED)
```

```
optimization_ai999x.solve('pure_nash_equilibria', A=[[2, 0], [0, 1]], B=[[1, 0], [0, 2]])
# -> count = 2, equilibria = [[1.0, 1.0], [2.0, 2.0]]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the pure-strategy Nash equilibria of the bimatrix game with row payoffs [[3, 0], [5, 1]] and column payoffs [[3, 5], [0, 1]].	SOLVED	count = 1, equilibria = [[2.0, 2.0]]
Find the pure-strategy Nash equilibria of the bimatrix game with row payoffs [[3, 0], [5, 1]] and column payoffs [[3, 5, 1], [0, 1, 2]].	INVALID_INPUT	Pure-strategy Nash equilibria of a bimatrix game: the row and column payoff matrices must have the same shape.

Mixed-strategy Nash equilibrium of a 2 x 2 game, `mixed_nash_2x2`

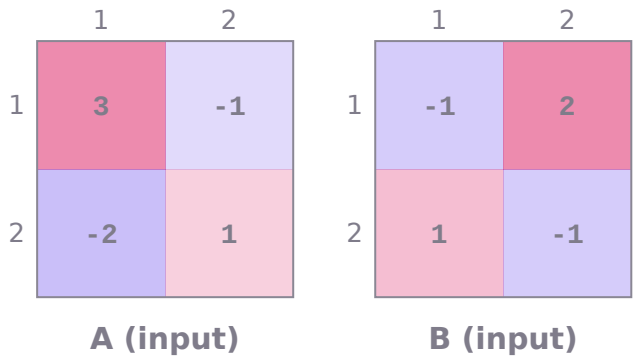
The equilibrium of a 2 x 2 two-player game without a pure equilibrium: each player's probability of the first strategy (chosen to make the OTHER player indifferent) and the expected payoffs.

$$q = \frac{a_{22} - a_{12}}{a_{11} - a_{12} - a_{21} + a_{22}}, \quad p = \frac{b_{22} - b_{21}}{b_{11} - b_{12} - b_{21} + b_{22}}$$

input	meaning	condition
`A`	row player's 2 x 2 payoffs	
`B`	column player's 2 x 2 payoffs	

output	meaning
`p_row1`	probability the row player plays row 1
`q_col1`	probability the column player plays column 1
`row_payoff`	row player's expected payoff
`col_payoff`	column player's expected payoff

- **Method:** indifference conditions in exact arithmetic: p from the column player's payoffs, q from the row player's
- **Assumes:** 2 x 2 matrices; no pure-strategy equilibrium (then the mixed one is unique)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): A (input); B (input). Returned with them by optimization-ai999x: p_row1 = 0.4, q_col1 = 0.285714, row_payoff = 0.142857, col_payoff = 0.2.

```
optimization_ai999x.solve('mixed_nash_2x2', A=[[3, -1], [-2, 1]], B=[[-1, 2], [1, -1]])
# -> p_row1 = 0.4, q_col1 = 0.285714, row_payoff = 0.142857, col_payoff = 0.2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the mixed-strategy Nash equilibrium of the game with row payoffs [[3, -1], [-2, 1]] and column payoffs [[-1, 2], [1, -1]].	SOLVED	p_row1 = 0.4, q_col1 = 0.285714, row_payoff = 0.142857, col_payoff = 0.2
Find the mixed-strategy Nash equilibrium of the game with row payoffs [[3, 0], [5, 1]] and column payoffs [[3, 5], [0, 1]].	INVALID_INPUT	Mixed-strategy Nash equilibrium of a 2 x 2 game: the game has a pure-strategy equilibrium: its mixed equilibrium is not

TOPSIS ranking (closeness to the ideal solution), `topsis`

Rank alternatives scored on several criteria by their relative closeness to the ideal and distance from the anti-ideal solution (vector normalisation, weighted Euclidean distances).

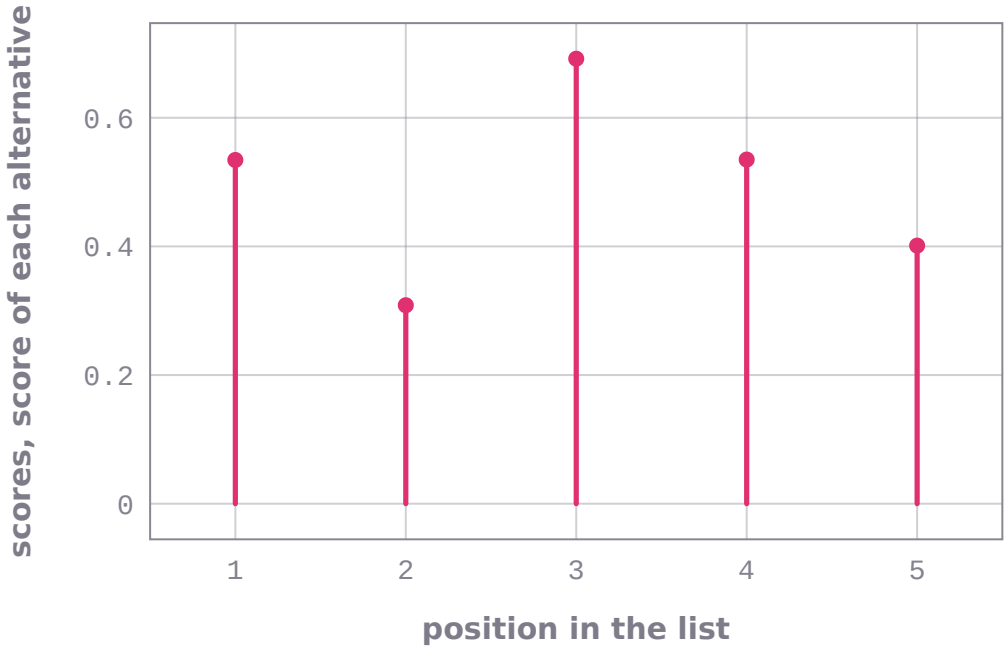
$$C_i = \frac{D_i^-}{D_i^+ + D_i^-}, \quad D_i^{\pm} = \left\| w \odot r_i - v^{\pm} \right\|_2$$

input	meaning	condition
`X`	decision matrix, rows = alternatives, columns = criteria (all values > 0)	
`w`	criterion weights (> 0; normalised to sum 1)	
`types`	criterion types: 1 = benefit (larger is better), -1 = cost (smaller is better)	

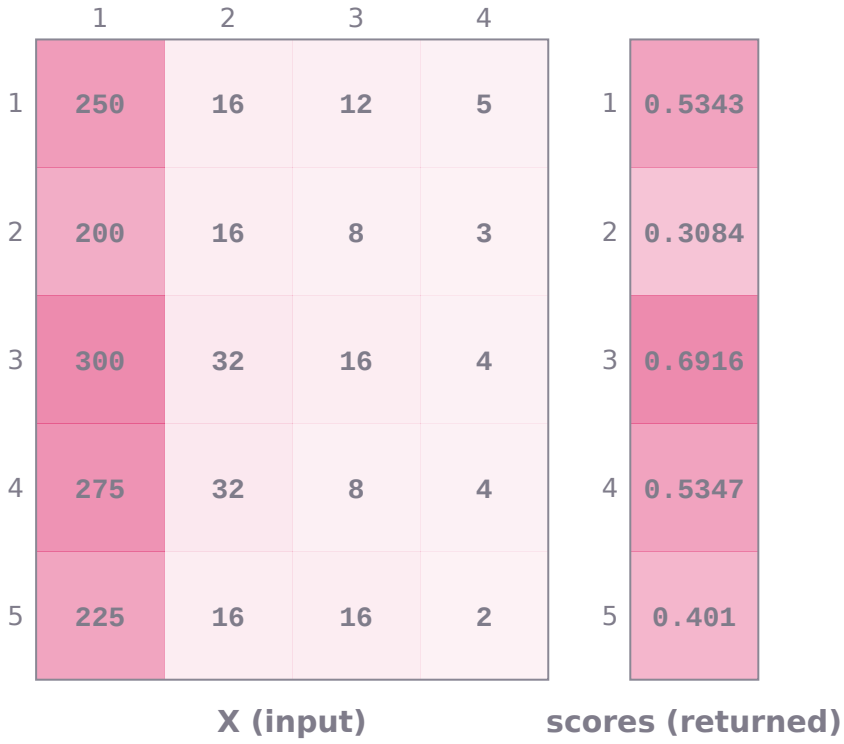
output	meaning
`scores`	score of each alternative
`best_score`	the highest score
`best`	best alternative, 1-based (on a tie the first listed)

- **Method:** vector normalisation $r = x$ / ||x_col||, weighted; ideal = best per criterion, anti-ideal = worst; $C = D$- / (D+ + D-)

- **Assumes:** values > 0; weights > 0 (normalised to sum 1); types 1 *or* −1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



score of each alternative (scores) returned by optimization-ai999x for the worked example, one value for each position in the list (5 values, first 0.534277, last 0.401046, largest 0.691632). Inputs: X = [[250, 16, 12, 5], [200, 16, 8, 3], [300, 32…, w = [0.25, 0.25, 0.25, 0.25], types = [-1, 1, 1, 1].



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): X (input); scores (returned). Returned with them by optimization-ai999x: best_score = 0.691632, best = 3. Other inputs: w = [0.25, 0.25, 0.25, 0.25], types = [-1, 1, 1, 1].

```
optimization_ai999x.solve('topsis', X=[[250, 16, 12, 5], [200, 16, 8, 3], [300, 32, 16, 4],
[275, 32, 8, 4], [225, 16, 16, 2]], w=[0.25, 0.25, 0.25, 0.25], types=[-1, 1, 1, 1])
# -> scores = [0.5342768571821003, 0.3083677687324685, 0.6916322312675315,
0.534736584486838, 0.40104612151678615], best_score = 0.691632, best = 3   (SOLVED)
```

```
optimization_ai999x.solve('topsis', X=[[1, 1], [2, 2]], w=[1, 1], types=[1, 1])
# -> scores = [0.0, 1.0], best_score = 1, best = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Rank with TOPSIS the alternatives with decision matrix [[1, 1], [2, 2]], weights [1, 1] and criteria types [1, 1].	SOLVED	scores = [0.0, 1.0], best_score = 1, best = 2
Rank with TOPSIS the alternatives with decision matrix [[1, 1], [2, 2]], weights [1, 1] and criteria types [1, 2].	INVALID_INPUT	TOPSIS ranking (closeness to the ideal solution): criterion types are 1 (benefit: larger is better) or -1 (cost: smaller

Weighted sum model (simple additive weighting), `weighted_sum_model`

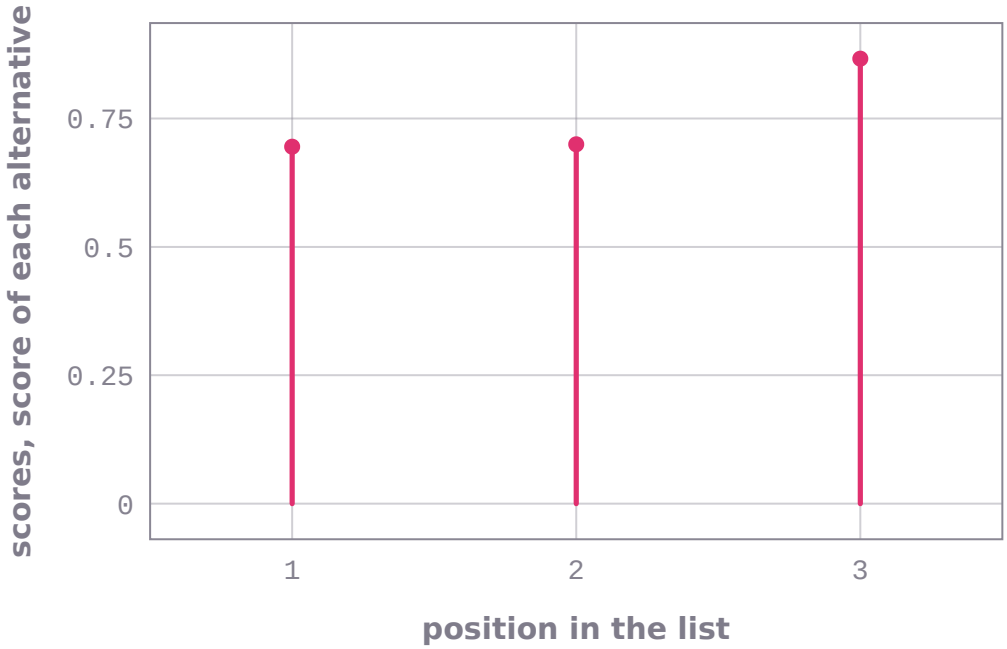
Score alternatives by the weighted sum of linearly normalised criterion values (benefit x / max, cost min / x) and pick the best.

$$S_i = \sum_j w_j n_{ij}, \quad n_{ij} = \frac{x_{ij}}{\max_i x_{ij}} \text{ or } \frac{\min_i x_{ij}}{x_{ij}}$$

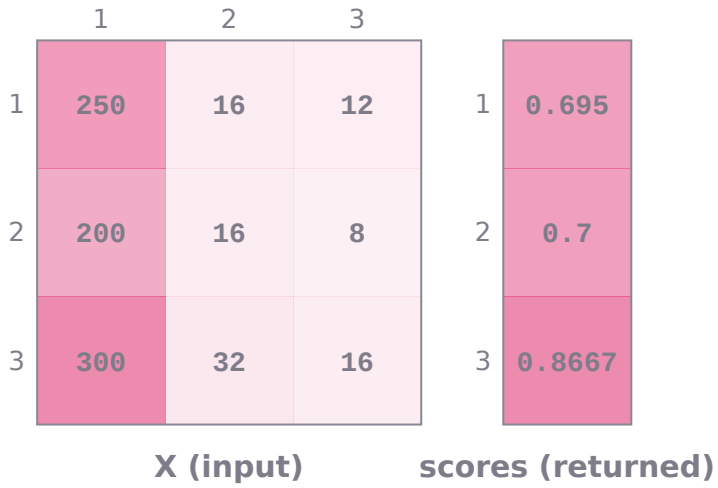
input	meaning	condition
`X`	decision matrix, rows = alternatives, columns = criteria (all values > 0)	
`w`	criterion weights (> 0; normalised to sum 1)	
`types`	criterion types: 1 = benefit (larger is better), -1 = cost (smaller is better)	

output	meaning
`scores`	score of each alternative
`best_score`	the highest score
`best`	best alternative, 1-based (on a tie the first listed)

- **Method:** linear normalisation (x / max for benefits, min / x for costs), then S_i = sum_j w_j n_ij
- **Assumes:** values > 0; weights > 0 (normalised to sum 1); types 1 $or - 1$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



score of each alternative (scores) returned by optimization-ai999x for the worked example, one value for each position in the list (3 values, first 0.695, last 0.866667, largest 0.866667). Inputs: X = [[250, 16, 12], [200, 16, 8], [300, 32, 16]], w = [0.4, 0.3, 0.3], types = [-1, 1, 1].



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): X (input); scores (returned). Returned with them by optimization-ai999x: best_score = 0.866667, best = 3. Other inputs: w = [0.4, 0.3, 0.3], types = [-1, 1, 1].

```
optimization_ai999x.solve('weighted_sum_model', X=[[250, 16, 12], [200, 16, 8], [300, 32, 16]], w=[0.4, 0.3, 0.3], types=[-1, 1, 1])
# -> scores = [0.6950000000000001, 0.7, 0.8666666666666667], best_score = 0.866667, best = 3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Score with the weighted sum model the alternatives with decision matrix [[250, 16, 12], [200, 16, 8], [300, 32, 16]], weights [0.4, 0.3, 0.3] and criteria types [-1, 1, 1].	SOLVED	scores = [0.6950000000000001, 0.7, 0.8666666666666667], best_score = 0.866667, best = 3

Weighted product model, `weighted_product_model`

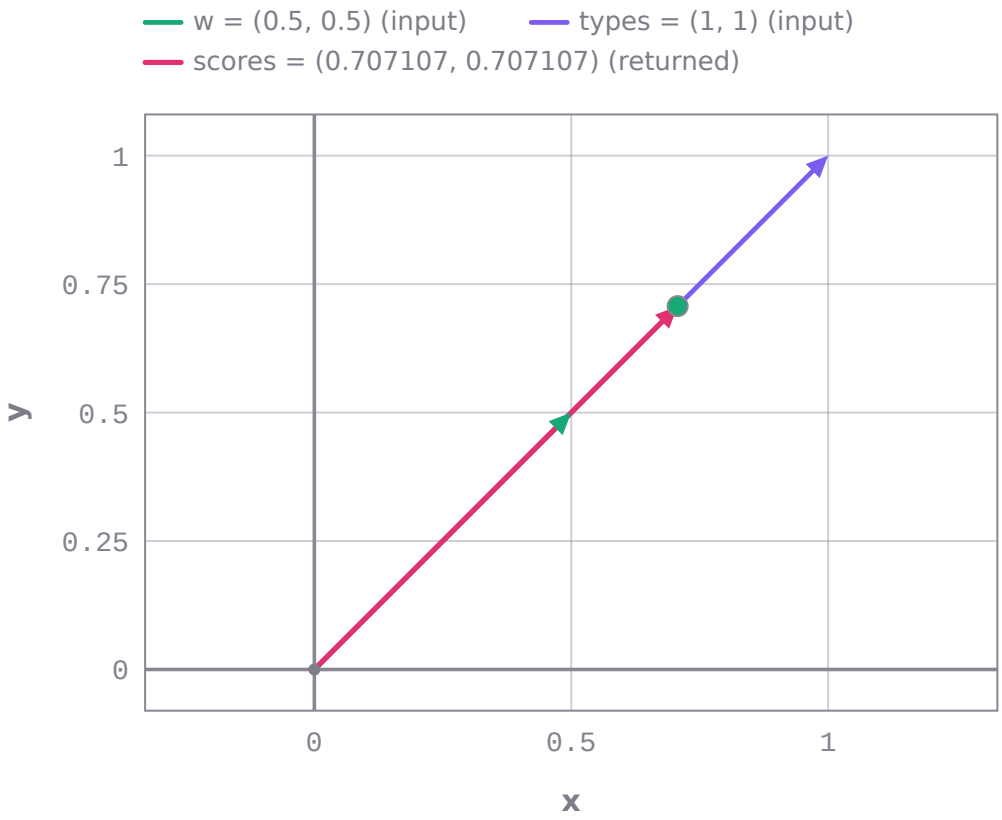
Score alternatives by the product of linearly normalised criterion values raised to the criterion weights.

$$P_i = \prod_j n_{ij}^{w_j}$$

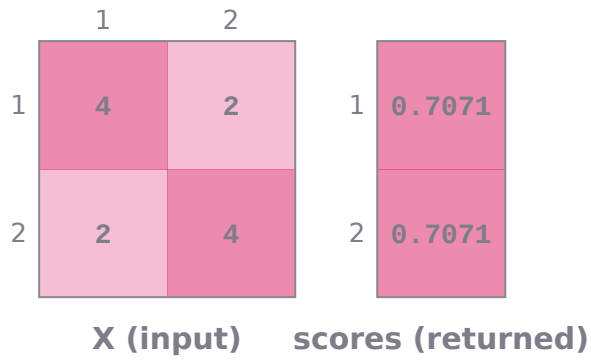
input	meaning	condition
`X`	decision matrix, rows = alternatives, columns = criteria (all values > 0)	
`w`	criterion weights (> 0; normalised to sum 1)	
`types`	criterion types: 1 = benefit (larger is better), -1 = cost (smaller is better)	

output	meaning
`scores`	score of each alternative
`best_score`	the highest score
`best`	best alternative, 1-based (on a tie the first listed)

- **Method:** linear normalisation (x / max for benefits, min / x for costs), then P_i = prod_j n_ij^{w_j}
- **Assumes:** values > 0; weights > 0 (normalised to sum 1); types 1 *or* -1
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The plane vectors of the worked example as arrows from the origin: input w = (0.5, 0.5), types = (1, 1); returned by optimization-ai999x: scores = (0.707107, 0.707107); also returned best_score = 0.707107, best = 1. Other inputs: X = [[4, 2], [2, 4]]. Marked: the tip of each returned vector.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): X (input); scores (returned). Returned with them by optimization-ai999x: best_score = 0.707107, best = 1. Other inputs: w = [0.5, 0.5], types = [1, 1].

```
optimization_ai999x.solve('weighted_product_model', X=[[4, 2], [2, 4]], w=[0.5, 0.5],
types=[1, 1])
# -> scores = [0.7071067811865476, 0.7071067811865476], best_score = 0.707107, best = 1
(SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Score with the weighted product model the alternatives with decision matrix [[4, 2], [2, 4]], weights [0.5, 0.5] and criteria types [1, 1].	SOLVED	scores = [0.7071067811865476, 0.7071067811865476], best_score = 0.707107, best = 1

AHP priority vector (principal eigenvector), `ahp_priority_vector`

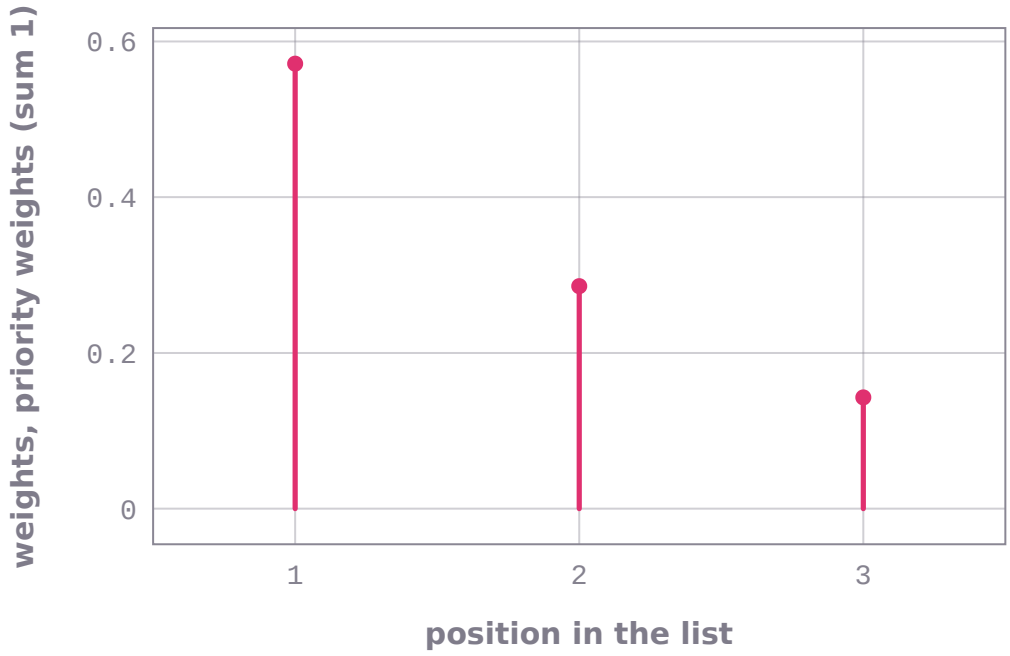
The priority weights of criteria compared in pairs (analytic hierarchy process): the normalised principal eigenvector of the reciprocal comparison matrix, and its eigenvalue lambda_max.

$$Pw = \lambda_{\max} w, \quad \sum_i w_i = 1$$

input	meaning	condition
`judgments`	upper-triangle pairwise judgments a12, a13, ..., a23, ... (Saaty scale; a_ji = 1/a_ij)	

output	meaning
`weights`	priority weights (sum 1)
`lambda_max`	principal eigenvalue
`best`	highest-priority criterion, 1-based

- **Method:** power iteration on the positive reciprocal matrix; certified as an eigenpair (P $w = \lambda$ w, $w > 0$: the Perron vector)
- **Assumes:** judgments > 0; 2 to 10 criteria
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



priority weights (sum 1) (weights) returned by optimization-ai999x for the worked example, one value for each position in the list (3 values, first 0.571429, last 0.142857, largest 0.571429). Inputs: judgments = [2, 4, 2].

```
optimization_ai999x.solve('ahp_priority_vector', judgments=[2, 4, 2])
# -> weights = [0.5714285714285714, 0.2857142857142857, 0.14285714285714285], lambda_max =
3, best = 1   (SOLVED)
```

```
optimization_ai999x.solve('ahp_priority_vector', judgments=[3])
# -> weights = [0.75, 0.25], lambda_max = 2, best = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the AHP priority vector for the pairwise judgments [2, 4, 2].	SOLVED	weights = [0.5714285714285714, 0.2857142857142857, 0.14285714285714285], lambda_max = 3, best = 1
Find the AHP priority vector for the pairwise judgments [2, 4].	INVALID_INPUT	AHP priority vector (principal eigenvector): give the upper-triangle judgments a12, a13, ..., a1n, a23, ... (3, 6, 10, .

AHP consistency ratio, `ahp_consistency_ratio`

How consistent a set of pairwise comparisons is: CI = (lambda_max - n)/(n - 1) and $CR = CI/RI$ (Saaty's random index); CR below 0.10 is conventionally acceptable.

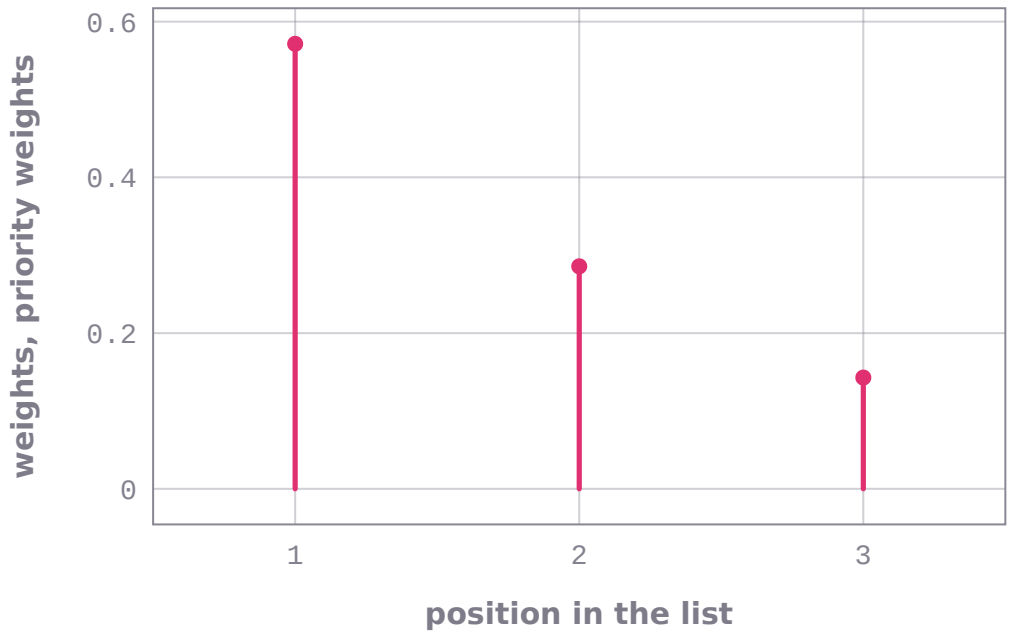
$$CI = \frac{\lambda_{\max} - n}{n - 1}, \quad CR = \frac{CI}{RI_n}$$

input	meaning	condition
`judgments`	upper-triangle pairwise judgments a12, a13, ..., a23, ... (Saaty scale; a_ji = 1/a_ij)	

output	meaning
`CR`	consistency ratio
`CI`	consistency index

output	meaning
`lambda_max`	principal eigenvalue
`weights`	priority weights
`acceptable`	yes when $CR < 0.10$

- **Method:** principal eigenpair by power iteration (certified as an eigenpair), then CI and CR
- **Assumes:** 3 to 10 criteria; $RI = 0.58$, 0.90, 1.12, 1.24, 1.32, 1.41, 1.45, 1.49 for $n = 3..10$ (Saaty)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



priority weights (weights) returned by optimization-ai999x for the worked example, one value for each position in the list (3 values, first 0.571429, last 0.142857, largest 0.571429). Inputs: judgments = [2, 4, 2].

```
optimization_ai999x.solve('ahp_consistency_ratio', judgments=[2, 4, 2])
# -> CR = 0, CI = 0, lambda_max = 3, weights = [0.5714285714285714, 0.2857142857142857,
0.14285714285714285], acceptable = yes   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the AHP consistency ratio for the pairwise judgments [2, 4, 2].	SOLVED	CR = 0, CI = 0, lambda_max = 3, weights = [0.5714285714285714, 0.2857142857142857, 0.14285714285714285], acceptable = yes
Find the AHP consistency ratio for the pairwise judgments [3].	INVALID_INPUT	AHP consistency ratio: the consistency ratio needs at least 3 criteria.

AHP weights by the geometric mean method, `ahp_geometric_mean`

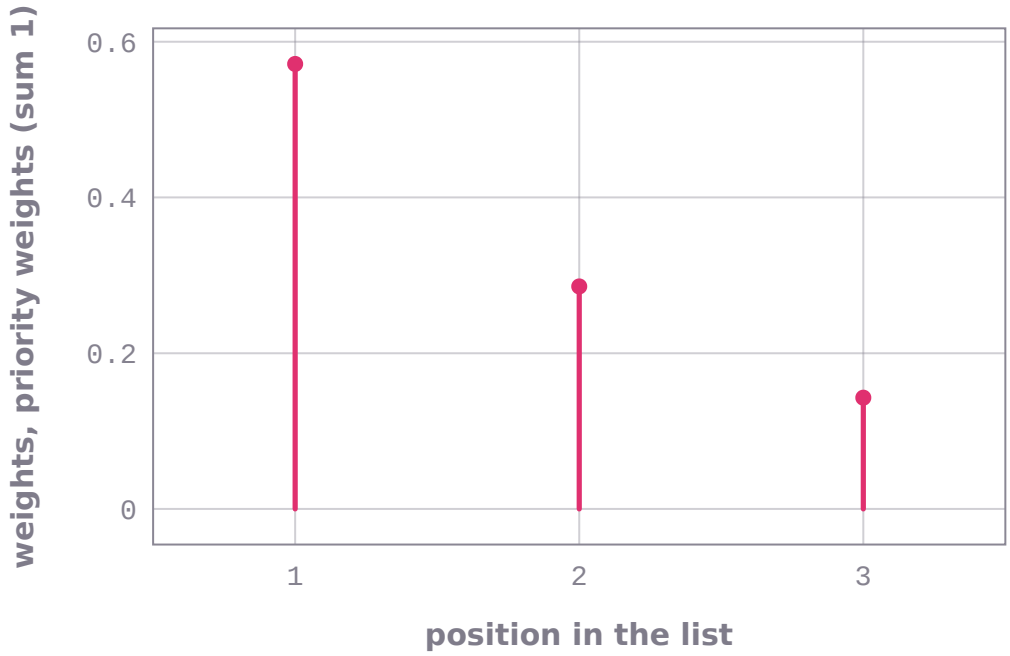
Priority weights from pairwise comparisons as normalised row geometric means (the logarithmic least-squares solution).

$$w_i = \frac{(\prod_j a_{ij})^{1/n}}{\sum_k (\prod_j a_{kj})^{1/n}}$$

input	meaning	condition
`judgments`	upper-triangle pairwise judgments a12, a13, ..., a23, ... (Saaty scale; a_ji = 1/a_ij)	

output	meaning
`weights`	priority weights (sum 1)
`max_weight`	largest weight
`best`	highest-priority criterion, 1-based

- **Method:** row geometric means, normalised
- **Assumes:** judgments > 0; 2 to 10 criteria
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



priority weights (sum 1) (weights) returned by optimization-ai999x for the worked example, one value for each position in the list (3 values, first 0.571429, last 0.142857, largest 0.571429). Inputs: judgments = [2, 4, 2].

```
optimization_ai999x.solve('ahp_geometric_mean', judgments=[2, 4, 2])
# -> weights = [0.5714285714285714, 0.2857142857142857, 0.14285714285714285], max_weight =
0.571429, best = 1   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the AHP weights by the geometric mean method for the pairwise judgments [2, 4, 2].	SOLVED	weights = [0.5714285714285714, 0.2857142857142857, 0.14285714285714285], max_weight = 0.571429, best = 1

Unconstrained quadratic minimisation, `unconstrained_quadratic_min`

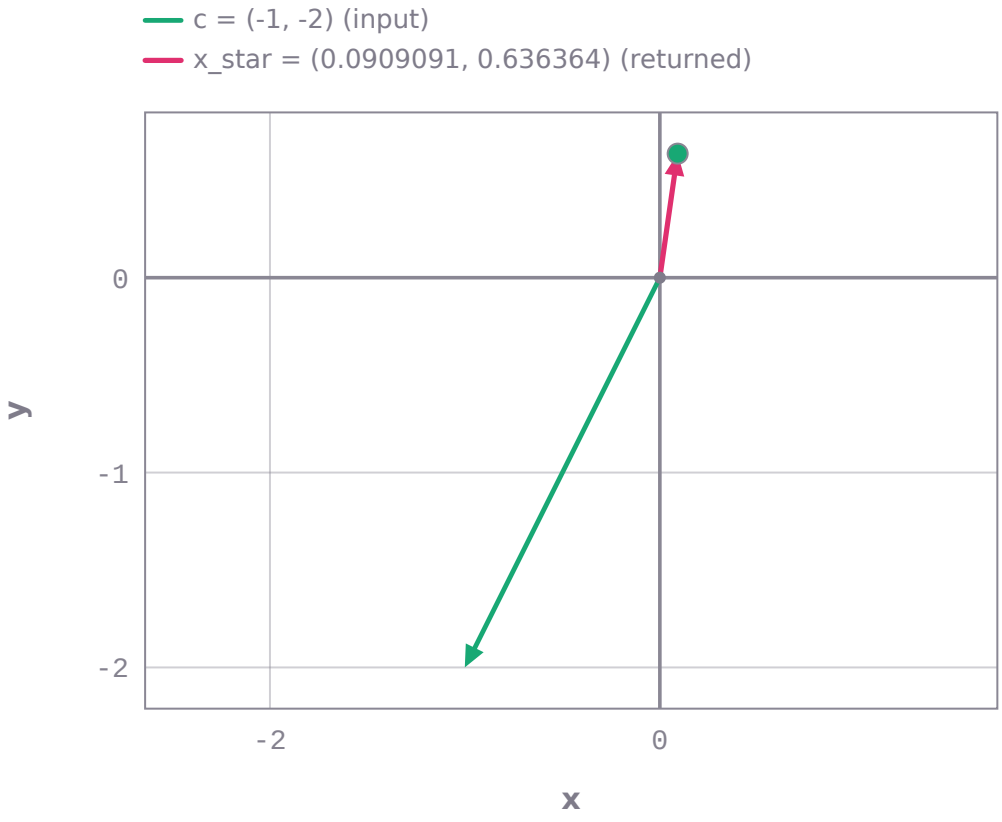
Minimise f(x) = 1/2 x^T Q $x + c^T$ x for a symmetric positive definite Q: the minimiser and the minimum.

$$x^* = -Q^{-1}c, \quad f^* = \tfrac{1}{2}c^\top x^*$$

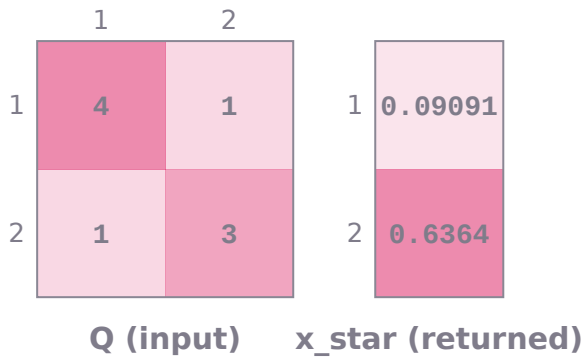
input	meaning	condition
`Q`	symmetric positive definite Hessian Q (n x n)	
`c`	linear term c (n entries)	

output	meaning
`f_min`	minimum value
`x_star`	the minimiser x* = -Q^{-1} c (unique)

- **Method:** positive definiteness by symmetric elimination, then Q $x = -c$ solved exactly; certified by the gradient
- **Assumes:** Q symmetric positive definite (convention: $f = 1/2\, x^T$ Q $x + c^T$ x); $n \leq 10$
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The plane vectors of the worked example as arrows from the origin: input c = (-1, -2); returned by optimization-ai999x: x_star = (0.0909091, 0.636364); also returned f_min = -0.681818. Other inputs: Q = [[4, 1], [1, 3]]. Marked: the tip of each returned vector.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): Q (input); x_star (returned). Returned with them by optimization-ai999x: f_min = -0.681818. Other inputs: c = [-1, -2].

```
optimization_ai999x.solve('unconstrained_quadratic_min', Q=[[4, 1], [1, 3]], c=[-1, -2])
# -> f_min = -0.681818, x_star = [0.09090909090909091, 0.6363636363636364]   (SOLVED)
```

```
optimization_ai999x.solve('unconstrained_quadratic_min', Q=[[2, 0], [0, 2]], c=[-4, 2])
# -> f_min = -5, x_star = [2.0, -1.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Minimise the unconstrained quadratic with Hessian Q = [[4, 1], [1, 3]] and linear term c = [-1, -2].	SOLVED	f_min = -0.681818, x_star = [0.09090909090909091, 0.6363636363636364]
Minimise the unconstrained quadratic with Hessian Q = [[1, 2], [2, 1]] and linear term c = [-1, -2].	INVALID_INPUT	Unconstrained quadratic minimisation: Q must be positive definite (otherwise there is no unique minimum).

Equality-constrained quadratic program (Lagrange multipliers / KKT),

`equality_constrained_qp`

Minimise $1/2\, x^T$ Q $x + c^T$ x subject to A $x = b$: the minimiser, the minimum and the Lagrange multipliers.

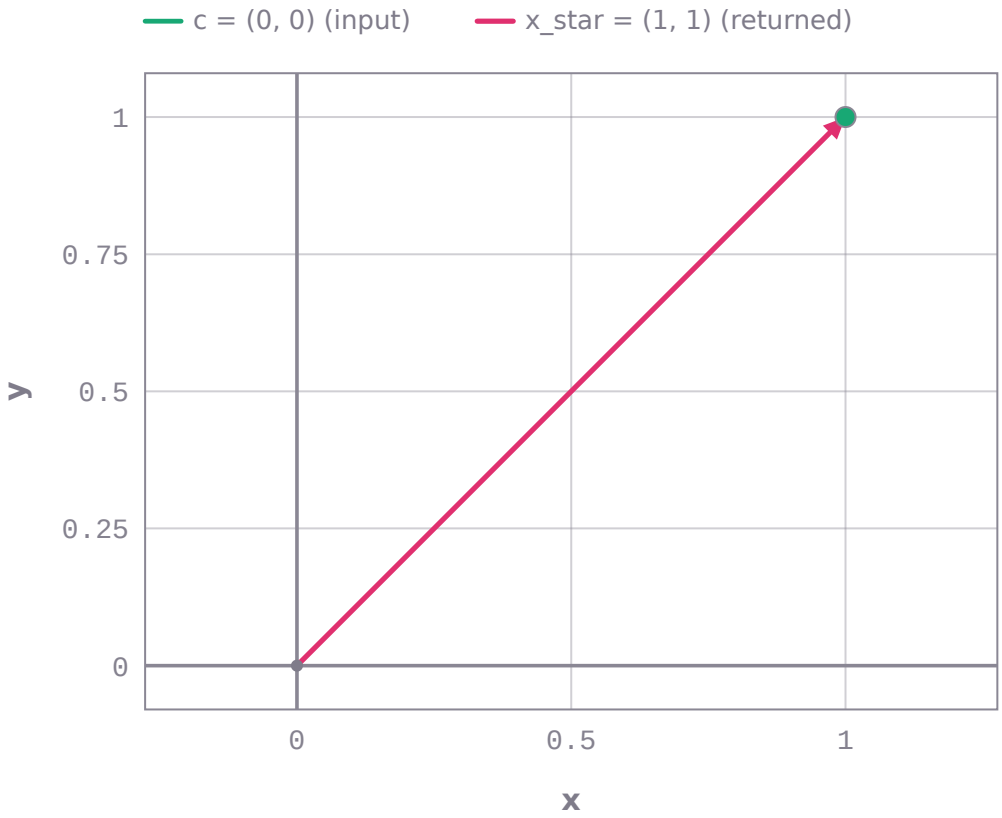
$$\begin{bmatrix} Q & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} -c \\ b \end{bmatrix}$$

input	meaning	condition
`Q`	symmetric positive definite Hessian Q	
`c`	linear term c	
`A`	constraint matrix A (rows = constraints)	
`b`	right-hand side b	

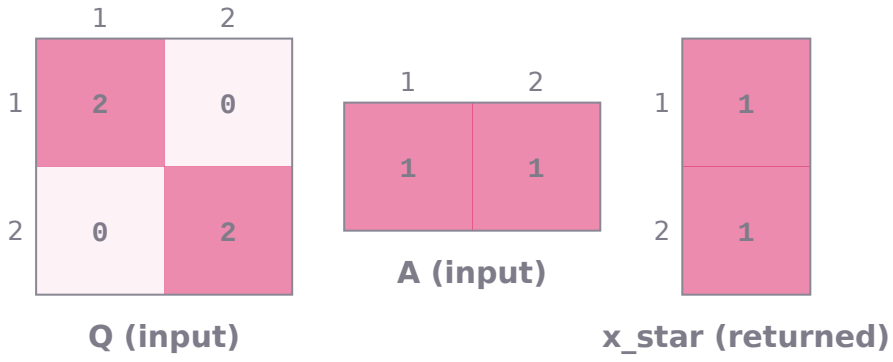
output	meaning
`f_min`	minimum value
`x_star`	the minimiser (unique)

output	meaning
`multipliers`	Lagrange multipliers lambda (convention Q $\boldsymbol{x} + \boldsymbol{c} + \boldsymbol{A}^{\boldsymbol{T}}$ lambda = 0)

- **Method:** the KKT system [Q $\boldsymbol{A}^{\boldsymbol{T}}$; A 0][x; lambda] = [-c; b] solved exactly; certified by stationarity and feasibility
- **Assumes:** Q symmetric positive definite; rows of A linearly independent
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The plane vectors of the worked example as arrows from the origin: input c = (0, 0); returned by optimization-ai999x: x_star = (1, 1); also returned f_min = 2. Other inputs: Q = [[2, 0], [0, 2]], A = [[1, 1]], b = [2]. Marked: the tip of each returned vector.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): Q (input); A (input); x_star (returned). Returned with them by optimization-ai999x: f_min = 2. Other inputs: c = [0, 0], b = [2].

```
optimization_ai999x.solve('equality_constrained_qp', Q=[[2, 0], [0, 2]], c=[0, 0], A=[[1, 1]], b=[2])
# -> f_min = 2, x_star = [1.0, 1.0], multipliers = [-2.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Minimise the equality-constrained quadratic with Hessian Q = [[2, 0], [0, 2]], linear term c = [0, 0], constraint matrix A = [[1, 1]] and right-hand side b = [2].	SOLVED	f_min = 2, x_star = [1.0, 1.0], multipliers = [-2.0]
Minimise the equality-constrained quadratic with Hessian Q = [[2, 0], [0, 2]], linear term c = [0, 0], constraint matrix A = [[1, 1], [2, 2]] and right-hand side b = [2, 4].	INVALID_INPUT	Equality-constrained quadratic program (Lagrange multipliers / KKT): the constraint rows must be linearly independent.

Steepest descent step with exact line search (quadratic), `steepest_descent_step`

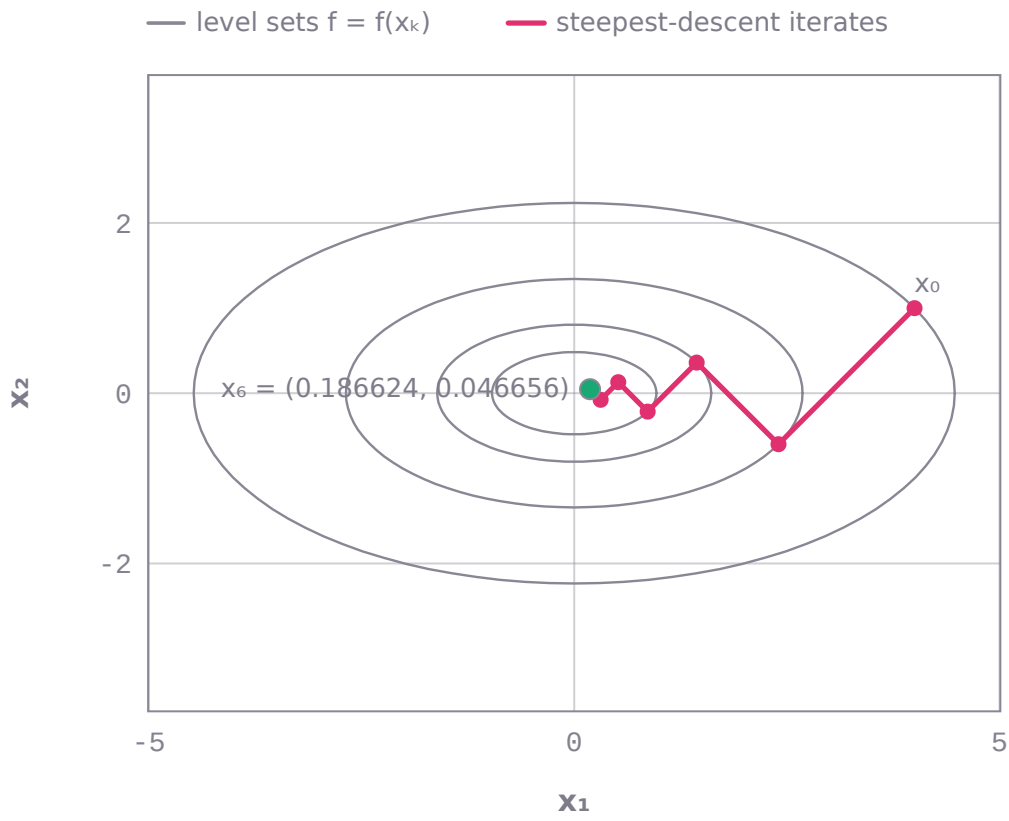
One iteration of steepest descent on f(x) = 1/2 x^T Q $x + c^T$ x from x0: the gradient, the exact step length, the new point and its function value.

$$\alpha = \frac{g^\top g}{g^\top Q g}, \quad x_1 = x_0 - \alpha g$$

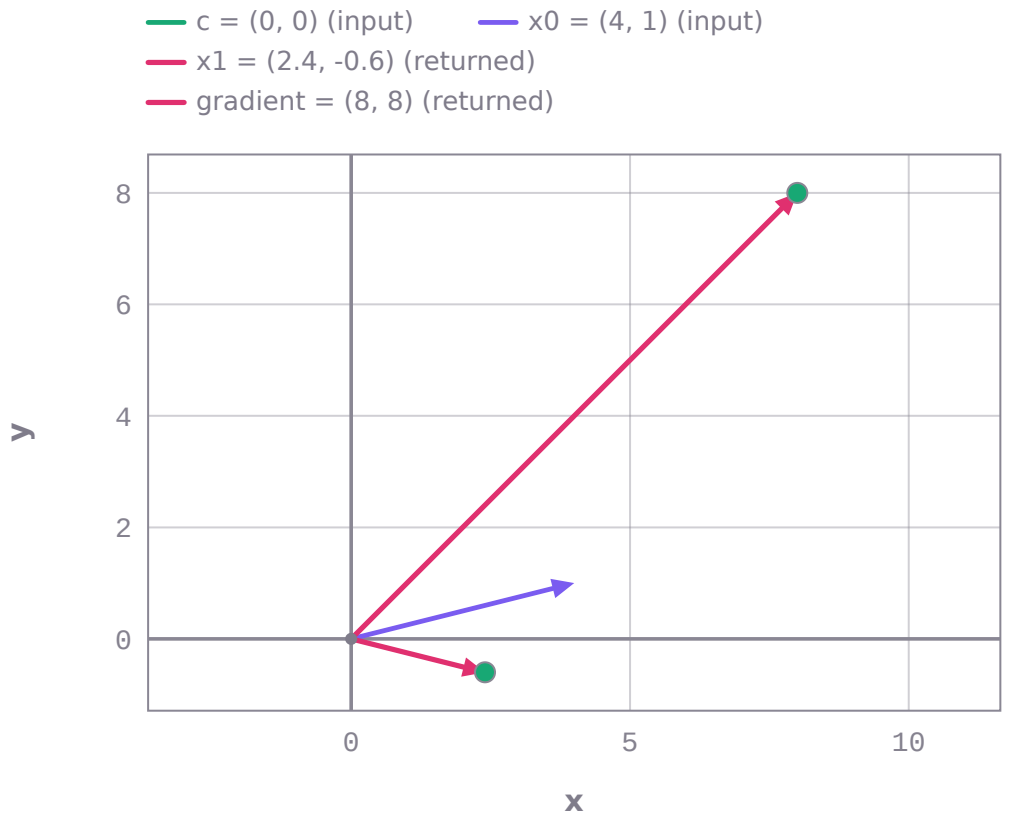
input	meaning	condition
`Q`	symmetric positive definite Hessian Q	
`c`	linear term c	
`x0`	starting point x0	

output	meaning
`alpha`	exact step length g^T g/g^T Q g
`x1`	new point $x_0 - \alpha$ g
`f1`	f at the new point
`gradient`	gradient at x0

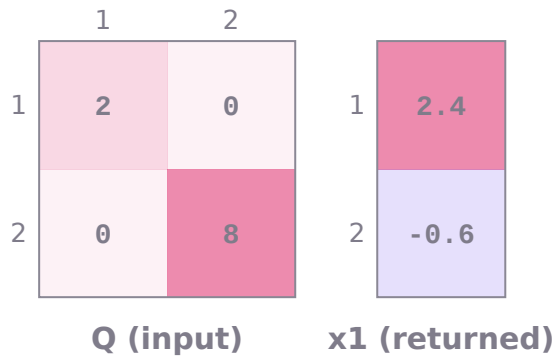
- **Method:** $g = Q$ $x_0 + c$; $\alpha = g^T$ g/g^T Q g in exact arithmetic; certified by g^T grad f(x1) = 0
- **Assumes:** Q symmetric positive definite; x0 is not the minimiser
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



Steepest descent with exact line search on f(x) = ½ xᵀQx, Q = [[2, 0], [0, 8]], from x_0 = (4, 1): six steps computed by the package, each perpendicular to the level set through its start (f = 20, 7.2, 2.592, 0.93312, 0.335923, 0.120932, 0.0435356); marked: x_6 = (0.186624, 0.046656).



The plane vectors of the worked example as arrows from the origin: input c = (0, 0), x0 = (4, 1); returned by optimization-ai999x: x1 = (2.4, -0.6), gradient = (8, 8); also returned alpha = 0.2, f1 = 7.2. Other inputs: Q = [[2, 0], [0, 8]]. Marked: the tip of each returned vector.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): Q (input); x1 (returned). Returned with them by optimization-ai999x: alpha = 0.2, f1 = 7.2. Other inputs: c = [0, 0], x0 = [4, 1].

```
optimization_ai999x.solve('steepest_descent_step', Q=[[2, 0], [0, 8]], c=[0, 0], x0=[4, 1])
# -> alpha = 0.2, x1 = [2.4, -0.6], f1 = 7.2, gradient = [8.0, 8.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Take one steepest descent step with exact line search for Hessian Q = [[2, 0], [0, 8]] and linear term c = [0, 0] from x0 = [4, 1].	SOLVED	alpha = 0.2, x1 = [2.4, -0.6], f1 = 7.2, gradient = [8.0, 8.0]
Take one steepest descent step with exact line search for Hessian Q = [[2, 0], [0, 8]] and linear term c = [0, 0] from x0 = [0, 0].	INVALID_INPUT	Steepest descent step with exact line search (quadratic): x0 is already the minimiser (the gradient is zero).

Critical point of a two-variable quadratic and its classification,

`critical_point_two_variables`

For f(x, y) = a x^2 + b xy + c y^2 + d x + e y + f_0: the critical point, its value and whether it is a local minimum, a local maximum or a saddle point (second-derivative test).

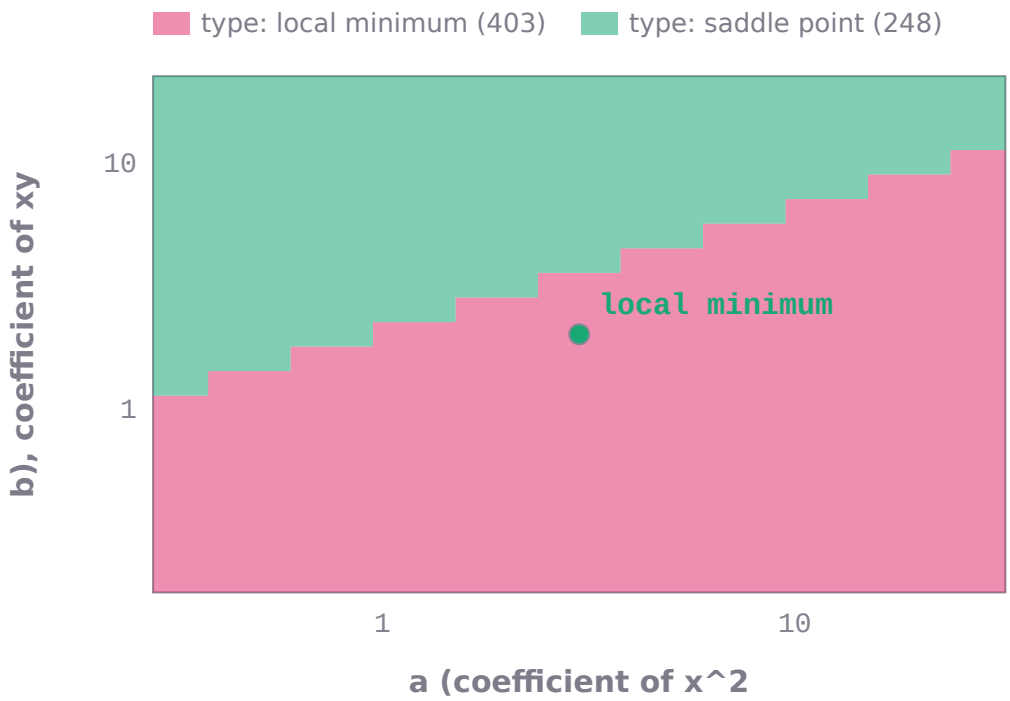
$$D = f_{xx} f_{yy} - f_{xy}^2 = 4ac - b^2$$

input	meaning	condition
`a`	coefficient of x^2	
`b`	coefficient of xy	
`c`	coefficient of y^2	
`d`	coefficient of x	
`e`	coefficient of y	
`f0`	constant term	

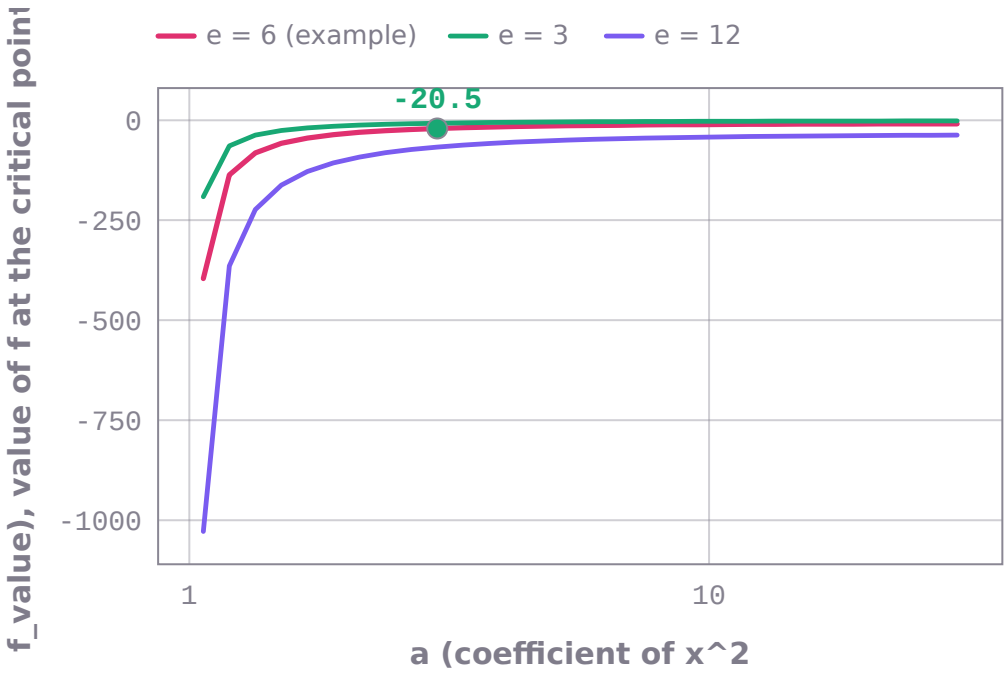
output	meaning
`f_value`	value of f at the critical point
`x`	x of the critical point

output	meaning
`y`	y of the critical point
`type`	local minimum, local maximum or saddle point

- **Method:** grad $f = 0$ solved exactly; $D = f_{xx}\, f_{yy} - f_{xy}^2 = 4ac - b^2$ classifies it
- **Assumes:** $4ac - b^2 \neq 0$ (an isolated critical point)
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



local minimum, local maximum or saddle point (type) over a (coefficient of x^2) from 0.3 to 30 and b (coefficient of xy) from 0.2 to 20 (logarithmic axes): each of the 31 × 21 cells is one run of optimization-ai999x, coloured by the verdict it returned, local minimum: 403 cells, saddle point: 248 cells. Other inputs as in the worked example: c = 1, d = -4, e = 6, f0 = 1. Marked: the worked example, a = 3, b = 2 → local minimum.



f_value (value of f at the critical point) against a (coefficient of x^2) from 1.06444 to 30 on a logarithmic axis. One curve for each value of e (coefficient of y): 6, 3, 12. Other inputs as in the worked example: b = 2, c = 1, d = -4, f0 = 1. 30 points per curve computed by optimization-ai999x; marked: the worked example, a = 3, f_value = -20.5.

```
optimization_ai999x.solve('critical_point_two_variables', a=3, b=2, c=1, d=-4, e=6, f0=1)
# -> f_value = -20.5, x = 2.5, y = -5.5, type = local minimum   (SOLVED)
```

```
optimization_ai999x.solve('critical_point_two_variables', a=1, b=0, c=-1)
# -> f_value = 0, x = 0, y = 0, type = saddle point   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find and classify the critical point of f(x, y) = 3x^2 + 2xy + y^2 - 4x + 6y + 1.	SOLVED	f_value = -20.5, x = 2.5, y = -5.5, type = local minimum
Find and classify the critical point of f(x, y) = x^2 + 2xy + y^2.	INVALID_INPUT	Critical point of a two-variable quadratic and its classification: 4ac - b^2 = 0: no isolated critical point (the second

Absolute extrema of a polynomial on a closed interval, `polynomial_extrema_interval`

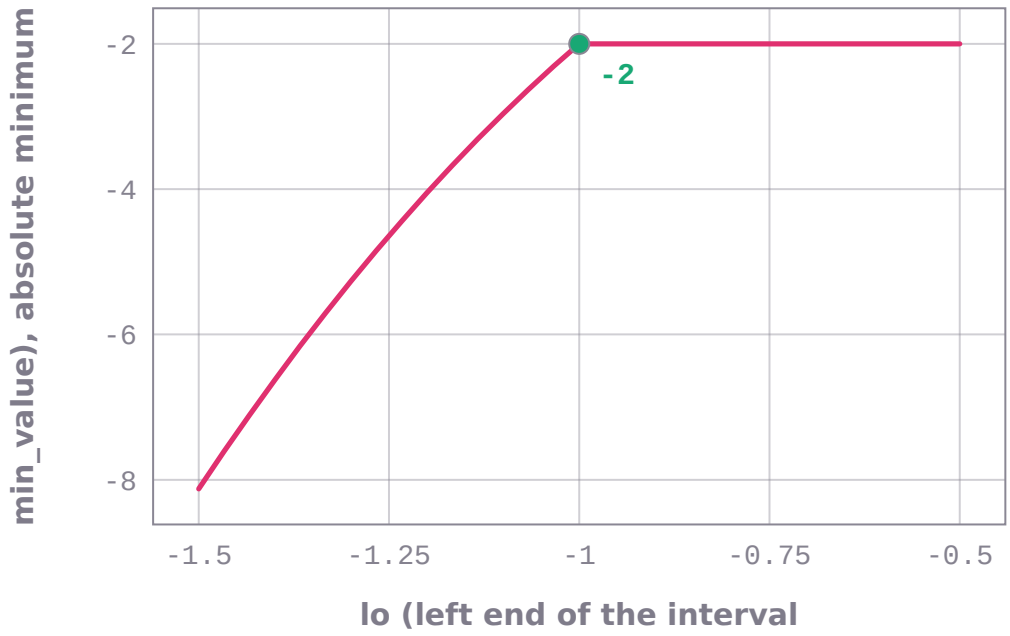
The absolute minimum and maximum of a polynomial on [lo, hi] (closed-interval method: endpoints and critical points), and where they are attained.

$$\min_{[a,b]} f = \min\{f(a), f(b), f(c) : f'(c) = 0,\ a < c < b\}$$

input	meaning	condition
`coefficients`	polynomial coefficients, highest power first	
`lo`	left end of the interval	
`hi`	right end of the interval	

output	meaning
`min_value`	absolute minimum
`max_value`	absolute maximum
`argmin_example`	a point attaining the minimum (the smallest one on a tie)
`argmax_example`	a point attaining the maximum (the smallest one on a tie)

- **Method:** real roots of f' in [lo, hi] (monotone pieces between the roots of f'' bisected to the last float), compared with the endpoints
- **Assumes:** $lo < hi$; degree 1 to 5
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



min_value (absolute minimum) against lo (left end of the interval) from -1.5 to -0.5. Other inputs as in the worked example: coefficients = [1, -3, 0, 2], hi = 3. 31 points computed by optimization-ai999x; marked: the worked example, lo = -1, min_value = -2.

```
optimization_ai999x.solve('polynomial_extrema_interval', coefficients=[1, -3, 0, 2], lo=-1, hi=3)
# -> min_value = -2, max_value = 2, argmin_example = -1, argmax_example = 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the absolute minimum and maximum of the polynomial with coefficients [1, -3, 0, 2] on the interval [-1, 3].	SOLVED	min_value = -2, max_value = 2, argmin_example = -1, argmax_example = 0
Find the absolute minimum and maximum of the polynomial with coefficients [1, -3, 0, 2] on the interval [3, -1].	INVALID_INPUT	Absolute extrema of a polynomial on a closed interval: the interval needs lo < hi.

Dynamic lot sizing (Wagner-Whitin), `wagner_whitin_lot_sizing`

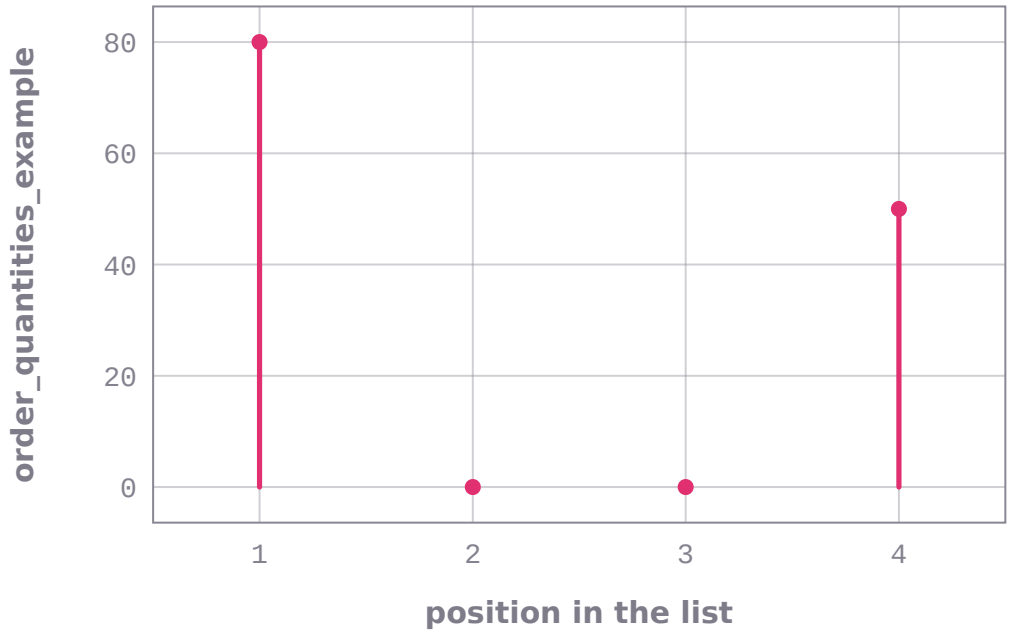
The cheapest ordering plan for known demands over several periods with a fixed cost per order and a holding cost per unit carried to the next period (no shortages).

$$C_t = \min_{1 \le s \le t} \left(C_{s-1} + K + h \sum_{u=s}^{t} (u - s)\, d_u \right)$$

input	meaning	condition
`demand`	demand in each period	
`K`	setup (ordering) cost per order	
`h`	holding cost per unit per period	

output	meaning
`min_cost`	minimum total setup plus holding cost
`order_quantities_example`	one optimal plan: quantity ordered in each period
`orders`	number of orders in that plan

- **Method:** Wagner-Whitin dynamic programme (each order covers whole periods) in exact arithmetic
- **Assumes:** demands, K, $h \ge 0$; no initial stock, no shortages, holding charged on end-of-period stock
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



one optimal plan: quantity ordered in each period (order_quantities_example) returned by optimization-ai999x for the worked example, one value for each position in the list (4 values, first 80, last 50, largest 80). Inputs: demand = [20, 50, 10, 50], K = 100, h = 1.



min_cost (minimum total setup plus holding cost) against K (setup (ordering) cost per order) from 10 to 1000 on a logarithmic axis. One curve for each value of h (holding cost per unit per period): 1, 0.5, 2. Other inputs as in the worked example: demand = [20, 50, 10, 50]. 41 points per curve computed by optimization-ai999x; marked: the worked example, K = 100, min_cost = 270.

```
optimization_ai999x.solve('wagner_whitin_lot_sizing', demand=[20, 50, 10, 50], K=100, h=1)
# -> min_cost = 270, order_quantities_example = [80.0, 0.0, 0.0, 50.0], orders = 2
(SOLVED)
```

```
optimization_ai999x.solve('wagner_whitin_lot_sizing', demand=[10, 0, 10], K=50, h=5)
# -> min_cost = 100, order_quantities_example = [10.0, 0.0, 10.0], orders = 2   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Wagner-Whitin lot sizing with demands [20, 50, 10, 50], setup cost 100 and holding cost 1 per unit per period.	SOLVED	min_cost = 270, order_quantities_example = [80.0, 0.0, 0.0, 50.0], orders = 2
Wagner-Whitin lot sizing with demands [20, 50, 10, 50] and setup cost 100.	NEEDS_INPUT	wagner_whitin_lot_sizing needs h (holding cost per unit per period). Example: solve('wagner_whitin_lot_sizing', demand=[

Discounted Markov decision process (optimal values and policy), `mdp_discounted`

The optimal expected discounted reward from every state of a finite Markov decision process and an optimal stationary policy.

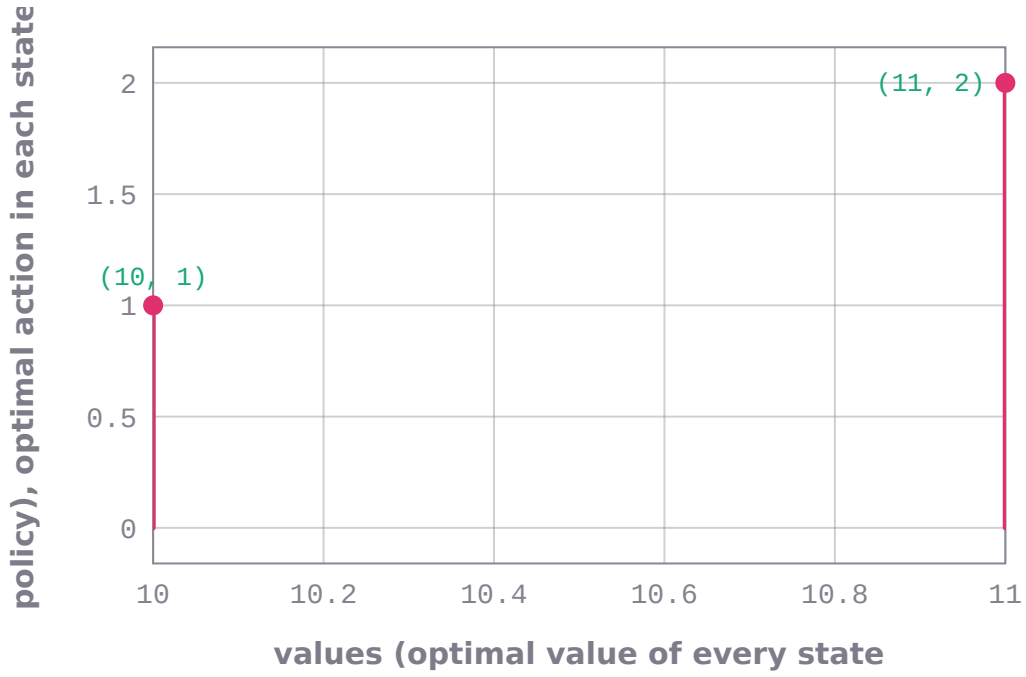
$$V^*(s) = \max_a \left(R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^*(s') \right)$$

input	meaning	condition
`R`	rewards, rows = states, columns = actions	
`T`	transition probabilities: one row per (state, action) in the order s1a1, s1a2, ..., s2a1, ...; columns = next state	

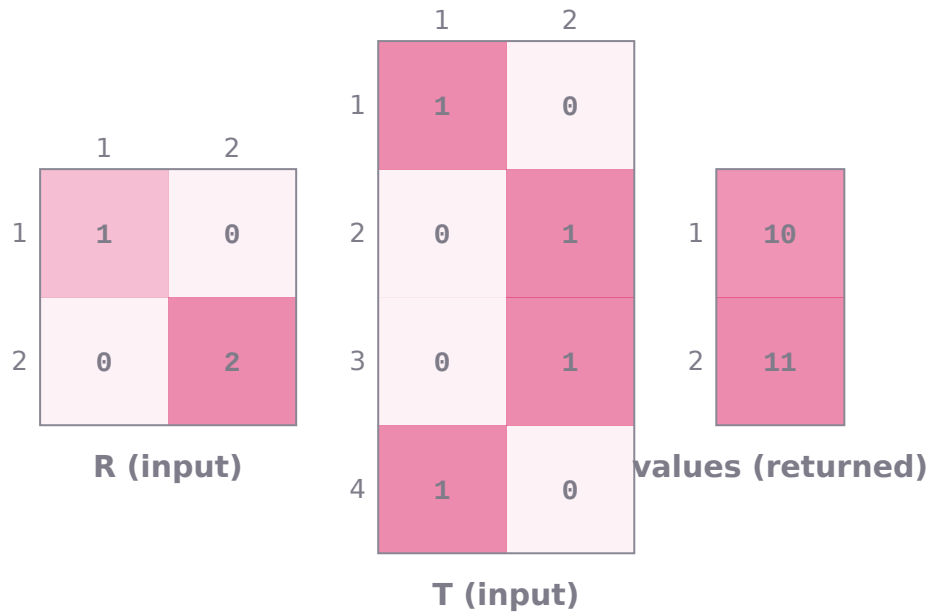
input	meaning	condition
`gamma`	discount factor $(0 < \gamma < 1)$	

output	meaning
`value_state_1`	optimal value of state 1
`values`	optimal value of every state
`policy`	optimal action in each state, 1-based (the lowest-numbered one on a tie)

- **Method:** policy iteration in exact arithmetic; certified by the Bellman optimality equation
- **Assumes:** $0 < \gamma < 1$; rows of T sum to 1; at most 10 states and 5 actions
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



optimal action in each state, 1-based (the lowest-numbered one on a tie) (policy) at each optimal value of every state (values), both returned by optimization-ai999x for the worked example (R = [[1, 0], [0, 2]], T = [[1, 0], [0, 1], [0, 1], [1, 0]], gamma = 0.9): 10 → 1; 11 → 2.



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): R (input); T (input); values (returned). Returned with them by optimization-ai999x: value_state_1 = 10. Other inputs: gamma = 0.9.

```
optimization_ai999x.solve('mdp_discounted', R=[[1, 0], [0, 2]], T=[[1, 0], [0, 1], [0, 1],
[1, 0]], gamma=0.9)
# -> value_state_1 = 10, values = [10.0, 11.0], policy = [1.0, 2.0]   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Solve the discounted Markov decision process with rewards [[1, 0], [0, 2]], transitions [[1, 0], [0, 1], [0, 1], [1, 0]] and discount factor 0.9.	SOLVED	value_state_1 = 10, values = [10.0, 11.0], policy = [1.0, 2.0]
Solve the discounted Markov decision process with rewards [[1, 0], [0, 2]], transitions [[1, 0], [0, 1], [0, 1], [1, 0]] and discount factor 1.	INVALID_INPUT	Discounted Markov decision process (optimal values and policy): the discount factor must be between 0 and 1 (exclusive).

Chromatic number (minimum graph colouring), `graph_coloring`

The fewest colours needed so that adjacent nodes get different colours (scheduling exams, assigning frequencies), and one such colouring.

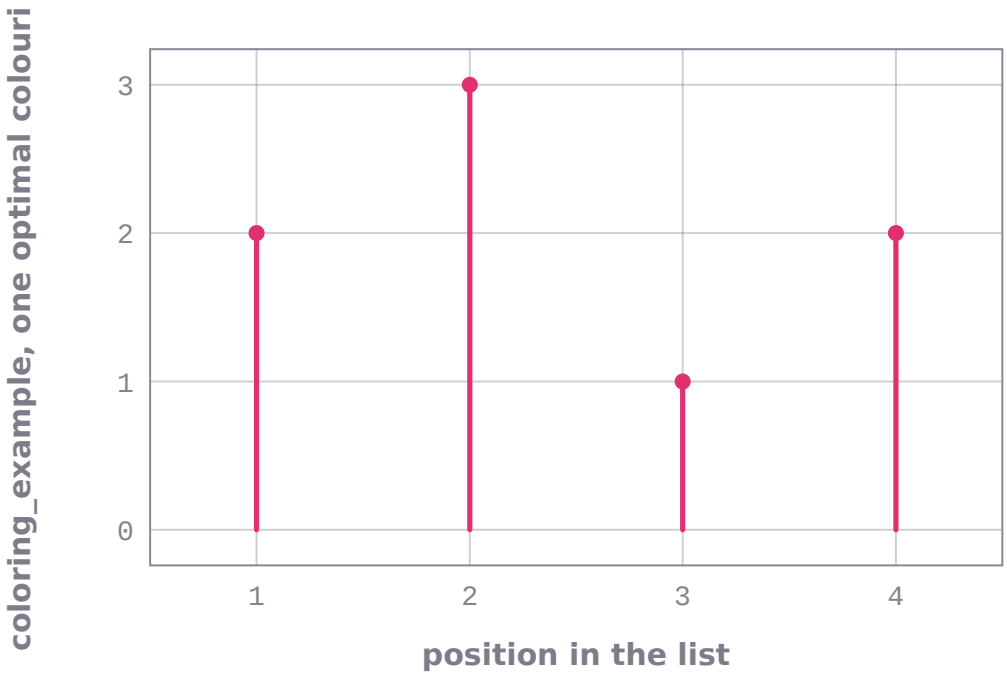
$$\chi(G) = \min\{k : \exists\, c : V \to \{1..k\},\ c(u) \neq c(v)\ \forall uv \in E\}$$

input	meaning	condition
`edges`	edges 'A-B, B-C, ...' (undirected)	

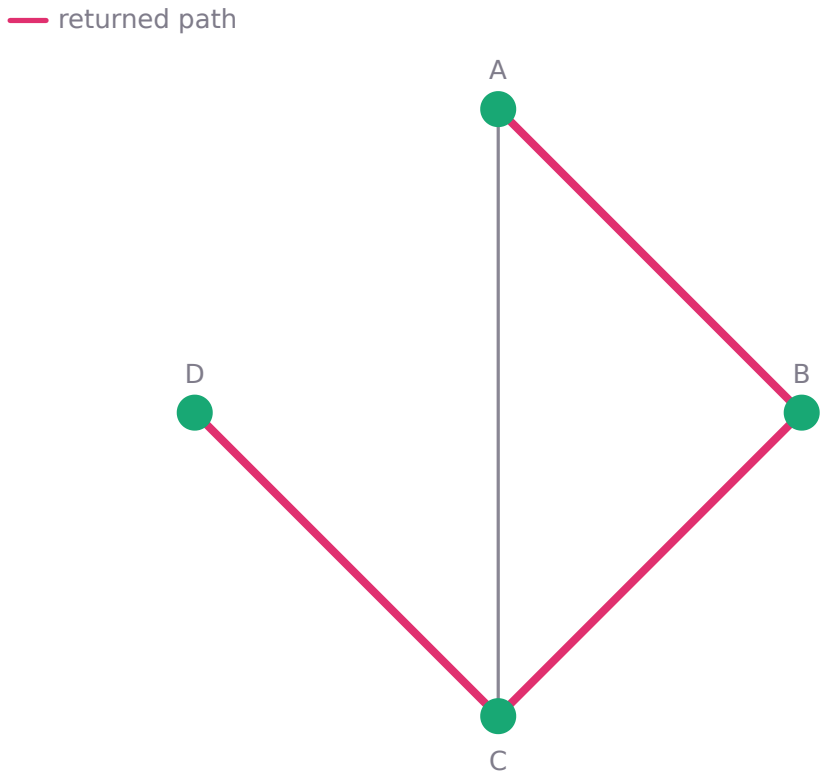
output	meaning
`chromatic_number`	minimum number of colours
`coloring_example`	one optimal colouring, in the order of 'nodes'

output	meaning
`nodes`	node order (first appearance)

- **Method:** exact backtracking for $k = 1$, 2, ... colours (largest degree first, symmetric colours skipped)
- **Assumes:** at most 14 nodes; no self-loops
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



one optimal colouring, in the order of 'nodes' (coloring_example) returned by optimization-ai999x for the worked example, one value for each position in the list (4 values, first 2, last 2, largest 3). Inputs: edges = A-B, B-C, C-A, C-D.



The network edges (edges 'A-B, B-C, ...' (undirected)) of the worked example (4 nodes, 4 edges, laid out on a circle); highlighted: the returned path A → B → C → D. Returned: chromatic_number = 3.

```
optimization_ai999x.solve('graph_coloring', edges='A-B, B-C, C-A, C-D')
# -> chromatic_number = 3, coloring_example = [2.0, 3.0, 1.0, 2.0], nodes = ['A', 'B', 'C',
'D']   (SOLVED)
```

```
optimization_ai999x.solve('graph_coloring', edges='A-B, B-C, C-D, D-A')
# -> chromatic_number = 2, coloring_example = [1.0, 2.0, 1.0, 2.0], nodes = ['A', 'B', 'C',
'D']   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the chromatic number of the graph with edges A-B, B-C, C-A, C-D.	SOLVED	chromatic_number = 3, coloring_example = [2.0, 3.0, 1.0, 2.0], nodes = ['A', 'B', 'C', 'D']
Find the chromatic number of the graph with edges A-A, A-B.	INVALID_INPUT	Chromatic number (minimum graph colouring): a node cannot be joined to itself.

Minimum vertex cover (and maximum independent set), `minimum_vertex_cover`

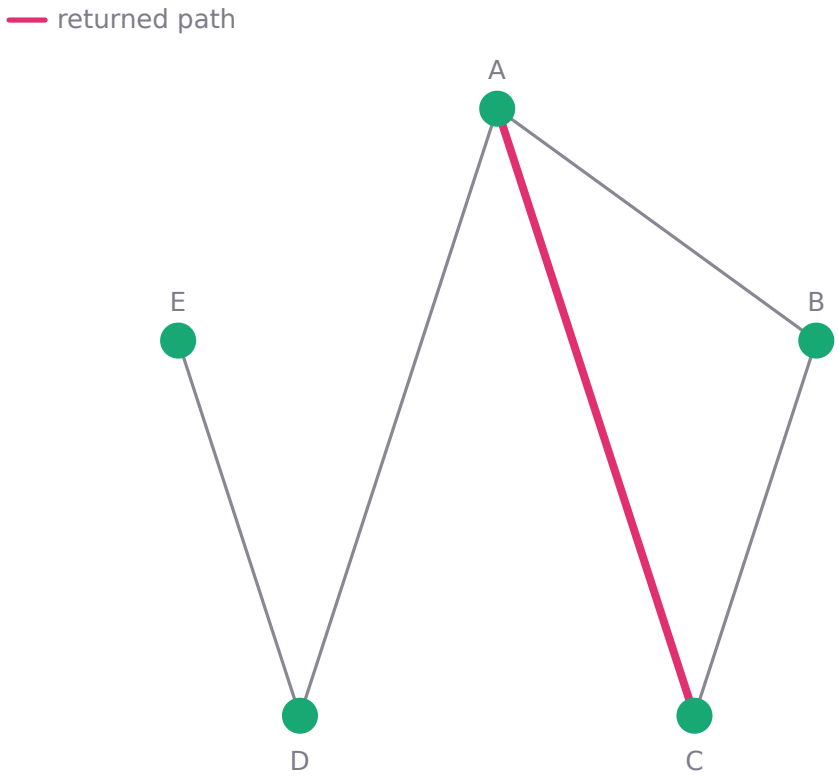
The fewest nodes that touch every edge of a graph (placing guards or sensors), one such set, and the size of a largest set of pairwise non-adjacent nodes.

$$\tau(G) = \min\{|C| : C \cap \{u, v\} \neq \emptyset \; \forall uv \in E\}, \quad \alpha(G) = |V| - \tau(G)$$

input	meaning	condition
`edges`	edges 'A-B, B-C, ...' (undirected)	

output	meaning
`min_cover_size`	size of a minimum vertex cover
`cover_example`	one minimum vertex cover
`max_independent_set_size`	size of a maximum independent set (nodes minus cover)

- **Method:** exact branch and bound (a node of largest degree in the cover, or all its neighbours)
- **Assumes:** at most 30 nodes; no self-loops; only nodes that appear in an edge count
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



The network edges (edges 'A-B, B-C, ...' (undirected)) of the worked example (5 nodes, 5 edges, laid out on a circle); highlighted: the returned path A → C → E. Returned: min_cover_size = 3, max_independent_set_size = 2.

```
optimization_ai999x.solve('minimum_vertex_cover', edges='A-B, A-C, A-D, B-C, D-E')
# -> min_cover_size = 3, cover_example = ['A', 'C', 'E'], max_independent_set_size = 2
(SOLVED)
```

```
optimization_ai999x.solve('minimum_vertex_cover', edges='A-B, A-C, A-D')
# -> min_cover_size = 1, cover_example = ['A'], max_independent_set_size = 3   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the minimum vertex cover of the graph with edges A-B, A-C, A-D, B-C, D-E.	SOLVED	min_cover_size = 3, cover_example = ['A', 'C', 'E'], max_independent_set_size = 2
Find the minimum vertex cover of the graph with edges A-B, B-B.	INVALID_INPUT	Minimum vertex cover (and maximum independent set): a node cannot be joined to itself.

LP duality and sensitivity ranges, `lp_duality_sensitivity`

The dual of a linear program and its optimal solution (the shadow prices), the range of optimality of every objective coefficient and the range of feasibility of every right-hand side, for an optimum with a unique nondegenerate basis.

$$\max c^\top x \text{ s.t. } Ax \le b,\ x \ge 0 \iff \min b^\top y \text{ s.t. } A^\top y \ge c,\ y \ge 0; \quad x_B + \delta B^{-1} e_i \ge 0, \quad d_k - \Delta (i$$

input	meaning	condition
`sense`	maximize or minimize	

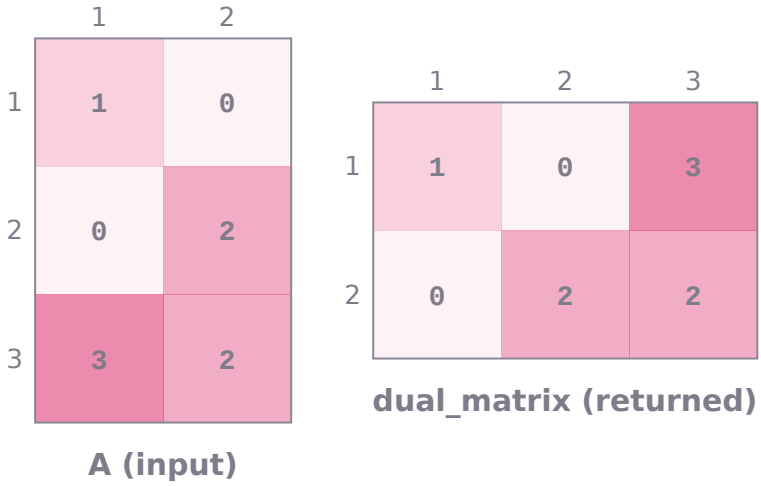
input	meaning	condition
`c`	objective coefficients, one per variable	
`A`	constraint coefficients, one row per constraint	
`b`	right-hand sides, one per constraint	
`relations`	relation of each constraint: <=, >= or = (default all <=)	

output	meaning
`cost_lower`	lowest value of each objective coefficient keeping the solution optimal (None: no limit)
`cost_upper`	highest value of each objective coefficient keeping the solution optimal (None: no limit)
`rhs_lower`	lowest value of each right-hand side keeping the shadow prices valid (None: no limit)
`rhs_upper`	highest value of each right-hand side keeping the shadow prices valid (None: no limit)
`optimal_value`	optimal value of the objective
`x`	the optimal solution, variables in order of first appearance
`dual_solution`	optimal dual variables = shadow prices, one per constraint
`dual_value`	optimal value of the dual (equal to the primal's: strong duality)
`dual_sense`	the dual's objective sense
`dual_objective`	the dual's objective coefficients (the primal right-hand sides)
`dual_matrix`	the dual's constraint rows (the primal's columns: A transposed)
`dual_relations`	relation of each dual constraint
`dual_signs`	sign of each dual variable (>= 0, <= 0 or free)

- **Method:** exact two-phase simplex; with the optimal basis B: $y = c_B\, B^{-1}$, reduced costs $d_j = c_j - y$ A_j; range of a nonbasic cost from its reduced cost, of a basic cost from the ratio test d_k / (B^-1 A_k)_r over the nonbasic columns; range of b_i from the ratio test -x_B / (B^-1 e_i)
- **Assumes:** all variables >= 0; a feasible LP with a finite optimum; a nondegenerate optimum (every basic variable > 0); a unique optimum (every reduced cost > 0); no redundant equality constraint; at most 12 variables and constraints
- **Evidence:** certified procedure (gates: conditions, control, examples, fast_enough, questions, reference, sensitivity, templates)



optimal dual variables = shadow prices, one per constraint (dual_solution) returned by optimization-ai999x for the worked example, one value for each b (right-hand sides, one per constraint) (3 values, first 0, last 1, largest 1.5); with rhs_lower (lowest value of each right-hand side keeping the shadow prices valid (None: no limit)), dual_objective (the dual's objective coefficients (the primal right-hand sides)). Inputs: sense = maximize, c = [3, 5], A = [[1, 0], [0, 2], [3, 2]], b = [4, 12, 18].



The matrices of the worked example as heatmaps (rows and columns numbered; each cell shaded by its magnitude, negative values in the second colour; each panel shaded by its own largest magnitude): A (input); dual_matrix (returned). Returned with them by optimization-ai999x: optimal_value = 36, dual_value = 36. Other inputs: sense = maximize, c = [3, 5], b = [4, 12, 18].

```
optimization_ai999x.solve('lp_duality_sensitivity', sense='maximize', c=[3, 5], A=[[1, 0],
[0, 2], [3, 2]], b=[4, 12, 18])
# -> optimal_value = 36, x = [2.0, 6.0], dual_solution = [0.0, 1.5, 1.0], dual_value = 36,
cost_lower = [0.0, 2.0], cost_upper = [7.5, None], rhs_lower = [2.0, 6.0, 12.0], rhs_upper
= [None, 18.0, 24.0], dual_sense = minimize, dual_objective = [4.0, 12.0, 18.0],
dual_matrix = [[1.0, 0.0, 3.0], [0.0, 2.0, 2.0]], dual_relations = >=, >=, dual_signs = >=
0, >= 0, >= 0   (SOLVED)
```

```
optimization_ai999x.solve('lp_duality_sensitivity', sense='minimize', c=[2, 3], A=[[1, 1], [1, 3]], b=[4, 6], relations='>=, >=')
# -> optimal_value = 9, x = [3.0, 1.0], dual_solution = [1.5, 0.5], dual_value = 9, cost_lower = [1.0, 2.0], cost_upper = [3.0, 6.0], rhs_lower = [2.0, 4.0], rhs_upper = [6.0, 12.0], dual_sense = maximize, dual_objective = [4.0, 6.0], dual_matrix = [[1.0, 1.0], [1.0, 3.0]], dual_relations = <=, <=, dual_signs = >= 0, >= 0   (SOLVED)
```

Questions in words (documented forms), as executed:

question	outcome	result
Find the ranges of optimality and feasibility and the dual program: maximize 3x + 5y subject to x <= 4, 2y <= 12, 3x + 2y <= 18, x, y >= 0.	SOLVED	optimal_value = 36, x = [2.0, 6.0], dual_solution = [0.0, 1.5, 1.0], dual_value = 36, cost_lower = [0.0, 2.0], cost_upper = [7.5, None], rhs_lower = [2.0, 6.0, 12.0], rhs_upper = [None, 18.0, 24.0], dual_sense = minimize, dual_objective = [4.0, 12.0, 18.0], dual_matrix = [[1.0, 0.0, 3.0], [0.0, 2.0, 2.0]], dual_relations = >=, >=, dual_signs = >= 0, >= 0, >= 0
Do the sensitivity analysis: minimize 2x + 3y subject to x + y >= 4, x + 3y >= 6, x, y >= 0.	SOLVED	optimal_value = 9, x = [3.0, 1.0], dual_solution = [1.5, 0.5], dual_value = 9, cost_lower = [1.0, 2.0], cost_upper = [3.0, 6.0], rhs_lower = [2.0, 4.0], rhs_upper = [6.0, 12.0], dual_sense = maximize, dual_objective = [4.0, 6.0], dual_matrix = [[1.0, 1.0], [1.0, 3.0]], dual_relations = <=, <=, dual_signs = >= 0, >= 0
Find the ranges of optimality: maximize 2x + 4y subject to x + 2y <= 8, x <= 5, x, y >= 0.	INVALID_INPUT	LP duality and sensitivity ranges: the optimum is not unique (alternative optimal solutions): the ranges depend on which
Find the dual program: maximize x + y subject to x + y <= 2, x + y >= 5, x, y >= 0.	INVALID_INPUT	LP duality and sensitivity ranges: the constraints are infeasible: no point satisfies all of them.
Find the ranges of optimality: maximize 3x + 5y subject to x + 2y <= 14.	OUT_OF_SCOPE	Out of scope: variables without a stated sign (state x >= 0 for every variable; free variables are not supported).